

# Self-Aware Networks: A Falsifiable Theory-and-Methods Framework for Phase-Gradient Routing and Recurrent Oscillatory Reconstruction

*Oscillatory Computational Agency*

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## Abstract

Neural systems face a routing and reconstruction problem. It is not enough to know which neurons are active: an account must explain how the timing and location of activity change a particular receiver, select a downstream path, and update a distributed model of the body and world. Phase-of-firing codes, frequency-selective effective connectivity, transient synchrony, recurrent processing, and cortical traveling waves describe important parts of this problem. The missing operation is a receiver-relative difference that can convert those population dynamics into a specific change of access, routing, reconstruction, or action.

Self-Aware Networks (SAN) proposes that a neural receiver compares arriving activity with its own ongoing state, learned timing relationships, and local excitability. When the resulting difference changes what that receiver or its downstream pathway can do, SAN calls it a **Phase Wave Differential (PWD)**. Recurrent neural arrays can then use sequences of these receiver-relative changes to constrain and update a distributed self / world model. SAN calls this joined receive-transform-project-reconstruct operation **Neural Array Projection Oscillation Tomography (NAPOT)**. When repeated structural matches and phasic updates reshape the network's continuing context, the resulting constructive process is called **Neural Rendering**. These terms name successive operations within one ordered causal proposal.

I formalize this proposal with a receiver-relative coincidence window, a delayed coupled-field model, a cross-scale phase representation, and a candidate reconstruction operator. A relatively persistent sparse structural state is separated from a fast residual state in phase, instantaneous frequency, amplitude, and one explicitly defined transmission variable. Object-, body-, or task-relative reference frames may condition that state. The reconstruction claim becomes empirical when multiple distributed projections constrain a latent state through an identifiable operation that outperforms rate-, power-, coherence-, connectivity-, and recurrent-processing controls. I specify observational and causal experiments, including incremental decoding, phase-matched perturbation, receiver-window shifts, wave-intersection / non-additive-interaction tests, embodied visuomotor conflict, and artificial-network benchmarks. The framework is falsified or materially narrowed if phase-gradient variables add no reproducible information, if phase-specific perturbations fail under matched controls, or if the proposed reconstruction is not identifiable. To connect the mechanism to embodied behavior, I specify a loop in which motor command and efference copy generate predicted sensory consequences, visual and proprioceptive return signals update distributed body state, and slower contextual activity shapes faster sensory-motor updates. SAN calls this layered visual-motor-reafferent loop the **Gamma Consideration Sandwich**. A preparation-specific laminar timing pattern provides a discriminator. A dated provenance section traces the framework from a 2011 predictive-self and distributed-observer formulation, through 2012 recursive feedback and projector architecture and 2017 spatial, coincidence, difference-transmission, and recurrent-rendering atoms, to the mature NAPOT / PWD terminology and phase-specific composite published in 2022. The account locates consciousness-relevant computation in distributed recurrent transformation and informative differentiation, rather than in undifferentiated oscillation or global synchrony. Neural coherence is formalized in its classical signal-analysis sense; receiver-relative information means discriminable access and causal use; and scale invariance requires an explicit transformation and scaling

law. Phenomenal sufficiency and machine sentience remain further empirical questions rather than premises of the mechanism.

communication through coherence; predictive processing; binding; self-model; embodiment; body representation; visuomotor integration; neural dynamics; tomographic reconstruction

**Figure status:** Figures 1, 2A-2C, and 3-6 are author-directed explanatory schematics with two complementary jobs: make the proposed SAN operation visible and show how that operation could be measured or distinguished from alternatives. Figure 2A is the positive population-level NAPOT mechanism map; Figure 2B supplies a candidate local receive-transform-re-express motif; and Figure 2C shows how SAN proposes that motif may participate in nested recurrent organization. Figure 5 is a bounded neural preparation for testing one COT branch, not a map of the entire cellular theory. The figures are not anatomical atlases, recorded data, or empirical confirmation. Their literature-backed components and SAN-specific arrows remain governed by the definitions, claim classes, controls, and failure conditions in the manuscript.

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**Keywords:** neural oscillations; traveling waves; phase coding; recurrent processing;

## Scope and evidence boundary

This manuscript reports no new human or animal experiment and no completed empirical benchmark. Its release endpoint is an explicit theory, operational model, discriminating prediction set, and reproducible prospective analysis program. Statements are labeled so that established component mechanisms are not confused with evidence for the full Self-Aware Networks framework. This draft is also governed by a semantic-conservation rule. Every substantive construct, mechanism, example, genealogy branch, limitation, and source family in the 2025 *Oscillatory Computational Agency* First Draft must remain in this paper, be strengthened here, or be retained here with an explicit correction and claim-status boundary. Revision is permitted; silent deletion is not. Draft 11 remains the immutable rollback baseline while this integration is audited. Every correction inherited from Drafts 2-11 was rechecked through a retroactive source audit that compared the original wording, the first revision that changed it, and Micah's wider dated corpus. That audit treated an apparent error first as a possible typo or compressed legitimate mechanism, then accepted a recovered interpretation only when another source corroborated it. Corrections may separate variables, scales, evidence levels, or timescales; they may not silently replace the intended SAN narrative.

## 1. Introduction

A nervous system must continuously transform partial, delayed, noisy, and modality-specific signals into actions that remain coordinated with an organism and its surroundings. The anatomical substrate is distributed: visual features, auditory structure, interoceptive signals, memory, goals, and motor plans are not computed at a single point. Yet perception and action often exhibit a temporarily coherent organization. Any mechanistic account must therefore explain both specialization and coordination without inserting a hidden observer that reads a finished internal screen.

Several mature research programs constrain this problem. Recurrent and predictive-processing models describe how top-down state interacts with bottom-up input [4,22]. Synchronization and communication-through-coherence accounts describe how relative timing can regulate effective communication among neural populations [5-10]. Phase-of-firing experiments show that timing relative to an ongoing rhythm can carry information beyond spike count [11], while theta-gamma models organize ordered items by phase [12]. Dendritic work establishes that neurons are structured nonlinear integrators rather than point relays [13]. Traveling-wave studies show spatially organized propagation in several cortical regions, frequencies,

and behavioral contexts [14,15]. Together, these findings establish that neural communication is temporally and spatially structured. They leave open the paper’s central problem: how an arriving pattern acquires a receiver-specific consequence and how many such consequences become a recurrent reconstruction of the body, world, and available actions.

The memory-prediction, sparse-distributed-representation, active-dendrite, and Thousand Brains programs supply additional constraints. Active-dendrite models show how branch-local synaptic clusters can recognize many sparse input patterns and support sequence prediction [69,70]. Column-level models combine sensory evidence with object-relative location and use lateral interactions to accelerate disambiguation [23]. A later framework proposes location / reference-frame codes and multiple distributed object models across cortical columns [71]. Current Thousand Brains engineering documentation emphasizes sensorimotor learning, structured reference frames, modular column-like learning units, lateral consensus, and hierarchical composition while distinguishing that implementation from earlier HTM / SDR systems [72]. These programs supply components and strong comparators, but the receiver-relative conversion from an arriving dynamical pattern to a changed route remains the operation that must be specified.

Self-Aware Networks (SAN) connects those levels by treating neurons and neural groups as stateful receivers and projectors. An arriving pattern is evaluated against the receiver’s current and learned state; a consequential difference changes the accessible downstream route; and recurrent arrays use the resulting changes to refine a distributed reconstruction. SAN’s mature 2022 vocabulary names the consequential receiver-relative change a **Phase Wave Differential** and the joined recurrent array operation **Neural Array Projection Oscillation Tomography** [1,2]. The original paper expanded the proposal across molecules, cells, networks, agency, consciousness, and artificial intelligence. That breadth obscured the framework’s testable center. It also blurred three propositions that must be kept separate:

1. known biological mechanisms can support temporally structured communication;
2. SAN proposes a particular composite operation built from those mechanisms; and
3. a much stronger claim would be that this operation is necessary or sufficient for conscious experience.

This version concentrates on the second proposition and treats the third as open. The central hypothesis is:

**NAPOT hypothesis.** Local phase gradients, wave direction, and receiver-relative phase relationships provide content-specific routing variables beyond rate, power, static connectivity, and scalar synchrony; recurrent neural arrays use these variables to constrain and update a distributed latent model that guides report and action.

The hypothesis is deliberately conjunctive. Showing that traveling waves exist is insufficient. Showing that phase improves decoding is insufficient to establish causal routing. Showing causal routing is insufficient to establish tomography. A literal tomographic claim additionally requires multiple projections, a latent target, an identifiable measurement model, a reconstruction criterion, and performance that survives appropriate controls.

The paper makes four contributions. First, it defines the SAN terms operationally. Second, it gives a phenomenological formal model that can be fitted to multichannel neural data. Third, it presents falsifiable observational, perturbational, and engineering tests. Fourth, it separates the theory’s scientific content from its dated provenance. The provenance record matters for accurate citation, but chronology is not evidence that a mechanism is true.

The 2025 First Draft established SAN’s first formal DOI and remains the immutable citation anchor for that version [1]. This revision is a lossless scientific successor: it preserves that record while making the mechanism, comparisons, measurements, and failure conditions more explicit.

## 2. Scope and claim-status taxonomy

### 2.1 Four evidential classes

Every substantive statement in this paper belongs to one of four classes.

| Class                        | Meaning                                                                                        | Examples in this paper                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Established component</b> | Supported independently of SAN; scope remains that of the cited result.                        | Dendritic nonlinear integration; phase-of-firing information; transient synchrony; cortical traveling waves.                                |
| <b>SAN synthesis</b>         | A proposed composite or operational definition assembled from components.                      | Receiver-defined coincidence windows; PWD routing plus recurrent latent reconstruction.                                                     |
| <b>Prediction</b>            | A result that should occur if the synthesis is correct but has not yet been demonstrated here. | Phase-gradient features add out-of-sample information above matched controls; phase-preserving power manipulations alter a predicted route. |
| <b>Metaphor or heuristic</b> | Intuition that does not count as a mechanism unless mapped to observables.                     | “Canvas and ink,” “projector,” “volumetric television,” and tomography before an inverse operation is identified.                           |

Evidence for a component is compatible with SAN but does not validate the SAN synthesis. A finding can also support one arrow in the causal chain while leaving every downstream arrow open.

### 2.2 From an ordinary routing problem to the PWD solution

Consider a familiar event. You turn your eyes, yet the room remains stable; you reach for a cup that is lighter than expected, and your grip changes before the cup escapes your hand. In both cases, incoming activity does not enter a blank system. It reaches cells and circuits that are already active, inhibited, expectant, and shaped by recent history. The same sensory event can therefore have different consequences at different receivers or at different moments.

The central gap can be stated without comparing labels. Neuroscience can measure when an event occurs within an ongoing rhythm, how consistently two signals vary together, and how an organized pattern moves across tissue. Those measurements establish that timing, relation, and propagation matter. The unresolved operation is why the same arriving pattern affects one receiver, fails to affect another, or changes its effect when the receiver’s state changes.

A solution requires an edge-level variable. It must compare the delayed incoming pattern with the receiver’s ongoing phase, learned expectation, excitability, and coincidence window; it must remain sensitive to changes in timing, instantaneous frequency, amplitude, duration, inhibition, and transmission; and it must predict a downstream consequence. SAN names this receiver-relative, state-dependent consequential difference the **Phase Wave Differential (PWD)**. The public SAN record fixes the named construct in 2022 and joins it to a larger sequence in which neural arrays receive, transform, re-express, recurrently reconstruct, and act [43].

The proposed operation determines what an experiment must show. A PWD result must add explanatory or causal information beyond matched firing rate, band power, scalar coherence, static connectivity, waveform, and common-input controls. The theory predicts which differential should matter, where it should be received, what transformation should follow, and which selective perturbation should alter that transformation. These commitments make PWD a falsifiable computational and neurophysiological construct.



The model operates at the level of classical neurophysiological observables. LFP, ECoG, MEG, and spike-field measurements estimate population relations; receiver-specific functional claims are assigned only to differences that survive the stated controls and predict downstream effects. Stochastic dynamics, structured desynchronization, inhibition, and graded forms of biological agency remain part of the model. Distributed, reportable self / world modeling and action selection are its immediate endpoints; phenomenal sufficiency and artificial sentience are further empirical questions. Historical comparisons establish provenance and conceptual boundaries, while the experiments in this paper decide the mechanism's scientific adequacy.

## 2.3 Unit of analysis

The theory is intentionally compatible with multiple spatial scales, but a study must choose one before analysis. A node may be a neuron, electrode-local population, cortical parcel, or artificial unit. Results cannot be transferred automatically between scales. The primary near-term target is mesoscopic electrophysiology - multielectrode, ECoG, or source-resolved MEG - because it provides both temporal precision and a spatial arrangement from which phase gradients can be estimated. MEG gradients require source-forward simulations, leakage controls, and recovery of known synthetic waves before they are interpreted as local mesoscopic propagation.

## 3. Operational definitions

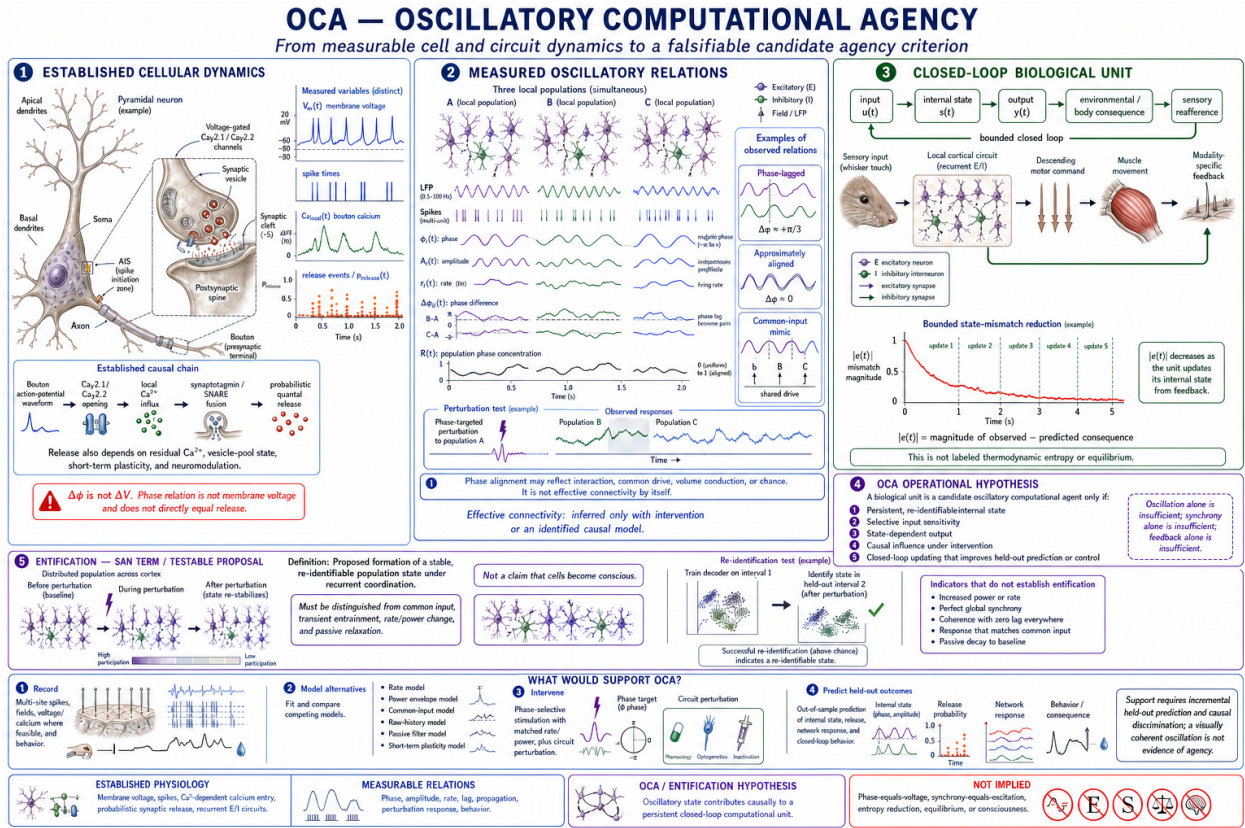
### 3.1 From stateful interaction to operational agency

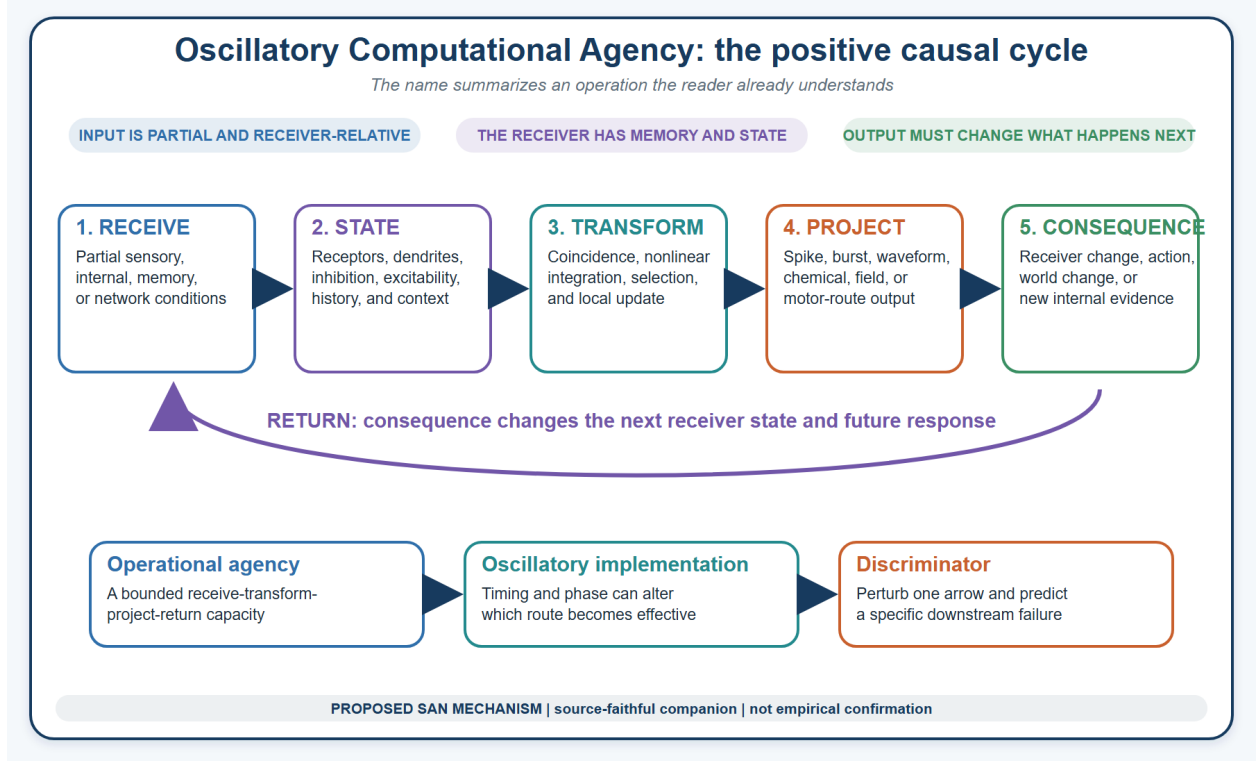
The same chemical, electrical, or sensory input need not produce the same response twice. A cell or network already has a membrane state, molecular history, learned structure, and recent pattern of activity when the input arrives. The relevant scientific problem is therefore to describe a bounded system whose present state changes how it transforms input and whose output can alter its own future conditions.

In this paper, a system qualifies as an *agent* when it (i) has a state, (ii) selectively transforms inputs as a function of that state, (iii) changes an output or environment, and (iv) can modify future transformations through feedback. This is an operational and graded definition. When timing-dependent state and feedback implement those operations in biological or artificial networks, SAN calls the resulting hypothesis **Oscillatory Computational Agency (OCA)**. The measurements must establish the boundary, state dependence, input-output transformation, and feedback effect; mere change or oscillation is insufficient.

**Figure 1.** OCA evidence-to-hypothesis ladder. The cellular and population panels separate established physiology from measured phase relations; closed-loop agency and the composition of many bounded agents into coordinated behavior remain operational SAN hypotheses. A phase difference is not membrane voltage, causal direction, entropy, equilibrium, or consciousness. Support requires held-out prediction or control, causal discrimination under intervention, and re-identification beyond common-input, rate, power, raw-history, and passive-decay baselines.

**Figure 1B.** Positive OCA mechanism companion. Partial conditions meet a receiver whose learned structure and current state change what the input can do; the receiver transforms those conditions, projects an output, and receives consequences that alter the next cycle. OCA names the bounded receive-transform-project-return capacity after the operation has been stated in ordinary functional terms. Timing and phase are candidate implementation variables, not substitutes for the causal cycle. Each arrow remains a proposed, measurable relation and does not establish that every biological scale performs the same computation or supports consciousness.





**Figure 2:** Positive Oscillatory Computational Agency causal cycle

### 3.2 How learned structure changes what a receiver can recognize

A neuron does not respond only to the signal that arrives now. Synaptic weights and locations, dendritic geometry, receptor and channel composition, excitability, and recent plasticity determine which features it detects and how strongly it responds. Established plasticity and active-dendrite accounts therefore require any model of reception to include a comparatively persistent learned configuration alongside faster membrane and oscillatory state.

SAN calls that relatively persistent response-shaping configuration an **expert data structure (EDS)**. “Expert” means tuned by history rather than infallible. The construct is a parameterized transformation, not a stored picture. It earns explanatory value only if measured structural state improves held-out prediction of receiver selectivity beyond current input and fast state alone.

For unit  $i$ , write its slower state as  $M_i(t)$ . Its response to incoming signals  $X_{\mathcal{N}_i}(t)$  is

$$y_i(t) = F_i(X_{\mathcal{N}_i}(t), M_i(t), q_i(t)), \quad (1)$$

where  $q_i(t)$  collects faster variables such as membrane potential, local phase, adaptation, and inhibitory context. The distinction between  $M_i$  and  $q_i$  is timescale-dependent.

For a bounded engineering abstraction, let  $s(t) \in \{0, 1\}^n$  be a sparse population activity vector and  $D_{ibh} \in \{0, 1\}^n$  the learned synaptic mask of branch or segment  $b$  of unit  $i$  under reference-frame hypothesis  $h$ . Define

$$m_{ibh}(t) = \mathbf{1}[\langle D_{ibh}, s(t) \rangle \geq \vartheta_{ibh}]. \quad (2)$$

This abstraction follows sparse-pattern / active-dendrite models [69,70] and belongs to the slower state  $M_i$ . It does not assert that biological dendrites are binary, that one segment stores a complete object, or that every detectable difference creates a new category. A candidate category is admitted only under a preregistered novelty rule requiring low overlap with existing structural codes, held-out separability, recurrence across contexts, predictive or policy value above added capacity alone, and stability under matched nulls.

### 3.3 How partial population states are transformed and passed onward

No single neuron carries every feature of a room, body, memory, or intended movement. The brain must combine partial population states while preserving where they came from, when they arrived, and which downstream populations can use them. Anatomical divergence, convergence, and recurrent connectivity provide the established substrate; the remaining modeling task is to specify the population boundary, its organized output, and the state-dependent transformation at the receiver.

SAN calls a selected population with jointly organized spatial and temporal output a **neural array**. Its structured output  $P_a(t)$  is a **projection**, and the downstream population’s state-dependent transformation is **reception**. The population need not be a regular anatomical grid. These definitions are tested by independently fixed parcellations and receiver-specific predictions rather than by resemblance to a visual screen.

Array boundaries and parcellations must be defined independently of the confirmatory outcome. Data-driven arrays are learned inside training data only, and results must be tested across a bounded set of preregistered alternative parcellations. A population may not be repartitioned after outcome inspection merely to make its projections appear nonredundant.

Because axons diverge and dendrites converge, a downstream unit receives partial contributions from many upstream units. Its result depends on connection topology, delays, current excitability, and the timing of those contributions. SAN’s claim is not merely that activity is transmitted, but that the receiver-relative temporal geometry helps select which distributed pattern becomes effective.

### 3.4 From the receiver-relative timing problem to the Phase Wave Differential

An incoming oscillatory event has no fixed consequence independent of its receiver. The same event can arrive while a downstream population is excitable, refractory, inhibited, phase-aligned, or outside its learned coincidence window. Rate, power, coherence, or the presence of a traveling wave alone cannot specify that edge-level consequence. The missing quantity must compare the delayed incoming event with the receiver’s current state and expected timing, then test whether the difference changes effective transmission. SAN calls that quantity the **Phase Wave Differential**.

The formal definition begins with measurable analytic phase. For a band-limited signal  $x_i(t)$ , define its analytic representation

$$a_i(t) = x_i(t) + i \mathcal{H}[x_i(t)] = A_i(t)e^{i\phi_i(t)}, \quad (3)$$

where  $\mathcal{H}$  is the Hilbert transform,  $A_i(t)$  is instantaneous amplitude, and  $\phi_i(t)$  is instantaneous phase. This definition is an estimator, not an assumption that every neural signal is a narrowband oscillator. Sensitivity to filtering, waveform shape, reference, and low amplitude must be measured explicitly.

Recorded LFP, ECoG, or MEG phase is not automatically the phase of receiver excitability. Where the theory interprets phase as an open or closed receiver window, an independent localizer or standardized probe must estimate a receiver-specific phase-response curve and demonstrate enough stability to predict transfer.

Otherwise  $\psi_{ij}$  may absorb reference, polarity, waveform, or source-leakage artifacts rather than a biological preferred offset.

For an edge assigned as  $j \rightarrow i$  with delay  $\tau_{ij}$ , the receiver-relative *phase-wave differential* (PWD) is

$$\Delta\phi_{ij}(t) = \text{Arg} \left[ e^{i(\phi_j(t-\tau_{ij})-\phi_i(t)-\psi_{ij})} \right], \quad (4)$$

where  $\psi_{ij}$  is an expected phase offset, estimated only from training data or specified by a mechanistic model. PWD is therefore not simply “phase.” It is a wrapped, edge-conditioned, receiver-relative deviation. The phase difference does **not** infer causal direction. “Directed PWD” may be used only when the edge direction is supplied independently by known anatomy, intervention, or a leakage-safe causal / effective-connectivity analysis.

In a single narrow frequency band, delay and expected phase offset are generally confounded: several  $(\tau_{ij}, \psi_{ij})$  pairs can produce the same observed phase relation. A confirmatory study must therefore use independently estimated delays, constraints shared across multiple frequencies, or report the resulting equivalence class and propagated uncertainty rather than presenting one pair as identified.

For a spatial phase field  $\phi(\mathbf{x}, t)$ , the local phase gradient is  $\mathbf{k}(\mathbf{x}, t) = \nabla\phi(\mathbf{x}, t)$ . For a locally planar wave, a phase-velocity estimate is

$$\mathbf{v}_\phi(\mathbf{x}, t) = -\frac{\partial_t \phi}{\|\nabla\phi\|^2} \nabla\phi, \quad (5)$$

The derivative of a wrapped phase cannot be taken naively. The analysis must use circular local regression or validated unwrapping on the cortical tangent plane or recording graph, with preregistered phase-quality, neighborhood-conditioning, and gradient thresholds. The expression above is the minimum-norm normal phase velocity for a locally planar wave, not a full material-flow velocity. Curved, rotating, source, sink, and spiral patterns require topology-aware estimators.

### 3.5 Tonic baseline and phasic update

*Tonic* and *phasic* describe relative timescales within a specified recording and task. Tonic state is the slower background configuration that changes receiver gain, timing, or context; phasic activity is a faster deviation that can update that state or be routed within it. These terms do not map one-to-one onto fixed frequency bands across all brain regions. Delta, theta, alpha, beta, or a stable gamma regime may serve as tonic context in a bounded preparation; gamma, high gamma, a burst, or another transient may be phasic relative to that context. A delta-, theta-, or other slower-band event can also be phasic if it is a preregistered transient deviation from the preparation’s current baseline. Broadband high gamma may largely track population spiking, and a nonsinusoidal slow waveform can generate apparent high-frequency harmonics. Frequency rank and band name alone therefore cannot assign the role.

SAN’s historical **tonic canvas / phasic ink** model is more specific than a generic slow / fast split [43,79]. The tonic state is an active predictive reference containing maintained expectations, receiver readiness, and currently available patterns. A phasic or high-phasic event is a sparse deviation that can introduce, amplify, suppress, or revise content. Recurrent propagation then attenuates, disperses, amplifies, or incorporates that deviation into a changed tonic state. SAN calls this cycle **Neural Rendering**: learned structural patterns are re-expressed as wave-pattern updates, distributed arrays receive and transform partial updates, and their recurrent interaction changes the active self / world latent without a separate viewer.

The information-theoretic reading is conditional rather than absolute. A familiar tonic state can have low event-wise surprise under a fitted receiver model while still carrying substantial structural, predictive, or mutual information. A phasic deviation can have greater self-information relative to that model because it is less expected, but rarity alone does not make the deviation meaningful or conscious. Its functional significance depends on whether the receiver can discriminate it, whether it changes a later state or action, and whether that change improves a declared task or prediction. Coherence and phase dispersion are therefore complementary routing conditions, not universal synonyms for information loss and information gain.

For scale  $s$ , signal  $X_s(\mathbf{x}, t)$ , and a preregistered slow operator  $\mathcal{L}_{\tau_s}$ , define a tonic reference

$$T_s(\mathbf{x}, t) = \mathcal{L}_{\tau_s}[X_s](\mathbf{x}, t). \quad (6)$$

If amplitude, phase, and instantaneous frequency are reliable, a dynamic differential may be represented as

$$D_s(\mathbf{x}, t) = [\Delta \log A_s, \text{wrap}(\phi_s - \Phi[T_s]), \omega_s - \Omega[T_s]]. \quad (7)$$

The operators, window, band, spatial scale, and reliability thresholds must be fixed without test labels. The differential is not automatically content. It becomes a candidate content-bearing update only if it predicts an independently defined latent or action above tonic state, rate, power, raw history, and equal-input recurrent controls.

The older inhibited / tonic / phasic / high-phasic four-state vocabulary is retained only as a preregistered event partition. It is not a claim that every synapse has four literal information levels or releases exactly zero, one, two, or three vesicles. Likewise, lower-frequency power often has greater magnitude in aperiodic neural spectra, but greater power does not by itself prove wider causal reach, binding, or synchronization.

Receivers with similar mean firing rates can nevertheless differ in the variability of their event timing, morphology, connectivity, and phase relations. The scientific problem is whether a stable multivariate response profile can help identify which receiver or route is active when mean rate alone is ambiguous. The ordinary coefficient of variation supplies one established component. For positive inter-event intervals  $\Delta t_{ik}$  in a fixed window, define

$$CV_i = \frac{\text{sd}_k(\Delta t_{ik})}{\text{mean}_k(\Delta t_{ik})}. \quad (8)$$

SAN proposes combining this statistic with phase, frequency, amplitude, morphology, and connectivity to form a candidate cell or population fingerprint. The historical name for this joint receiver descriptor is the **addressable coefficient of variation**. It becomes addressable only if held-out tests show that the joint fingerprint identifies or routes to the receiver beyond firing rate and cell identity. Failure to re-identify the receiver or predict its route under matched conditions rejects the addressing claim while leaving the ordinary CV measurement intact.

“Canvas,” “ink,” “brushstroke,” “voxel,” and “volumetric television” remain heuristics. The operational claim is a nested, measurable multiscale phase field and a recurrent state-update operator. The term *tomography* remains conditional on the stronger inverse and perturbation tests in Section 3.7.

### 3.6 How continuous integration yields a consequential event

Biological integration is continuous, nonlinear, and history-dependent. A postsynaptic or population event can nevertheless become discrete after integration crosses a threshold. Let  $e_j(t) = \sum_k \delta(t - t_{jk})$  be the event

train from source  $j$ , let  $r_{ij}(t) = (\kappa_{ij} * e_j)(t)$  be its normalized postsynaptic trace, and let the receiver's state-dependent integration window be  $W_i(q_i, t)$ . A minimal coincidence statistic that requires at least two sources is

$$C_i(t) = \sum_{j < k} \int_{t-W_i(q_i, t)}^t \frac{r_{ij}(s)r_{ik}(s)}{W_i(q_i, t)} ds. \quad (9)$$

Define a discrete receiver event only on a threshold **rising edge** outside a refractory interval:

$$b_i(t) = \mathbf{1} \left[ C_i(t) \geq \eta_i, C_i(t - \Delta t) < \eta_i, t - t_i^{\text{last}} \geq \rho_i \right]. \quad (10)$$

Kernel normalization keeps the scale interpretable as  $W_i$  changes; a concrete study must state the units and threshold convention. A richer  $G_i$  may include single-source drive and higher-order interactions, but a result called *coincidence* must isolate the interaction from a single large input. The phrase *coincidence is a bit* refers to the binary outcome  $b_i$  after continuous integration. It does not say that the inputs, the internal state, or all neural information are binary.

Receiver excitability windows, inhibitory timing, and delay-sensitive phase relations are established precursor ideas in temporal coding and communication-through-coherence. SAN uses those atoms but does not claim them as its delta. Its specific hypothesis is that receiver-conditioned timing combines with spatial wave geometry to make an incremental, causal contribution to recurrent reconstruction of a predefined latent state.

### 3.7 Reconstruction and tomography

The hypothesized neural computation and the experimenter's observation model must be separate. Let  $h_t$  be a downstream neural state and let each upstream array generate a biological projection  $p_a(t) = Q_a(q_a(t))$ . NAPOT proposes a neural update

$$h_{t+1} = U(h_t, p_1(t), \dots, p_A(t), a_t), \quad (11)$$

where  $a_t$  denotes action or efference-copy context. The experimenter does not observe that update directly. A recording modality instead supplies

$$m_a(t) = O_a(q_a(t)) + \epsilon_a(t), \quad (12)$$

and an external decoder estimates an independently observed target  $z(t)$ :

$$\tilde{z}(t) = D(m_1(t), \dots, m_A(t), m_{1:A}(< t)). \quad (13)$$

Successful decoding of  $z$  shows that information is available in the measurements; it does not show that the brain applies the experimenter's decoder or that the neural latent state is uniquely identified. A candidate linear forward model may be written as

$$\begin{aligned} \mathbf{m}(t) &= \mathbf{A} s(t) + \boldsymbol{\epsilon}(t), \\ \tilde{s}(t) &= \arg \min_s \|\mathbf{m}(t) - \mathbf{A} s\|_{\Sigma^{-1}}^2 + \lambda \Omega(s). \end{aligned} \quad (14)$$

For a  $d_s$ -dimensional admissible linear latent space, literal inversion requires at least  $d_s$  independent constraints,  $\text{rank}(\mathbf{A}) = d_s$ , and a reported condition number and noise-stability analysis. For a nonlinear forward map, the manuscript must state its gauge constraints, demonstrate local full-column-rank Jacobians on the tested region, and disclose global equivalence classes. The regularizer  $\lambda$  and  $\Omega$  are selected inside training data only. These checks can establish an experimenter’s identifiable inverse; the neural-computation claim additionally requires intervention on the hypothesized update  $U$ .

NAPOT earns the term *tomography* only if:

1. multiple distributed projections meet the relevant rank, conditioning, and stability conditions;
2. a latent state and an independently observed target are defined without circularly using the neural decoder;
3. spatial and phase relationships improve reconstruction beyond nonspatial controls and an equal-information flexible model receiving the same raw signals, geometry, and history;
4. the mapping generalizes to held-out trials, sessions, or subjects as preregistered; and
5. perturbing one neural projection changes a downstream neural state or behavior in the specific latent dimension that projection is predicted to constrain, beyond feature dropout in the experimenter’s decoder; and
6. the intervention result cannot be explained by matched local rate, power, evoked response, common input, or nonspecific disruption.

Otherwise, *oscillatory recurrent integration* is the more accurate description.

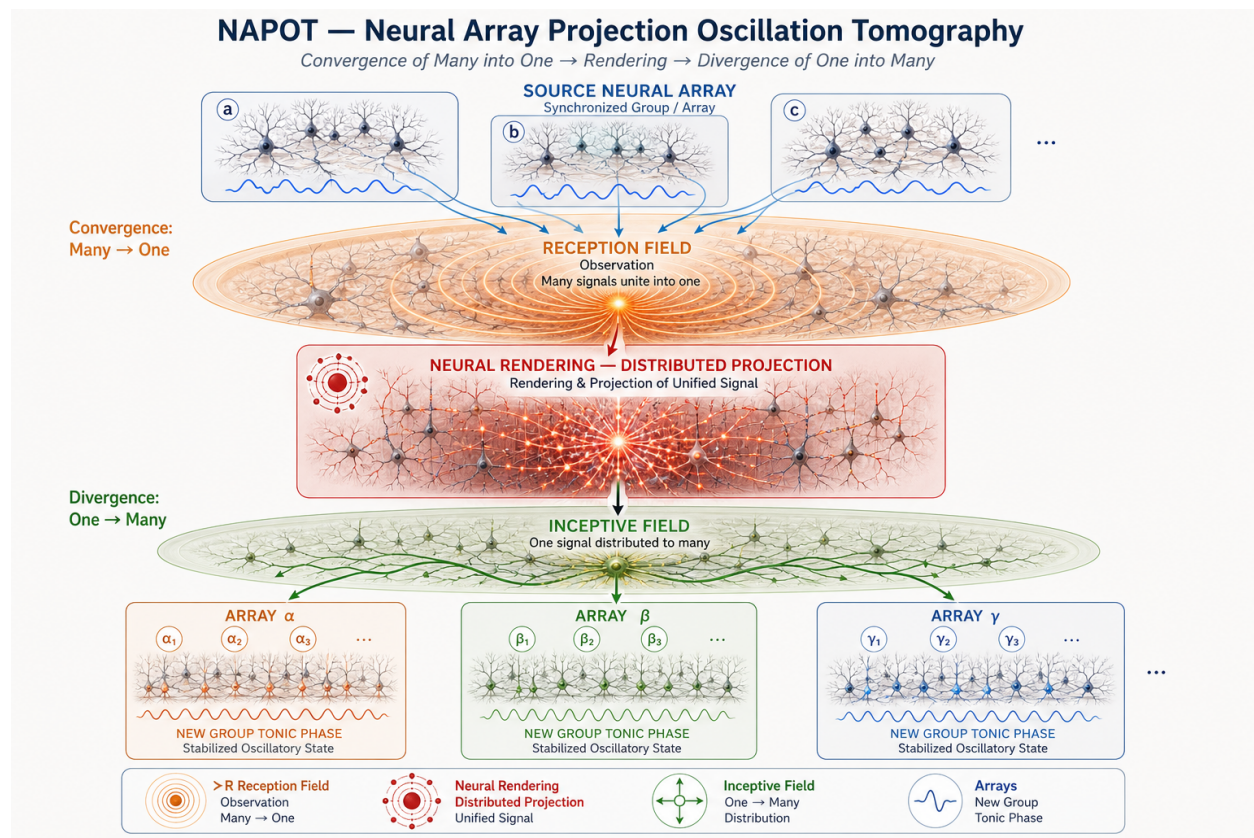
The First Draft described this operation as aligning and fusing many oscillatory *slices* into a dynamic three- or four-dimensional volume [1]. In operational language, a slice is one partial, time-indexed constraint supplied by a declared neural projection. Shared timing, geometry, receiver relations, and recurrent history determine whether several constraints can be aligned; the update operator  $U$  determines how they jointly revise the latent state. Fusion therefore means many-to-one constraint integration across arrays and time, not pasting literal image planes into an internal screen. SAN’s stronger consciousness proposal is that such a distributed, continuously revised volume can become the content used by the same system for report, memory, and action. Reconstruction alone does not establish phenomenal sufficiency; content-specific perturbation, loss, and rescue remain necessary discriminators.

The three NAPOT panels form one proposed operation at different explanatory resolutions. Figure 2B begins locally: a receiver samples partial inputs through its dendritic and synaptic organization, transforms their joint effect according to its present state, and re-expresses the resulting change through its output. Figure 2A then shows many such partial contributions converging on population states, participating in a reconstruction, and diverging as changed projections. Figure 2C places that receive-transform-re-express cycle inside recurrent micro-, meso-, and macro-scale organization, where a new projection becomes a new partial condition for another receiver.

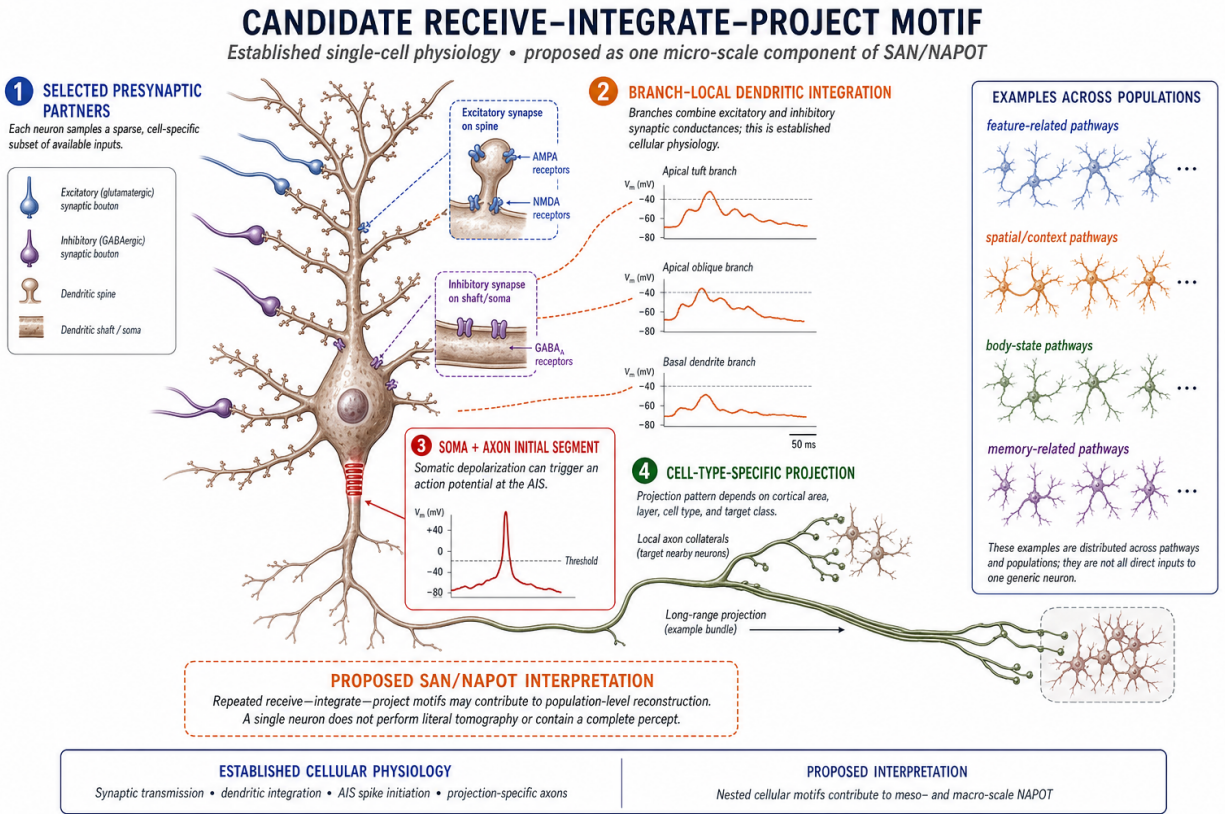
Synaptic transmission, branch-local dendritic integration, spike initiation, and projection-specific axons are established cellular mechanisms [13,60,73]. SAN / NAPOT adds the testable claim that their organized repetition contributes to an identifiable, recurrent population reconstruction. That added claim does not require one neuron to contain a complete percept or receive every information class in the schematic, and it does not make a central renderer necessary.

**Figure 2A.** Positive NAPOT convergence-reconstruction-projection mechanism map. Distributed arrays contribute partial conditions; receiver and population states transform those conditions; and changed projections carry the updated state into further recurrent processing. The illustrated arrays and neuron denote functional population and receiver roles rather than a literal tract map or central renderer. The stronger *tomography* interpretation is supported only when the rank, conditioning, held-out reconstruction, equal-information comparison, and causal-intervention requirements above are satisfied.



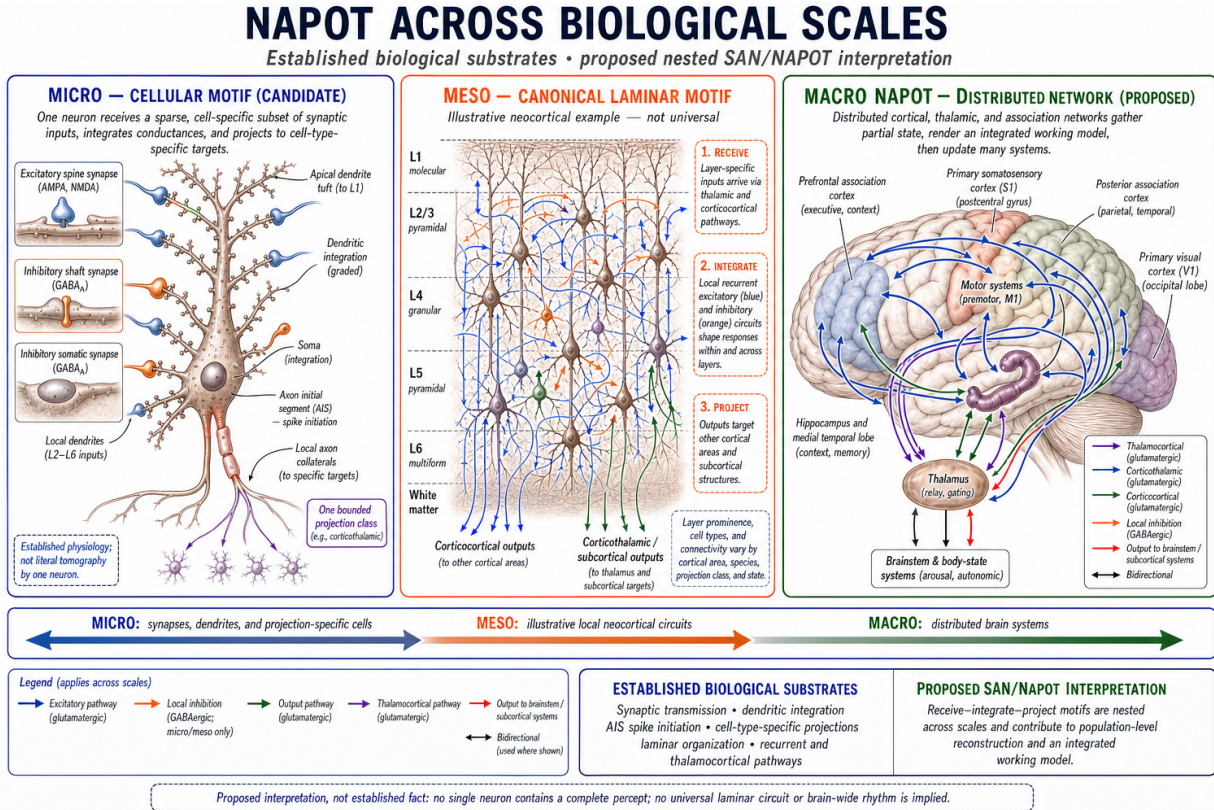


**Figure 3:** Neural Array Projection Oscillation Tomography concept map



**Figure 4:** Candidate cellular receive-integrate-project motif

**Figure 2B.** Candidate local receive-transform-re-express motif. Dendritic and synaptic structure samples partial conditions; the cell's current state transforms their combined effect; and spike initiation plus projection-specific output re-expresses the change for downstream receivers. Those stages summarize established single-cell physiology. SAN / NAPOT proposes that organized repetition of this motif contributes to population-level reconstruction. Feature, context, body-state, and memory labels therefore denote distributed pathway roles across cell types and populations, not four obligatory direct inputs to one generic neuron or a complete percept inside that neuron.



**Figure 5:** Proposed NAPOT nesting across biological scales

**Figure 2C.** Proposed NAPOT nesting across biological scales. At the micro scale, local receivers transform partial conditions; at the meso scale, laminar and recurrent populations route and recombine them; at the macro scale, distributed circuits repeatedly exchange the resulting updates. SAN proposes that this nesting helps maintain and revise an integrated working model. The micro panel remains a candidate cellular motif, the meso panel an illustrative laminar organization, and the macro panel a summary of recurrent distributed anatomy. The proposed recurrence does not require a universal rhythm, identical computation at every level, or unmeasured mathematical scale invariance.

The same measurement problem can be extended beyond one neural route. Consider an organism healing a wound, changing metabolism, recruiting an immune response, or adjusting behavior after an internal disturbance. No one cell receives the complete situation. Local cells encounter partial chemical, electrical, mechanical, and molecular conditions; tissues constrain which local responses remain possible; organs exchange changed signals; and nervous, endocrine, immune, and circulatory systems help coordinate what happens next. The organism-wide result is assembled through repeated local interactions rather than read from one central biological screen.



Established physiology, systems biology, homeostasis, morphogenesis, bioelectric regulation, and neuroimmune signaling already describe many parts of this nested control problem [26,89]. They show that biological state is distributed, history dependent, and organized through feedback. They do not by themselves show that every level performs the same computation, that biological rhythms form one scale-invariant field, or that a larger system reconstructs a latent state from the projections of smaller systems. Observing oscillation at several scales is therefore not enough.

The proposed operation can first be stated without a new label. At each declared boundary, a component samples only part of a time-varying condition; its existing state transforms that sample; its changed output becomes a new condition for other components; recurrent exchange constrains an organism-level estimate, policy, or repair trajectory; and the resulting action changes what will be sampled next. A cross-scale claim must identify each sampled variable, state transformation, emitted output, receiving system, delay, and jointly constrained behavior. It must also show that the nested model predicts or controls the declared outcome better than separate within-scale models supplied with matched information.

Only after that receive-transform-project-return chain is specified does SAN call the widest version of the hypothesis **Biological Oscillatory Tomography (BOT)**. The word *biological* marks the extension beyond a neural route; *oscillatory* requires measured recurrent or cyclic state change, not a presumed universal frequency; and *tomography* remains conditional on multiple informative projections, an independently specified target, held-out reconstruction, and perturbation. A strong test records at least two nested scales, perturbs one predeclared local operation, and asks whether the model predicts the direction, timing, and location of the downstream organism-level change after common-input, rate, power, volume-conduction, and raw-history controls. If the cross-scale model adds no reproducible information or causal leverage, BOT narrows to ordinary multiscale feedback rather than being protected by the metaphor.

**Figure 3.** One measurement-and-perturbation test of BOT. Distributed samples describe partial local conditions; a perturbation changes the system’s evolving state estimate; and a spatially ordered delay tests whether that update propagates into predicted downstream activity. Phase concentration  $R(t)$ , prediction mismatch  $E(t)$ , and model-state change  $\Delta s(t)$  are distinct observables rather than alternate names for entropy, information content, equilibrium, or consciousness. The proposed tomographic interpretation requires distributed recording, a declared latent target, held-out prediction, perturbational discrimination, and controls for common input, volume conduction, rate, power, and raw history.

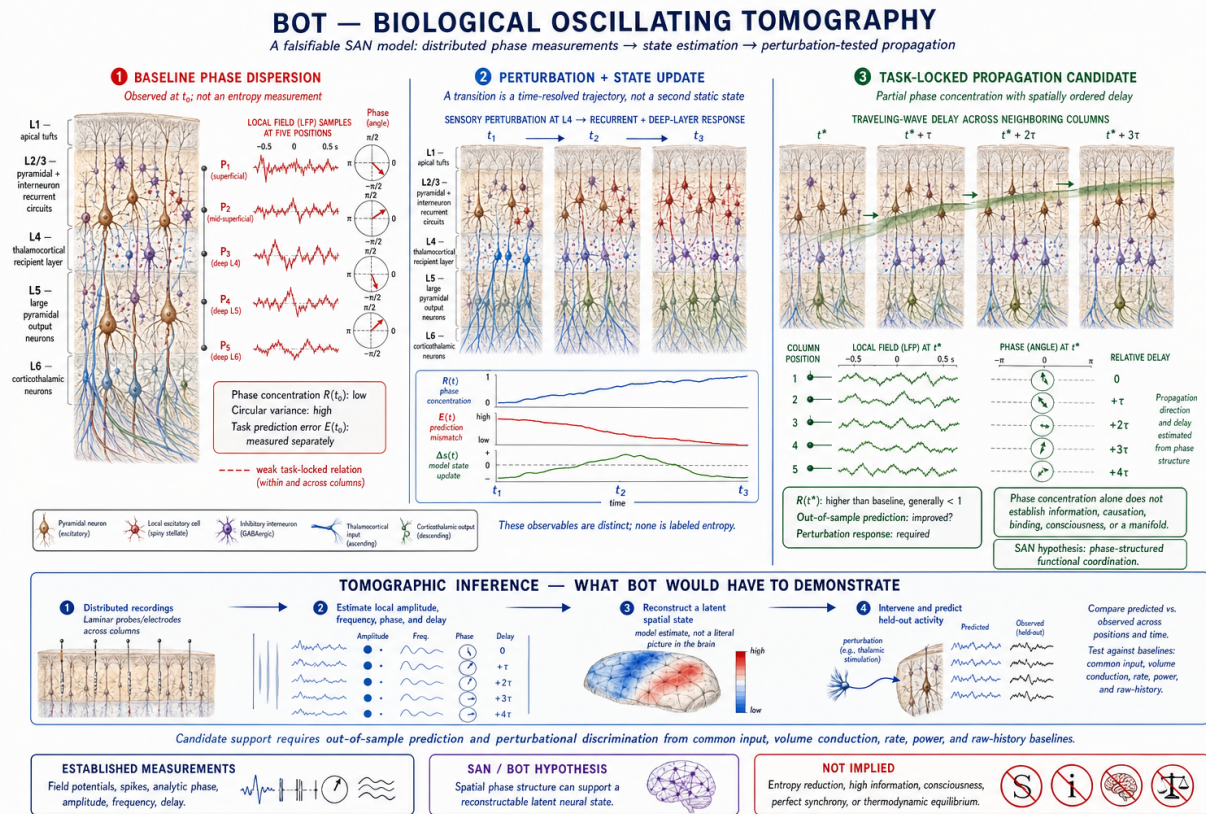
### 3.8 Distributed self / world model

A *self / world model* is an internal state that jointly predicts variables attributed to the body or agent and variables attributed to the environment, and that influences report or action. It need not be explicit, linguistic, or centralized. Operational evidence requires successful prediction or control of independently measured variables. A decoder that identifies a stimulus without any self-related variable demonstrates content representation, not a self-model.

A SAN self-model test additionally requires an independently measured self-related variable - for example self / other attribution, agency judgment under perturbed action feedback, interoceptive prediction, or metacognitive sensitivity - and must outperform matched perceptual, decisional, mnemonic, and motor accounts. Even a successful self-model test would not establish phenomenal self-awareness.

### 3.9 Embodiment and vision: the Gamma Consideration Sandwich

One concrete self-related latent is the organism’s current body state: pose, expected sensory consequences, agency, ownership, self-location, and action possibilities. These variables are not one thing and need not occupy one location. Somatotopic motor and sensory maps, visual body-selective areas, posterior



**Figure 6:** Multiscale biological measurement and perturbation test

parietal and premotor circuits, cerebellar predictions, insular / interoceptive state, and thalamic relays are candidate contributors to a distributed embodied state. Motor imagery can recruit body-part-specific motor representations [52]; visual body representations show topographic and motor-related organization [53]; and spatially and temporally congruent visual and somatosensory signals can alter experienced limb ownership [51,54]. In primary motor cortex (M1), the continuous textbook motor homunculus is incomplete: precision mapping finds effector-specific zones interleaved with regions that Gordon and colleagues interpret as an integrative somato-cognitive action network [55]. This result does not by itself revise the organization of primary somatosensory cortex.

The anatomical *cortical homunculus* is therefore a map family, not a little observer. Thinking about a hand does not switch on one literal hand pixel. The bounded prediction is that seeing, moving, imagining movement of, or imagining sensation in a body part can partially reinstate distributed, body-part-biased population states. Evidence cited here supports motor imagery; imagined sensation remains a separate SAN prediction unless independently localized. The conditions must be separated experimentally from verbal labeling, attention, eye movement, covert muscle activity, motor preparation, and report.

At the subcellular level, apical and basal dendrites are arbor systems of a neuron, not neural arrays made of neurons. SAN uses **dendritic input array** for the spatially arranged synaptic sites inside a declared arbor domain. A biological implementation must subdivide that domain into branch-local or electrotonic cable compartments [13,60,73]. Synaptic organization, receptor and channel state, morphology, inhibition, and plasticity provide relatively persistent pattern selectivity; expected input rate and phase provide a dynamic tonic context. To avoid mixing voltage, conductance, calcium, and timing units, the following  $d_i^c$  is a training-standardized, dimensionless latent drive rather than any one of those observables:

$$d_i^c(r, t) = \sum_{k \in S_i^c} K_c(\delta_c(r, r_{ik})) w_{ik} x_k(t - \tau_{ik}) g(\Delta\phi_{ik}(t)) - I_i^c(r, t), \quad (15)$$

where  $c$  identifies a non-overlapping input domain,  $S_i^c$  its excitatory synaptic sites,  $r_{ik}$  their arbor locations,  $\delta_c$  a declared physical, graph-geodesic, or electrotonic distance,  $w_{ik} \geq 0$  their learned efficacies,  $x_k$  the standardized incoming drive,  $\tau_{ik}$  an independently constrained delay, and  $I_i^c$  a separately standardized inhibitory contribution. The receiver-relative  $\Delta\phi_{ik}$  already contains the preferred offset  $\psi_{ik}$  under the definition in §3.4;  $g$  is a bounded dimensionless receiver-window function, so the offset is not subtracted twice.  $K_c$  is a preregistered localized kernel with its peak or integral normalization held fixed across model comparisons. The phasic pattern is  $\Delta d_i^c(r, t) = d_i^c(r, t) - \bar{d}_i^c(r, t)$  relative to a training-defined or independently localized tonic expectation.

This is the restricted sense in which a dendritic arbor can be compared with a screen or grid: its electrical, chemical, excitatory, inhibitory, and temporal state is spatially patterned. No one views that pattern. It is physically available to local dendritic nonlinearities and is re-expressed through somatic spike / burst timing and axonal projection. The safe causal phrase is therefore **receive**  $\rightarrow$  **transform**  $\rightarrow$  **re-express**, not “a picture is displayed to an internal spectator.” A single synaptic firing rate is not assumed to store the representation.

“Transmission magnitude” must name an observable. It may refer, for example, to spike or burst count, analytic amplitude / power, action-potential width, repolarization slope, presynaptic membrane potential, calcium-current integral, vesicle-release probability, or postsynaptic response. These quantities are not interchangeable. Experimental work shows in bounded preparations that presynaptic action-potential waveform can affect calcium-current and release timing [74], and that subthreshold potential can modulate calcium-dependent glutamate release [75]. Modeling of layer-5 pyramidal neurons further shows that timed excitation, inhibition, and cholinergic modulation can change dendritic  $\text{Ca}^{2+}$ -spike duration and thereby alter the somatic input-output relation and plasticity window [84]. The defensible bridge is therefore presynaptic

or dendritic state  $\rightarrow$  declared waveform / repolarization variable  $\rightarrow$  calcium entry  $\rightarrow$  release or somatic-output endpoint. SAN does not posit a universal action-potential “decay code.”

For a baseline-corrected waveform  $x(t) - x_0(t)$  over a preregistered event window  $W$ , candidate summaries include

$$M_{\text{peak}} = \max_{t \in W} |x(t) - x_0(t)|, \quad M_{\text{signed}} = \int_W [x(t) - x_0(t)] dt, \quad M_{\text{abs}} = \int_W |x(t) - x_0(t)| dt, \quad (16)$$

and the energy-like signal summary  $E_x = \int_W [x(t) - x_0(t)]^2 dt$ . Duration requires a declared threshold or event model. Peak amplitude times duration is only a rectangular-pulse approximation to an area; amplitude plus duration is dimensionally undefined, and no one summary preserves a signal’s biological “weight” across receivers. The endpoint must be chosen from the hypothesized mechanism and validated against the measured calcium, release, or postsynaptic consequence.

For an engineering realization,  $K_c$  can be a localized anisotropic **Gaussian radial-basis field**, with “dendritic splat” used only as a visualization analogy. Its covariance must be symmetric positive definite; its normalization and the physical, graph-geodesic, or electrotonic metric must be fixed before comparison. Signed coefficients can replace, rather than duplicate, the separate inhibitory term. The superposition is a smooth partial field rather than a square-pixel image. It is not evidence that dendrites implement the projection, depth sorting, view dependence, or alpha compositing of a computer-graphics Gaussian rasterizer.

The proposed re-expression may include a change from a tonic timing regime to a different spike, burst, or population-frequency regime. A candidate codebook such as 20 $\rightarrow$ 40/60/80 Hz must be preregistered rather than read from a favorable spectrum. Because 40, 60, and 80 Hz are harmonics of 20 Hz, a nonsinusoidal 20-Hz waveform can generate all three peaks without a new carrier or symbol [61]. Confirmatory analysis must preregister the relevant  $n : m$  phase relation, minimum cycle count and burst duration, narrowband-gamma versus broadband high-gamma / spiking and aperiodic-slope controls, and 50/60-Hz mains, display-refresh, sampling / aliasing, hardware, and stimulation-artifact checks [62,63]. Cycle-by-cycle timing, waveform-preserving surrogates, source separation, and carrier-specific perturbation are required. Bispectral or cross-frequency structure is diagnostic but not decisive because the same nonsinusoidal waveform can produce it [61].

The **Gamma Consideration Sandwich (GCS)** is the working name for SAN’s more specific embodiment hypothesis. The functional “consideration sandwich” is public by 2024-10-21, and the exact capitalized name is public by 2025-02-21 local / 2025-02-22 UTC [59,67]. In a recurrent visuomotor loop, a goal and current body estimate generate a motor command; an efference-copy / corollary-discharge signal enters a forward model that generates a predicted sensory consequence; and movement changes whichever retinal, proprioceptive, tactile, vestibular, or interoceptive signals apply to that action. Distributed circuits compare predicted and observed consequences, and the updated body state changes the next action. Corollary discharge is established in specific visuomotor pathways, including the sequential-saccade preparation in [50], but that result is not a universal embodiment circuit. GCS proposes - not demonstrates - that phase-structured laminar activity regulates when sensory / thought evidence and motor / reafferent context become mutually effective in this update.

The author-held 2025 diagram-derived assignment is frozen below so it can be falsified [66]. The two primary working diagrams have been recovered, hashed, and privately verified, but their working-file metadata does not establish a public publication date. The table is therefore a current falsifiable model, not Grade-A public-priority evidence; the public GCS concept and name are independently anchored by [59,67,68]. A confirmatory study must choose one cortical target  $T$  with a known granular laminar organization and declare

its anatomical source areas and time windows; the table must not be transferred unchanged to agranular motor cortex.

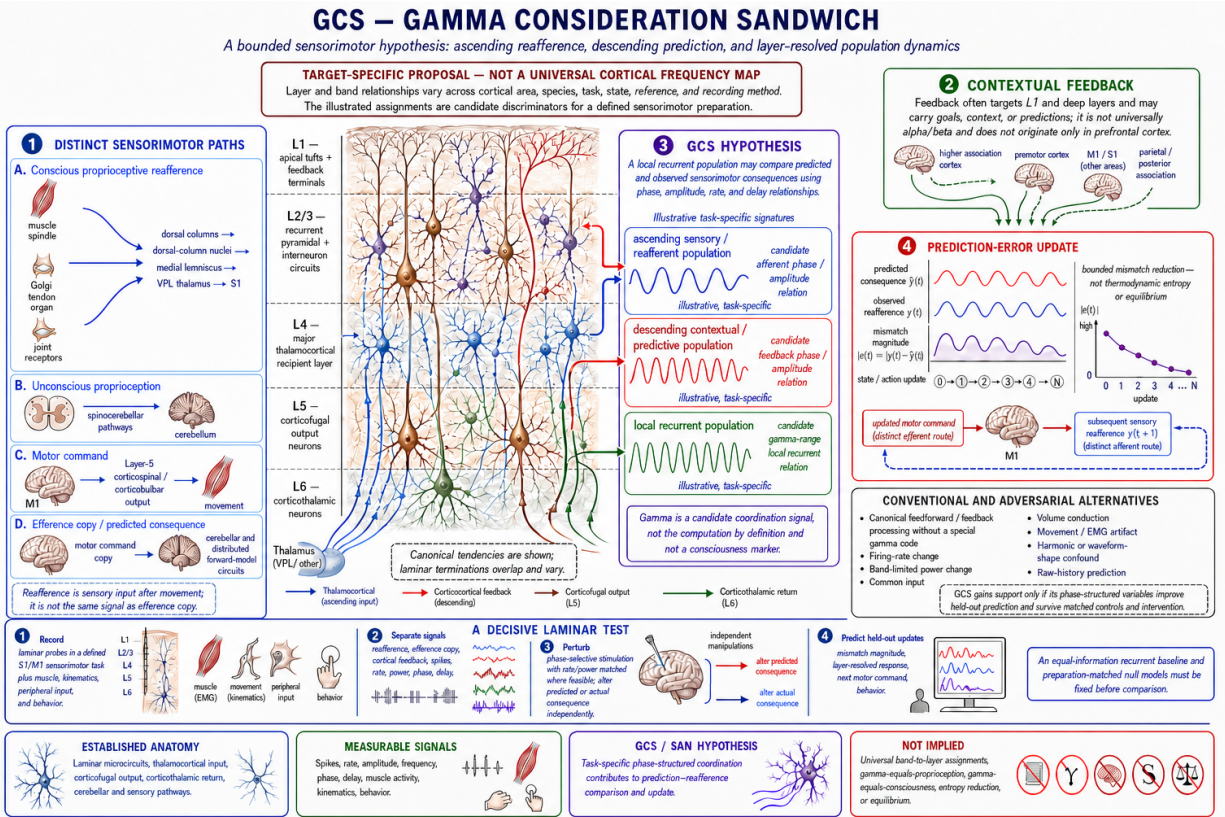
| Role in the provisional GCS                              | Candidate target population in $T$          | Candidate band / relation                           | Required temporal and causal signature                                                                                                                                                                      |
|----------------------------------------------------------|---------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensory / thought context from declared source $S$       | Layer 4 input population                    | Alpha / beta; negative relation to Layer-2/3 gamma  | $S \rightarrow T$ context precedes or overlaps the superficial comparison, and perturbing it shifts the comparison and body-state error in the preregistered direction.                                     |
| Motor-context / deep recurrence from declared source $M$ | Layer 5 excitatory population               | Theta / gamma; positive relation to Layer-2/3 gamma | $M \rightarrow T$ context follows the motor command / efference-copy event and changes with imposed sensorimotor delay. Peripheral proprioceptive afference still follows its established ascending relays. |
| Local inhibitory timing                                  | PV and other inhibitory interneurons in $T$ | No afferent carrier band assumed                    | Local inhibition may modulate gamma timing; PV interneurons are explicitly <b>not</b> treated as the ascending proprioceptive channel.                                                                      |
| Comparison / update                                      | Layer 2/3 excitatory population             | Tonic gamma with phasic departures                  | The interaction of the two contexts precedes and predicts a delay-sensitive body-state update beyond each input, power, rate, and raw-history baselines.                                                    |

This exact assignment is a discriminator, not a fact inherited from the literature. Primate visual studies provide a preparation-specific comparator in which feedforward influence tends toward theta / gamma and feedback toward alpha / beta [9,57]. The GCS assignment survives only if a layer-resolved recording and intervention in the declared target outperform that comparator under matched rate, power, visual feature, arousal, movement, and common-input controls.

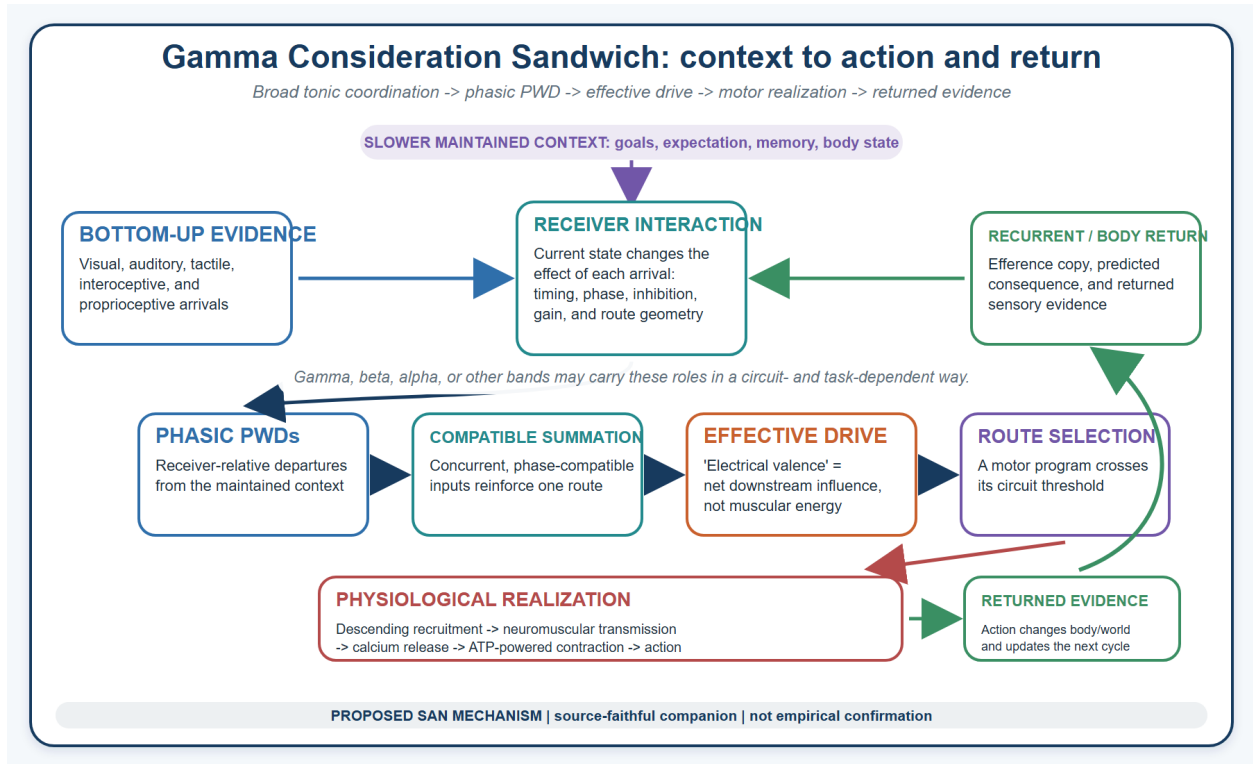
**Figure 4.** Bounded Gamma Consideration Sandwich sensorimotor loop. Maintained contextual and body state meets incoming sensory evidence in a recurrent target population; phase-compatible activity can increase the effective drive of one candidate route; motor output changes the body or world; and the returning reafference updates the next comparison. The afferent, motor, efference-copy, thalamocortical, corticofugal, corticothalamic, and contextual-feedback routes summarize established anatomical constraints, while their task-specific phase relations, building-PWD / electrical-valence interpretation, and prediction-reafference comparison are SAN hypotheses. The assignments are not a universal cortical frequency map and must outperform preparation-matched feedforward / feedback, common-input, rate, power, waveform, artifact, and raw-history alternatives under intervention.

**Figure 4B.** Positive Gamma Consideration Sandwich mechanism companion. Slower maintained context, incoming evidence, and returned body or world evidence meet in state-dependent receivers. Phasic receiver-relative departures can sum when their timing is compatible, increasing the effective downstream influence of one candidate route until a motor program crosses its circuit threshold. Here *electrical valence* means that accumulated effective drive; it is not the metabolic energy of muscle contraction. Descending recruitment, neuromuscular transmission, calcium release, and ATP-powered contraction realize the selected action,





**Figure 7: Bounded Gamma Consideration Sandwich sensorimotor hypothesis**



**Figure 8:** Positive Gamma Consideration Sandwich context-to-action cycle

whose consequences provide the next evidence. The relevant bands and phase relations must be identified for each circuit and task rather than assumed from a universal beta-gamma assignment.

## 4. Established component mechanisms and the SAN delta

### 4.1 Structured local computation

Hebb proposed that repeated co-activation could strengthen later joint response [3]. Modern dendritic physiology adds nonlinear compartmental operations and context-dependent integration [13]. These findings support the EDS premise that a receiver's physical configuration selectively transforms input. They do not establish NAPOT, and SAN does not claim origination of dendritic computation, coincidence detection, long-term potentiation, or predictive processing.

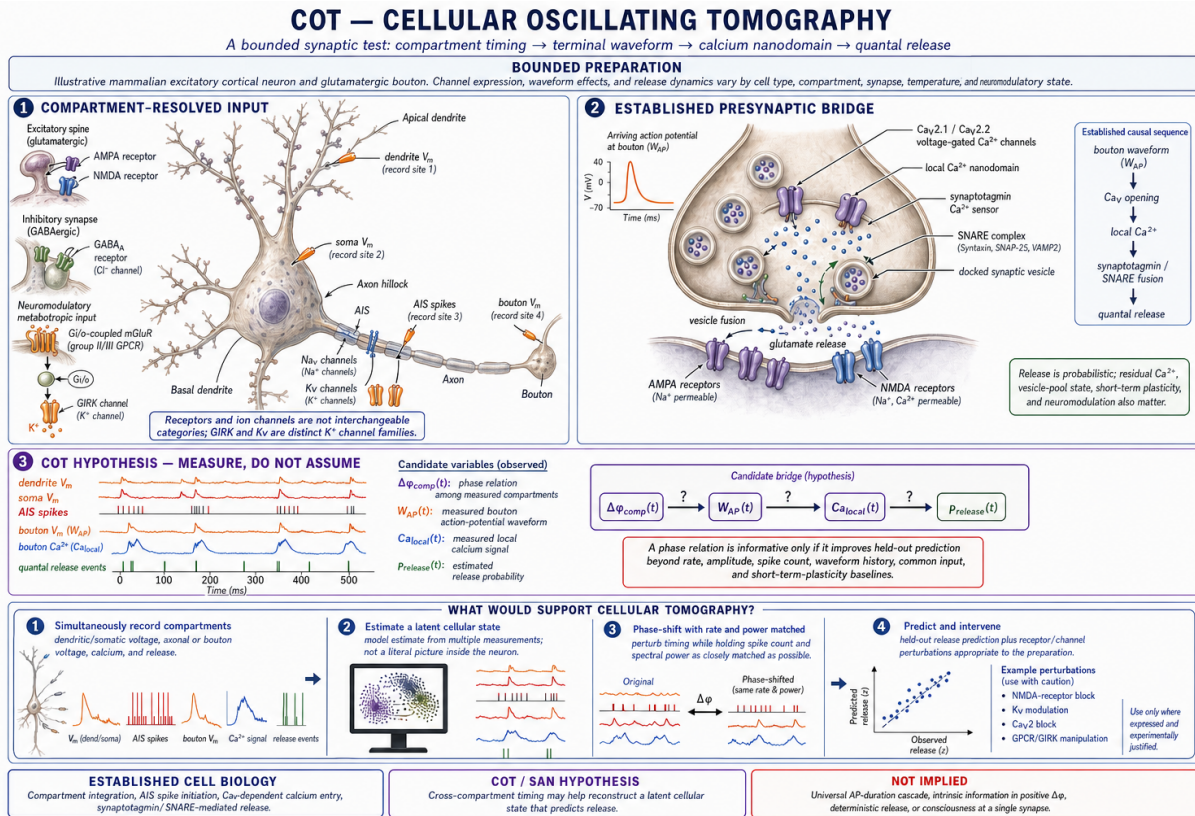
The review-copy book and repository also route to two molecular persistence results omitted from the previous draft. CaMKII-dependent potentiation can survive replacement of synaptic CaMKII through activity transfer from pre-existing to newly synthesized kinase [80]. KIBRA-PKM $\zeta$  complexes can anchor a potentiating action at activated synapses, and disrupting the interaction reverses established late LTP and long-term spatial memory in the reported mouse preparations [81]. These results strengthen the plausibility of a slower, continually maintained structural memory state. They do not demonstrate that the maintained state has a particular oscillatory frequency, encodes a phase-wave differential, or performs Neural Rendering.

Receiver state, excitability windows, inhibition, delays, and expected phase relationships all have pre-SAN precedents. The SAN-specific hypothesis is not receiver-relative gating by itself. It is the conjunction in which delay-aware receiver gating and spatial wave geometry make an incremental, causal contribution to recurrent reconstruction of a predefined latent state. The same upstream event may therefore be effective for

one receiver and ineffective for another, but that observation supports the SAN delta only when it is linked to the predicted projection and latent update.

At the cellular scale, the concrete problem is to predict how temporally distributed receptor, channel, voltage, calcium, and release conditions are combined into a changed cellular output. Established synaptic physiology supplies each measured component. SAN proposes testing the joined operation by asking whether the cell's time-varying internal state improves held-out prediction of its output beyond rate, power, waveform, and standard plasticity controls. SAN calls this bounded cellular receive-transform-output proposal **Cellular Oscillating Tomography (COT)**.

This synaptic preparation tests one tractable branch of the wider COT proposal; it does not define the theory's full scope. The 2022 COT sources also ask whether learned receptor configuration, cellular morphology and history act as memory, whether coincident spatial-temporal inputs alter transcriptomic and other generative responses, and whether analogous receive-transform-output operations can be tested in non-neural cells and multicellular organization [2]. Each branch needs its own measured variables, controls, and failure conditions. Restricting Figure 5 to one preparation makes that experiment interpretable without deleting the wider receptor-memory, transcriptomic, non-neural, behavioral, morphological, and evolutionary research program.

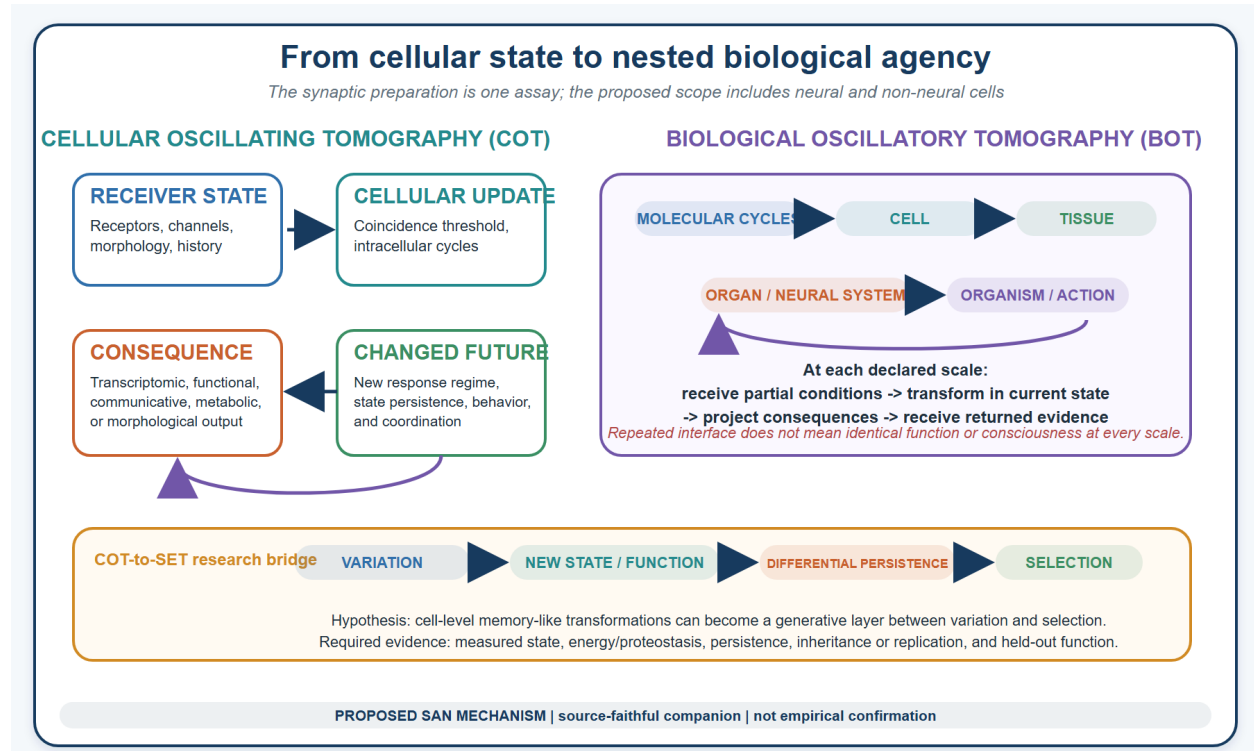


**Figure 9:** Bounded cellular receiver-timing and synaptic-output test

**Figure 5.** One bounded neural COT preparation. The proposed operation is a stateful path from temporally distributed receptor and channel conditions through compartment voltage, spike waveform, local calcium, and quantal release. Those measurements are kept distinct so the joined transformation can be tested rather than assumed. This plate addresses only the synaptic-output branch of COT; the wider cellular program is stated in the preceding paragraph. In this preparation, the proposed cross-compartment phase predictor gains



support only if phase-aware models improve held-out release prediction and survive matched rate, power, waveform, common-input, short-term-plasticity, and receptor / channel interventions.



**Figure 10:** Positive COT, BOT, and COT-to-SET mechanism companion

**Figure 5B.** Positive COT and BOT scope companion. The bounded synaptic preparation in Figure 5 is one assay within the wider proposal: receptor and channel conditions meet a cell’s morphology and history, undergo an intracellular update, produce functional, communicative, metabolic, transcriptomic, or morphological consequences, and thereby alter the cell’s future response. Biological Oscillatory Tomography (BOT) asks whether the same declared interface can be measured across molecular, cellular, tissue, organ, neural-system, and organism-action scales without assuming identical functions or consciousness at every level. The COT-to-SET bridge is a separate hypothesis: persistent cell-level transformations may form a generative layer between variation and selection. It requires measured state, energetic and proteostatic accounting, persistence, inheritance or replication where claimed, and held-out functional prediction.

## 4.2 Phase coding and communication through coherence

Synchronization has long been proposed as a means of dynamically linking distributed populations [5,6,27]. Communication-through-coherence theories explain how aligned excitability windows can selectively regulate effective communication [7-10]. Phase-of-firing experiments demonstrate information beyond spike count [11], while theta-gamma models organize ordered items by phase [12]. Frequency-dependent short-term synaptic effects also vary gradually along the rat hippocampal dorsoventral axis [82], and visual-cortex slice experiments found LTP for inputs paired near peaks of imposed 20/40-Hz membrane oscillations and LTD near troughs [83]. These are direct precursors and component mechanisms: they establish frequency- and phase-sensitive synaptic transformations in bounded preparations, not the SAN PWD definition or a global spectral address.

SAN’s proposed addition is the joint use of independently oriented, delay-constrained PWDs and spatial phase gradients as variables in a recurrent reconstruction. A scalar coherence value does not specify where a wave is going, how its local gradient changes, which receiver window reads it, or which latent-state constraint it carries. NAPOT predicts that those structured variables improve explanation after coherence, power, rate, and an equal-information raw sequence / graph baseline are already included.

Here *neural coherence* has its ordinary classical signal-analysis meaning: a frequency-resolved statistical relation among recorded or source-estimated neural time series. *Quantum coherence* is a different, basis-dependent property of a quantum state, commonly represented by off-diagonal terms of a density matrix and evaluated within a specified resource theory [85]; decoherence is the loss or redistribution of such quantum-state relations through system-environment interaction [86]. The shared word does not identify a shared mechanism. Nothing in an LFP, ECoG, MEG, spike-field, or behavioral coherence estimate establishes quantum coherence, entanglement, or a quantum contribution to cognition. Any future quantum-neural claim would require a separately specified physical carrier, basis, preparation, observable, timescale, environmental model, and discriminating measurement beyond the classical neural account.

### 4.3 Traveling waves and structural constraints

Traveling waves organize phase over space. In macaque prefrontal cortex, theta-, alpha-, and beta-band activity can form planar and rotating waves modulated by working-memory performance [14]. Modeling based on human dMRI-derived connectomes and comparison with resting MEG suggests that structural instrength gradients and modeled delays can shape wave direction and frequency gradients [15]. It is not a causal demonstration of tract polarity in humans. These results make spatial propagation a plausible computational variable; they do not show that a wave contains a reconstructed percept or that traveling waves are sufficient for awareness.

The First Draft made the narrower directional proposal that a systematic gradient of local oscillators’ intrinsic frequencies can bias propagation from higher- toward lower-frequency regions [1]. That proposed rule is preserved as a candidate mechanism rather than inferred from a measured power gradient. Testing it requires estimating each population’s natural or perturbation-revealed frequency, manipulating the gradient while holding connectivity, delay, input power, and arousal as nearly matched as possible, and asking whether wave direction reverses when the intrinsic-frequency gradient reverses.

NAPOT predicts an interaction that is stronger than either result alone:

$$\begin{aligned} \text{connectivity and delay} &\rightarrow \text{wave geometry} \\ &\rightarrow \text{receiver-relative routing} \\ &\rightarrow \text{latent-state update.} \end{aligned} \tag{17}$$

The theory fails at this level if wave geometry has no incremental relation to effective routing or latent-state update after matched controls.

### 4.4 Recurrent and predictive processing

Predictive coding models use hierarchical recurrent exchange to explain how prior state and sensory error interact [4,22]. SAN accepts that broad architecture. It does not require the brain to store a literal image, nor does it assert that internal generation eliminates sensory constraint. Its more specific claim is that spatiotemporal phase structure participates in implementing the recurrent exchange and in selecting which prediction and error signals become mutually effective.

This distinction makes recurrent predictive processing a required baseline. If a conventional recurrent model - or a flexible graph / sequence model given the identical raw multichannel signals, geometry, and history - predicts the data as well as a PWD / gradient model, the SAN-specific addition has not been supported by prediction alone.

#### 4.5 From component mechanisms to a discriminating composite

The full SAN composite is not “the brain oscillates.” It is:

1. history-dependent receiver structure selects inputs;
2. receiver-relative timing converts some continuous relations into effective events;
3. local phase differentials and gradients regulate communication along independently oriented or experimentally assigned edges;
4. structural connectivity and delays constrain propagation;
5. recurrent arrays exchange multiple partial projections;
6. those projections jointly improve reconstruction of a predefined latent self / world variable;
7. the reconstructed state guides report or action.

Each item can be tested separately. Support for items 1-4 is not automatically support for items 5-7.

#### 4.6 Oscillatory Computational Agency across micro, meso, and macro scales

Touching a hot surface illustrates the scale problem. Ion channels and receptors change locally; cells integrate those changes; neural circuits route sensory and contextual activity; distributed brain and body systems coordinate withdrawal, learning, and later avoidance. Systems biology, control theory, cellular signaling, and network neuroscience each describe parts of this nesting. The missing theoretical operation is a common receive-transform-output-feedback pattern that can be specified at each scale without treating the scales as identical.

The First Draft names that graded multiscale proposal **Oscillatory Computational Agency (OCA)** [1]. At the micro scale, proteins, receptors, ion channels, synapses, dendritic branches, neurons, glia, and immune cells alter their state in response to local conditions. At the meso scale, arrays, interneuron-defined subnetworks, cortical columns, hippocampal-entorhinal circuits, and thalamocortical loops select and transform signals. At the macro scale, distributed brain rhythms, traveling waves, recurrent communication, body-state signals, memory, and action constrain one another. A component is an operational agent when it has a boundary, receives state-dependent inputs, transforms them, produces outputs, and can change its future response. The prediction is scale-specific: each claimed agent must expose those four measurable operations at its declared boundary.

Before naming the scale variants, their common operation can be stated plainly. A bounded component samples partial time-varying conditions, transforms them according to its current and learned state, projects a changed output, and participates in recurrent comparison or correction. SAN retains three historical names for increasingly broad applications of that operation:

- **Cellular Oscillating Tomography (COT)** concerns how a cell samples temporally distributed molecular, receptor, ionic, mechanical, electrical, and chemical conditions and changes its internal and output state.
- **Neural Array Projection Oscillation Tomography (NAPOT)** concerns how structured neural arrays receive partial conditions, transform them, and project phase-structured updates into recurrent circuits.
- **Biological Oscillatory Tomography (BOT)** is the broadest scale claim: nested biological agents may integrate multiple partial, time-varying projections into organism-level state estimates and coordinated behavior.

The common operator is not the word *tomography*. It is repeated partial sampling followed by state-dependent transformation, projection, recurrent comparison, and update. The medical-imaging analogy remains useful only if a study identifies the sampled variables, projection operator, receiver, latent target, and reconstruction test. Otherwise, tomography remains a metaphor.

OCA also retained two wider propositions. First, biological control is distributed: no single executive homunculus is required when nested local agents cooperate and compete through feedback. Second, the same receive-transform-project motif can recur at several scales. The First Draft called this self-similar organization a *fractal sensor-transmitter loop* and used *entification* for the process by which many bounded agents become one behaviorally coherent entity [1]. In this revision, those terms name hypotheses about compositional organization; they are not measurements of a mathematical fractal dimension and do not prove phenomenal unity.

The resulting scale ladder is selective rather than globally synchronized. Molecular and receptor state constrain what a cell can receive; cells and local ensembles recognize partial patterns and emit state-dependent outputs; interneuron, thalamic, hippocampal, and recurrent cortical circuits regulate which outputs gain access; and distributed sensory, memory, body, and action systems temporarily coordinate a usable self / world state. Calling a neuron, column, or circuit *semi-autonomous* does not make it a miniature conscious person or the source of a complete motor directive. It means that its bounded receive-transform-output operation can bias the larger route, while the larger network and body change its next local state.

#### **4.7 From molecular persistence and dendritic coincidence to a PWD token**

The original molecular bridge is retained. Synaptic efficacy depends on receptor state, ion-channel kinetics, calcium entry, transmitter-release machinery, protein turnover, and plasticity-related molecular maintenance. KIBRA-PKM $\zeta$  interactions and other persistence mechanisms can help stabilize learned synaptic effects [80,81]. Potassium and calcium conductances can alter action-potential and dendritic-spike duration, and duration can affect presynaptic calcium entry and release probability [74,75,84]. These facts motivate, but do not by themselves prove, SAN's claim that waveform duration and phase-history can carry information beyond a binary spike label.

The First Draft joined two claims that must now be separated rather than allowing one to erase the other [1]. At the level of present measurements, vesicle release is appropriately modeled as a conditional stochastic process: action-potential waveform, voltage history, calcium entry, release-site and vesicle-pool state, receptor state, and recent network activity can change its probability distribution. At the mechanistic level, SAN retains the stronger hypothesis that part of the remaining trial-to-trial variability may be the causal result of microstate variables that have not yet been measured. Existing evidence establishes state-dependent probability modulation; it does not decide whether the residual variance is fundamentally stochastic or instead reflects hidden causal state.

This distinction produces a direct test. As progressively richer presynaptic and postsynaptic state variables are measured, a microstate account predicts improved out-of-sample prediction of release count, timing, and downstream effect. A reproducible residual-variance floor that survives adequate measurement and intervention would constrain the stronger deterministic interpretation. SAN does not require every vesicle event to be predictable with present instruments, but neither does an observationally stochastic model by itself refute the paper's physical-determinism hypothesis.

The dendritic arbor supplies the next stage. Its learned synaptic distribution can be represented as a high-dimensional, branch-structured receptive state. Dendrites perform nonlinear integration; coincident inputs can be amplified while other inputs are attenuated [13,60,73]. Softky and Koch showed that models driven by many small, independent random EPSPs did not reproduce the high firing variability observed in awake



primate visual cortex; fast dendritic nonlinearities or synchronized input allowed the simulated cell to behave more like a millisecond-scale coincidence detector [101]. This supplies a specific historical bridge for SAN’s coincidence account. It does not imply that every cortical neuron is exclusively a coincidence detector or that irregular firing alone establishes the later PWD operator. The First Draft called the resulting spike or burst a **compressed token** of the convergent dendritic pattern [1]. The current operational statement is:

$$\begin{aligned} \mathbf{z}_i(t) &= F_i(\{x_{ij}(t - \tau_{ij})\}_j, \mathbf{w}_i, \mathbf{m}_i(t), \mathbf{q}_i(t)), \\ u_i(t) &= G_i(\mathbf{z}_i(t)), \end{aligned} \tag{18}$$

where  $\mathbf{w}_i$  is learned dendritic / synaptic structure,  $\mathbf{m}_i$  is membrane and molecular state,  $\mathbf{q}_i$  is local oscillatory context, and  $u_i$  is an output event with timing, waveform, phase, and release consequences. *Compression* here means many receiver conditions converge on a lower-dimensional emitted event. It does not mean that the event is a lossless inverse of the dendritic state.

Coincidence is therefore both relational and receiver-defined. Two events constitute an effective coincidence when their arrival relationship, receptor state, and local integration window jointly alter the receiver. Repeated effective coincidence can contribute to Hebbian and behavioral- timescale plasticity, strengthening or weakening future routes [3,48]. The resulting output can become a **Phase Wave Differential (PWD)** when its receiver-relative timing, duration, phase, or frequency changes the accessibility of a downstream pathway. This restores the First Draft’s full chain:

```
molecular and receptor state
-> learned dendritic receptive geometry
-> nonlinear coincidence and inhibition
-> spike / burst / waveform and release change
-> receiver-relative PWD
-> local-field and population effect
-> recurrent array routing
```

The First Draft’s **addressable coefficient of variation** is also retained. For a measured feature  $X_i$  of cell or circuit  $i$ ,  $CV_i = \sigma_i / \mu_i$  can describe variability relative to its mean. SAN’s proposed addition is that a multivariate variability profile—not a scalar CV alone—may help identify a receiver’s learned response regime and therefore act as one component of an oscillatory address. A valid test must show that the profile predicts receiver or route identity beyond firing rate, anatomy, and state covariates. The phrase *unique cellular fingerprint* is a testable hypothesis, not an assumption of globally unique scalar CV values.

At every declared scale, a system receives a continuously changing state rather than a completed symbol. Its nonlinear transformation produces an approximation that becomes part of the next receiver’s conditions, so recurrent passes can progressively revise an estimate. Nonlinear state-space models and recurrent estimation provide the inherited mathematical vocabulary. SAN’s historical name for the cross-scale receive-update-pass-on proposal is **Non-linear Differential Continuous Approximation (NDCA)**. Section 6.6 states the bounded mathematical interpretation and the conditions under which this adds anything beyond an ordinary nonlinear recurrent model.

COT further proposes that repeated receptor coincidences, intracellular cycles, transcriptomic responses, and morphology changes can give ordinary cells memory-like and prediction-like behavior. Maintaining and changing those configurations consumes cellular energy, while configurations differ in their consequences for survival and control. The First Draft calls the hypothesis that energetic cost and functional consequence help select among maintained configurations the **ATP-based reinforcement analogy**. A test must measure energy use, state persistence, selection, and later functional consequence. ATP is the cell’s energy carrier in this account rather than a literal machine-learning reward token.

The mechanism can be stated without importing the psychology of reward or punishment. A receptor, protein, or regulatory configuration incurs measurable energetic and proteostatic costs, changes the cell's response to later conditions, and is then differentially stabilized, modified, degraded, or replicated by ordinary cellular processes. The bounded COT hypothesis is that the joined sequence

$$\begin{aligned}
 &\text{configuration} \\
 &\rightarrow \{\text{energy cost, proteostatic consequence, functional output}\} \\
 &\rightarrow \text{differential persistence} \\
 &\rightarrow \text{changed future response}
 \end{aligned} \tag{19}$$

can produce a history-dependent cellular selection process. Supporting evidence would require simultaneous measurement of ATP flux, the relevant proteostasis pathway, configuration persistence, and held-out functional outcome. Energy use alone, or survival alone, would not identify this operator, and no foresight by the cell is assumed.

The SAN genealogy contains a second, earlier ATP that must not be collapsed into that biochemical reinforcement analogy. The 2017 Neural Lace Podcast first asks an ordinary engineering question: does a biological signal always wait for confirmation, or can some updates be sent quickly even if one is missed? It contrasts a connection-and-feedback mode with a rapid loss-tolerant mode using TCP and UDP as analogies, proposes that a biological network may switch between them according to the task and route, and only then names the proposal **Alternating Transmission Protocol (ATP)** [39, approximately 14:00-23:30]. The name intentionally echoes adenosine triphosphate as cellular energy currency. It does not assert that neurons literally implement an Internet packet stack.

The same 2017 sequence supplies a direct ancestor of the later receiver-relative PWD operation. For a proposed BCI, it says not to send an entire desired sensory pattern on top of the pattern already present; instead, estimate the difference between the current pattern and the pattern that should be present, then transmit that difference [39, approximately 25:50-26:30]. In current notation, that design question is whether an intervention based on a target-minus-estimated-current residual outperforms full-pattern stimulation, while alternating between feedback-confirmed and loss-tolerant update modes. The notation and experiment are current formalizations; the dated 2017 source contains the plain-language operator. This protocol lineage and the later ATP-energy selection hypothesis are complementary but distinct.

Likewise, the historical suggestion that electrical, chemical, mechanical, magnetic, or spin-dependent processes might participate at different scales is retained as a search space; only carriers with independently measured causal effects can enter the active mechanism. The First Draft's further phrase *soliton waves* is also preserved as a nonlinear propagation hypothesis, not as a synonym for every traveling wave. It earns that description only if a declared neural or molecular medium supports a localized waveform whose dispersion is balanced by nonlinearity, whose form persists over the predicted distance and time, and whose perturbation produces the forecast receiver-relative PWD consequence beyond ordinary dispersive-wave and synaptic-propagation controls.

#### 4.8 Mesoscale projection, functional connectivity, binding, and Biological Oscillatory Tomography

At the mesoscale, NAPOT treats an array as both receiver and projector. Each array samples only part of a scene, body state, memory state, or upstream computation. Its history-dependent transformation changes the distribution of outputs offered to other arrays. Recurrent exchange allows multiple partial views to constrain

one another. A stable percept or policy is therefore not a picture stored at one address but a distributed state that can be decoded from the coordinated network and tested against its sensory and motor consequences.

The First Draft’s **volumetric television**, **canvas**, **ink**, and **brushstroke** language names the constructive side of this operation [1]. A learned synaptic trace constrains what one receiver can detect; a phasic difference changes that receiver; its output becomes a partial condition for other arrays; and recurrent multimodal exchange changes the next distributed estimate and action. The rendering is therefore active rather than a passive copy or a hidden screen. A NAPOT *slice* is a measurement-conditioned partial constraint on a declared latent state, not a literal image plane. *Volumetric* means that relational content can be decoded across spatially distributed, temporally ordered receivers; *television* and *brushstroke* remain memorable descriptions of the receive-transform-project-return process rather than additional biological premises.

The flow from molecule to rendered state is likewise explicit. Receptor, channel, voltage, calcium, release-site, and protein-maintenance states shape dendritic integration; coincidence and inhibition compress that history into changed spike, burst, waveform, and release consequences; cell-type-specific inhibition and projection determine which downstream receivers can be recruited; and recurrent network exchange returns the result to perception, memory, and action. This chain is bidirectional because population state and neuromodulation can change the cellular conditions that produce the next output. Cell-class anatomy prevents one historical compression from becoming a universal rule: PV basket cells principally target somatic and proximal-dendritic compartments, PV chandelier cells target axon initial segments, and many SST interneurons target dendritic compartments [111]. The First Draft’s functional insight survives the distinction: inhibition can regulate dendritic integration, spike initiation, route competition, and later plasticity at different sites and times.

The First Draft joined this idea to functional connectivity, BTSP, binding, and traveling waves [1]. That join remains central. Repeated coactivation can alter physical connectivity; phase alignment can transiently alter effective connectivity without rewiring; inhibition can prevent competing assemblies from becoming simultaneously effective; and traveling waves can order when distributed populations become mutually accessible. In SAN, binding is not global synchrony alone. Coherence supports communication among some components, while relative phase differences and structured dispersion preserve distinctions among others. A useful rendering requires both integration and differentiation.

Constructive and destructive interference belong first to the fast routing layer. Phase-compatible arrivals can sum and recruit a receiver, whereas phase-incompatible arrivals can reduce effective drive, inhibit a route, or preserve separation between competing assemblies. The 2022 SAN sources then add a relational coupling transition: a subgroup can become less synchronized with a parent ensemble while becoming more synchronized internally. SAN called those paired changes **Group Short-Term Depression (GSTD)** and **Group Short-Term Potentiation (GSTP)** [2]. Repetition under the right spike timing, dendritic, receptor, neuromodulatory, and plasticity conditions may subsequently bias LTD, LTP, spine stabilization, or spine loss. The First Draft phrase “destructive interference prunes connections” compressed this full chain into the wrong timescale. Interference changes effective recruitment first; anatomical pruning, when it occurs, is a slower plastic consequence that must be measured separately.

The original **global spectral frequency map and coordinate system** is retained as an empirical hypothesis. A cell, edge, or population may have a state-dependent spectral and phase-response profile that helps route signals. The address is relational: it depends on source, receiver, delay, frequency, phase, and context. Demonstration requires decoding or causal targeting that outperforms rate-, power-, anatomy-, and connectivity-matched controls.

Ephaptic coupling remains one candidate contributor. Endogenous extracellular fields can affect nearby membrane potentials under some conditions [76-78]. SAN does not treat every field as a message, nor

does it replace synaptic transmission with a wireless metaphor. The prediction is that measured local fields can sometimes change phase alignment or excitability enough to affect the next receiver-defined event after synaptic and common-input controls.

BOT generalizes the nesting claim beyond a single neural route: molecular cycles, cells, tissues, organs, and nervous systems can participate in coupled control processes. The First Draft included glia, immune-neural signaling, wound repair, morphogenesis, and cellular adaptation in this family [1,26]. These examples support graded biological agency; they do not establish that non-neural tissues perform NAPOT or are phenomenally conscious.

Two examples make the boundary concrete. Many characteristic brain rhythms and timing relations are conserved across species even as brain size changes [88], supporting oscillatory timing as a candidate multiscale scaffold; conservation alone does not prove the BOT reconstruction operator. Adaptive immune cells, including meningeal T cells, can alter neural and cognitive function, and the wider nervous and immune systems are reciprocally interdependent [89]. This neuroimmune coupling preserves the First Draft's cell-to-system example, but it does not make an immune cell and a neuron computationally identical.

Hippocampal sharp-wave ripples make the state-dependence of oscillatory routing more specific. The First Draft listed GABA<sub>A</sub> inhibition, acetylcholine, noradrenaline, and dopamine as modulators of ripple expression or of the encoding conditions that shape later ripples [1]. The positive SAN proposition is that inhibitory and neuromodulatory state changes which ripple-associated patterns can recruit downstream receivers and which encoding events alter later plasticity. Its statements that excess GABA<sub>A</sub> inhibition can suppress ripples and that dopamine can influence them indirectly through active encoding and plasticity are retained as transmitter-, dose-, circuit-, and state-specific hypotheses rather than one universal dose-response law. A test must jointly measure receptor or neuromodulatory state, ripple rate and content, receiver recruitment, behavioral state, and later plasticity; changing inhibition or dopamine alone does not establish a PWD.

The acetylcholine and noradrenaline branches require the same specificity. Hippocampal acetylcholine rises with theta / gamma-rich walking and REM states, reaches a minimum around SPW-Rs, and suppresses SPW-R incidence when medial-septal cholinergic neurons are stimulated during a working-memory delay [114]. Thus the First Draft's compressed wording that acetylcholine *facilitates* sleep ripples is recovered as a state-transition claim: reduced cholinergic tone permits ripple-dominated offline processing, whereas higher cholinergic tone favors active-state dynamics. Noradrenaline is not a one-directional ripple control either. In rat hippocampal slices, norepinephrine acting through  $\alpha_1$  adrenoreceptors suppressed SPW-R generation, while  $\beta$ -adrenergic activation increased SPW-R incidence and facilitated activity-dependent induction [115]. The measurable scientific claim is therefore receptor-, dose-, circuit-, and state-dependent neuromodulation, not a universal rule that either transmitter simply turns ripples on or off.

The established sharp-wave-ripple literature supplies the inherited memory mechanism: hippocampal SPW-Rs are brief, highly synchronous population events, commonly carrying time-compressed replay and influencing distributed circuits involved in episodic memory and planning [112]. SAN does not rename a ripple as a PWD. It asks the later receiver-relative question: which ripple-associated arrival changes a declared downstream receiver, under what state and delay, and with what measurable consequence for reconstruction, plasticity, or action?

Ordered local projections can encode high-dimensional structure without reducing that structure to simple periodic repetition. The First Draft calls this comparison the **quasicrystal analogy**. A literal quasicrystal claim requires a measured aperiodic order parameter; otherwise the analogy serves only to motivate a test of organized nonperiodic structure. Likewise, the combinatorial state space of many neurons and synapses can be enormous even when its effective task-relevant dimension is much smaller. The phrase

**four trillion dimensions per millisecond** was the First Draft’s illustration of that capacity. Its testable successor is an effective-dimensionality estimate and a comparison asking whether phase-structured features add reproducible latent information beyond matched models.

The First Draft also used Siegel, Donner, and Engel’s **spectral fingerprints** proposal to motivate BOT’s slice metaphor [102]. Their review proposes that frequency-specific correlations can index canonical computations in distributed networks. The preserved SAN question is therefore not whether one frequency band universally equals one function. It is whether a preregistered, frequency-resolved set of partial measurements adds reproducible information about a declared cross-scale state or transition beyond anatomy, broadband power, waveform shape, and task covariates. Without that incremental reconstruction result, a spectral fingerprint remains an interaction descriptor rather than evidence for BOT.

#### 4.9 How recurrent thalamocortical return sustains and reshapes cortical state

The thalamic account in the First Draft is restored as part of the main SAN mechanism [1]. The thalamus is not merely a passive relay and not an executive homunculus. Its relay nuclei, diffuse and focused projection systems, local inhibitory control, and reciprocal cortical inputs place it inside the computation. In the historical OCA vocabulary, **matrix** pathways provide comparatively diffuse cortical coordination, **core** pathways provide more focused driving projections, and the thalamic reticular nucleus (TRN) regulates which thalamic outputs are available and when. These are functional distinctions whose exact anatomy and terminology must be matched to the nucleus, species, task, and measurement.

The PFC-thalamic loop supplies a concrete example. A cortical pattern can influence mediodorsal and other thalamic activity; the thalamic system transforms that input through its own excitatory, inhibitory, and state-dependent dynamics; the resulting output returns to cortex; and repeated cycles can sustain, sharpen, redirect, or suppress the cortical pattern. Thus the thalamus does not only *gate* a completed thought. It participates in developing the thought. The First Draft’s working-memory claim is retained as the hypothesis that recurrent PFC-thalamic exchange helps maintain task-relevant activity while suppressing incompatible routes.

Primary circuit results now make that retained claim more specific. Mediodorsal-thalamic projections can sustain prefrontal activity during working-memory maintenance [106]; thalamic activity can amplify task-relevant cortical connectivity during attentional control [105]; and separable thalamic circuits can regulate prefrontal signal and noise [107]. These findings support a stateful selection-and-amplification role, not the universal claim that the thalamus simply overrides bottom-up gamma whenever prediction is unchallenged. SAN’s stronger, testable proposition is that a declared thalamic pathway uses its current internal state and return timing to stabilize one expected cortical route while admitting, amplifying, or redirecting mismatch-bearing evidence. That effect must be demonstrated pathway by pathway against matched cortical activity, sensory drive, arousal, and band-power controls.

SAN calls this a **neural heartbeat** or continuation pulse when emphasizing its temporal function:

cortical proposal or sensory evidence

- > thalamic reception
- > internal relay / TRN transformation
- > phase-compatible return to selected cortical targets
- > cortical refinement and renewed proposal
- > repeat, redirect, suppress, or terminate

The loop can concentrate a pattern through repetition without simply copying it. Phase-compatible return increases the chance that selected cortical receivers are excitable when the next input arrives. Incompatible return can interrupt or redirect the route. This yields the operational state update

$$\mathbf{c}_{k+1} = \mathcal{C}(\mathbf{c}_k, \mathcal{T}(\mathbf{c}_k, \mathbf{s}_k, \mathbf{h}_k), \Delta\phi_k), \quad (20)$$

where  $\mathbf{c}_k$  is cortical state,  $\mathcal{T}$  is the thalamic input-transform-output operation,  $\mathbf{s}_k$  is new sensory / body evidence,  $\mathbf{h}_k$  is thalamic internal state, and  $\Delta\phi_k$  is the return-phase relation at the receiving cortical population. The model predicts that disrupting the timing or internal transformation of the return should impair sustained or refined representation even when mean input strength is matched.

This is related to, but not identical with, pooling in deep learning. Spatial max or average pooling aggregates local activations and reduces representational resolution. Thalamocortical recurrence is closer to a composition of recurrent refinement, attention, routing, competition, state maintenance, and selective aggregation over time. The analogy is useful when it predicts compression or selection; it fails if it erases the thalamus's stateful transformation and bidirectional control.

The same architecture links hippocampal-entorhinal memory and context to cortical processing. Reciprocal hippocampal-entorhinal exchange, theta-structured timing, thalamic participation, and the Circuit of Papez were grouped in the First Draft as **agentic co-regulation**. SAN predicts that memory-derived context changes which returning cortical-thalamic cycles stabilize. The thalamus's own internal cycle simultaneously controls what it accepts, computes, inhibits, and emits. It is therefore an agent within a larger loop, not a transparent wire.

The First Draft further described the thalamus as directly connected with the hippocampal-entorhinal loop and described each participating region as a semi-autonomous sender, receiver, and transformer [1]. The receive-transform-return architecture is preserved. The word *directly* is now an anatomical question to resolve nucleus by nucleus and pathway by pathway rather than an undifferentiated claim about whole regions. The corresponding SAN test asks whether a declared thalamic pathway changes memory-derived cortical selection beyond matched cortical and hippocampal activity, and whether disrupting its return timing produces the predicted route-specific failure.

The First Draft called these repeated re-entrant routes **Sonic the Hedgehog loops**: sensory and contextual signals repeatedly pass through cortical layers, thalamic and other subcortical hubs, and back, with each pass able to sharpen, update, reaffirm, reroute, or suppress the state. The nickname is retained because it records the genealogy and makes the cyclic architecture memorable; the operational term is *re-entrant thalamocortical refinement loop*.

This sequence also preserves the original layered sensory account. Receptor arrays in eye, ear, skin, tongue, nose, and internal organs transduce physical conditions. Feedforward signals reach thalamic and cortical targets; cortical Layer 4 is a major recipient in many sensory pathways; Layers 2/3 support local and lateral integration; Layer 5 provides major descending and motor-related outputs; Layer 6 participates in corticothalamic feedback; and Layer 1/apical dendrites receive important contextual inputs. The actual route varies by system. SAN's claim is the recurring receive-transform-project-return architecture, not a universal one-direction layer diagram.

#### 4.10 Tonic canvas, phasic PWD ink, cognitive control, and choice

The original tonic / phasic distinction is retained. **Tonic** activity names the comparatively persistent, high-occupancy background state that maintains excitability, context, and broad coordination. A **phasic PWD** is a receiver-relative departure carrying new, inhibitory, corrective, or action-relevant information. The mental **canvas** is the maintained background; the **ink** is the difference that changes what the receiver can infer or do. Slow activity can be phasic and fast activity can participate in a tonic regime. Tonic and phasic are functional timescale roles, not fixed frequency labels.



The First Draft joined a real measurement problem to a broader scale-invariance conjecture [1]. At the population level, EEG, MEG, ECoG, and LFP spectra often contain a declining aperiodic component as well as periodic peaks; offset, exponent, center frequency, power, and bandwidth must be separated before interpretation [90]. At the single-cell and terminal level, spike amplitude, waveform width, repolarization, calcium-current integral, release distribution, and postsynaptic consequence are different measurements. Aperiodic spectral slope therefore cannot be derived merely from the claim that a slower event lasts longer, and amplitude and duration cannot be added without normalization or a declared physical quantity.

What survives from SAN's original formulation is the positive bridge that motivated the word *magnitude*: signal consequence is not exhausted by frequency or peak amplitude; persistence, waveform, spatial recruitment, and receiver response can matter. A tonic pattern may provide a persistent, comparatively expected context, while a phasic departure may carry greater conditional surprise and change a receiver. Neither low frequency nor rarity alone supplies those functional roles. The proposed scale-spanning relation is tested by defining, at each scale, a quantity such as waveform area, delivered charge, calcium-current integral, release distribution, field power, or downstream effect, then asking whether a common normalized law survives. Failure of a universal law would not erase the tonic-context / phasic-update mechanism; it would localize which measurements and scales support it.

This distinction supplies SAN's proposed mechanical account of qualitative content. An ongoing system first needs a state against which any particular content can differ. The tonic canvas is that maintained, comparatively high-occupancy and therefore conditionally low-information state of readiness. A phasic PWD is the content-bearing distinction: not merely "more gamma," but a learned, receiver-relative temporal, spatial, signed, and state-dependent pattern that changes what later arrays detect and re-express. Synchronization can help maintain the canvas or open a route, while structured separation, inhibition, and phase departure supply distinctions within it. On this account, consciousness cannot be reduced to synchrony alone because a perfectly undifferentiated state would contain no particular red, tone, body location, word, or object.

The 2022 **Theory of Redness** makes this operation concrete [79]. Wavelength-dependent receptor and retinal activity is transformed through learned visual routes; a particular receiver-relative pattern differs from the ongoing tonic state; distributed neural arrays detect, transform, and project parts of that pattern; and recurrent exchange sustains and relates it to other features, memories, body state, and action. SAN calls the resulting distributed operation a neural rendering of redness. The claim is not that an LFP trace, frequency band, or decoded stimulus label is itself the experience. It is that the recurrent system is both changed by and causally responsive to the content-specific differential it helps produce. That maker-and-receiver loop is the proposed difference between a merely displayed computer rendering and a conscious neural rendering.

A discriminating test must cross the gap between stimulus decoding and content. It should compare physically different inputs that yield the same reported color and matched inputs that yield different color experiences, estimate the receiver-relative differential after rate, power, waveform, eye-movement, attention, decision, and report controls, and perturb an independently localized re-entrant route. SAN predicts cross-condition generalization of the content-specific differential, a selective change in experienced / reported redness when that route is perturbed, and a phase- and route-specific rescue. Failure would constrain the proposed qualia operator without turning the tonic / phasic measurement distinction into an error.

Peter Tse's criterial-causation work remains a historical bridge [47]. SAN uses the bridge to ask how threshold-sensitive receivers convert physically continuous input histories into consequential route selections. When inhibition prevents one activity trajectory and thereby allows another to dominate, the route has divided without requiring a central chooser. SAN calls this distributed selection event a **choice bifurcation**. The term is tested by measuring the competing routes, inhibitory intervention, threshold crossing, and resulting action rather than inferred from behavior alone.

The original alpha / beta-gamma discussion and Earl K. Miller comparison joined a fixed anatomical proposal to a broader functional operation. Those layers must be separated rather than allowing the anatomical qualification to erase the operation. Feedforward and feedback influences can occupy different bands, and alpha / beta activity can oppose or constrain gamma in some tasks and circuits [9,57]. A universal assertion that beta and alpha originate in Layer 4, gamma is confined to Layers 2/3, or one band always suppresses another is therefore a historical hypothesis, not an inherited fact. The surviving SAN mechanism is positive and band-relative: a maintained goal-, memory-, and body-dependent context changes which phasic sensory or internal departures gain downstream effect. The First Draft's ink / canvas account is broader than one beta-stenciling result because the canvas is the receiver's maintained state and the ink is the consequential difference, regardless of which band carries either role in a particular circuit.

Lundqvist and colleagues supplied a more specific established comparator: in monkey prefrontal cortex, brief gamma bursts accompanied encoding and reactivation of remembered items, while beta bursts more often marked a default state interrupted during encoding and decoding [113]. This is evidence for task-dependent beta / gamma burst organization, not by itself a complete stencil or rendering mechanism. SAN's added operation is to specify the maintained receiver context, the particular phasic difference, the route it changes, and the returned evidence that revises the next state. That ordering lets the empirical burst result and the later SAN synthesis be compared without collapsing either into the other.

The First Draft also credited Miller's program on prefrontal rules and concepts, working memory, attention, and mixed selectivity as important background [1]. That background remains part of the paper's intellectual genealogy; it is not the SAN delta. SAN's additional question is how a maintained rule- or goal-dependent state changes the effective consequence of a particular arrival at a stateful receiver, and how that altered route participates in reconstruction, choice, action, and returned evidence. This separates respect for the established PFC program from the later receiver-relative tonic / phasic mechanism being proposed here.

A bat preparing to vocalize gives the operation a concrete biological shape. Fronto-striatal low-beta and high-gamma power before call onset predicted the later vocal class, while theta interareal coupling and pre-vocal spike-field locking also differed by vocal condition in the reported preparation [91]. A later fronto-auditory study found predominantly frontal-to-auditory information flow before vocalization and an auditory-to-frontal return after echolocation [92]. These studies already measure component variables needed for a pre-action -> act -> returned-evidence loop. They do not establish one universal beta-to-gamma command, and their directional analyses do not by themselves prove that an LFP initiated the call.

The remaining mechanistic problem is how activity distributed across many cells becomes effective at a particular downstream receiver. If signed inputs arrive within compatible receiver windows, their effects can reinforce one candidate route; incompatible timing, inhibition, or geometry can reduce or redirect it. A bounded candidate-drive model is

$$D_c(t) = \sum_i w_{ic} I_i(t) K(\Delta t_{ic}, \Delta \phi_{ic}, \mathbf{s}_c(t)),$$

$$\Delta D_c(t) = D_c(t) - \bar{D}_c(t),$$
(21)

where  $I_i$  is a declared signed current or validated proxy,  $w_{ic}$  contains projection efficacy and geometry,  $K$  is a receiver-specific timing / phase compatibility function, and  $\bar{D}_c$  is the expected drive under the maintained context. Only after this operation is explicit does SAN name  $\Delta D_c$  one electrical component of a **building PWD** and call its signed downstream influence **electrical valence**. The term does not mean bulk net charge or psychological pleasantness. The present paper uses it for the proposed coherence- and timing-weighted effective drive through which a represented candidate can gain control of a route. That 2026 operational

definition sharpens, rather than backdates, the earlier send-return, phase-valence, bat-integration, and PWD sources [93-96]. The joined hypothesis is therefore

maintained tonic context

- > candidate-specific phasic PWDs
- > receiver-compatible summation and route recruitment
- > distributed motor-program selection and execution
- > changed body or world
- > returned sensory evidence
- > updated tonic context and later criteria

The 2017 SAN source supplies an echolocation-like send / collect / send-again ancestor [93]. The explicit electrical / psychological-valence comparison appears in August 2022 [94]; the bat-paper integration and joined PWD shorthand follow in August and September 2022 [95,96]. The 2026 equation above is a current formalization and is not backdated into those earlier sources.

The **Gamma Consideration Sandwich (GCS)** retains the joined sensory-control-body loop. Bottom-up sensory evidence, top-down expectation or policy, and proprioceptive / interoceptive return jointly condition which gamma and non-gamma events become effective. One sentence in the first public GCS chapter compressed this into proprioceptive feedback being “channeled through” Layer-5 parvalbumin-positive (PV+) interneurons [59]. The same source later unpacks the intended circuit into three operations: ascending spinal, brainstem, and thalamic routes bring body-related evidence into motor and sensory cortex; Layer-5 pyramidal populations provide major descending motor output; and local PV+ interneurons regulate which excitatory activity gains influence. The compressed sentence should therefore be recovered, not simply deleted.

Current evidence supports the recovered local-gating role. Cutaneous and proprioceptive inputs can dynamically modulate excitatory and fast-spiking inhibitory neurons in motor cortex [97]. In mouse motor cortex, somatosensory-cortical and posterior-thalamic afferents recruit PV+ and pyramidal subnetworks in layer-specific patterns, including posterior-thalamic input to Layer-5 PV+ cells [98,99]. Thalamocortical input to Layer-5 corticospinal populations is also strong and changes with motor learning [100]. These results do not make a local PV+ interneuron the long-range afferent from muscle or joint. They make the stronger source-faithful statement plausible: PV+ circuits can help gate, time, and route body-related evidence after it reaches the relevant cortical circuit. The GCS hypothesis is that this locally gated body-state and efference-copy evidence returns to the network that selected an action, allowing mismatch to change the next route.

#### 4.11 From coordinated differentiation to biologically enriched artificial networks

The First Draft argued that coherent conscious function depends on widespread effective connectivity, including recurrent cortical, thalamic, hippocampal, sensory, and motor exchange [1]. Sleep, anesthesia, seizure, and lesions can alter different parts of this architecture [18-20]. SAN does not infer that gamma synchrony alone is consciousness. Excessive uniform synchrony can reduce differentiation or become pathological; too little coordination can prevent integration. The target is a structured regime in which coherent routes bind selected content while phase differences, inhibition, and splay preserve distinctions. Conscious rendering is therefore proposed to require both coordination and informative deviation from the tonic background.

The First Draft’s phrase **deterministic computation** is retained as a physical-mechanism claim: measured prior state constrains the distribution of later states. It does not require noiseless prediction, make every event presently predictable, or make wave coordination sufficient for consciousness. Koch, Massimini, Boly, and Tononi distinguish background conditions for consciousness, full neural correlates, and content-specific

correlates [110]. That distinction sharpens the SAN experiment without replacing SAN’s mechanism: tonic readiness, the content-bearing PWD, recurrent availability, and report or action must be measured separately. The stochastic term in Section 6.1 then represents unresolved and potentially irreducible variability rather than an escape from causal explanation.

The related First Draft phrase that conflicting patterns can *dissolve through desynchronization* is retained through intent recovery rather than read as literal destruction. Its operational meaning is loss of effective access: phase dispersion, inhibition, or incompatible arrival timing can reduce summation, route recruitment, recurrence, and persistence until a pattern no longer controls the current state. The physical connection need not disappear, and the same structure can be recruited later under a different timing relation. This complementary off-operation matters as much as coherence: structured coordination binds selected content, while structured separation keeps other content distinct or inactive.

The First Draft singled out the medial prefrontal cortex, hippocampus, and central lateral thalamus as contributors to this recurrent conscious-state architecture, and used the compressed phrase that the medial prefrontal cortex is *vital for consciousness* [1]. The intended systems proposition is retained positively: executive context, memory, sensory feedback, and thalamic state modulation can jointly shape which contents remain effectively available to the wider network. *Vital* is treated as a causal-necessity question, not erased and not assumed: a region-specific perturbation must abolish the declared conscious-state operation after arousal, task, motor, report, and network-disconnection controls. The SAN mechanism remains distributed even when one node is a critical contributor.

The First Draft also extended oscillatory coordination to conversation, proposing that listeners partly mirror a speaker’s emotional and conceptual dynamics and that temporary cross-person timing relations can support mutual understanding [1]. This interpersonal extension is preserved as a testable social-scale hypothesis, not as a claim that aligned signals merge two minds or guarantee shared meaning. A discriminating study must show that cross-brain timing predicts transfer of specific content beyond common stimulus timing, speech envelope, movement, turn-taking, and matched behavior, and must perturb or temporally disrupt the proposed coupling without merely degrading the conversation.

When a receiver encounters a mismatch, the difference can be incorporated into a revised coordinated state, inhibited, redirected, or allowed to decay. The scientific problem is to identify which dynamical pathway reduces one mismatch while preserving the distinctions needed for later inference and action. SAN calls this proposed family of relaxation and reconfiguration processes **oscillatory dissipation**. It predicts measurable state transitions and route changes; the term earns a thermodynamic interpretation only when energy, entropy production, boundary conditions, and the relevant conservation law are explicitly measured.

The deep-learning bridge also belongs in SAN. The historical arc from Turing and the perceptron to multilayer and deep networks provided a computational contrast: artificial units usually compress many inputs into a lower-dimensional activation, while biological dendrites add branch-local nonlinearity, recurrent timing, waveform dynamics, molecular state, inhibition, and plasticity. SAN therefore proposed a computational unit with multiple receptive compartments, temporal state, structured projection, and local feedback. The 2022 corpus calls the transition from a conventional unit to this biologically enriched unit **Perceptron-to-Metatron**. *Metatron* here names the proposed unit; the older SAN corpus also used *Metatron* as the name of an artificial being. Implementation, ablation, and matched-network comparison determine whether the proposed unit adds computational value.

Dendritic structure can be modeled as a learned high-dimensional receptive embedding; a spike or burst can serve as a context-dependent token offered to downstream receivers; and repeated recurrent updates can denoise or stabilize a latent estimate. The First Draft calls the structural part the **dendritic vector-embedding model** and the iterative-update correspondence the **Latent Diffusion Analogy**. A biological

test must use measured neurons and states; an artificial-neurology test must demonstrate an advantage over matched recurrent, attention, diffusion, and graph / sequence baselines.

The First Draft made this bridge concrete by placing cortical semantic maps beside interpretable features in artificial networks. Huth, Gallant, and colleagues used voxel-wise models to recover a distributed semantic atlas from responses to natural speech [103]. Anthropic’s dictionary-learning work extracted sparse, partly interpretable features from a transformer [104]. These results show that high-dimensional biological and artificial systems can support distributed, decodable feature structure. They do not establish that the structures are mechanistically identical or that either system performs COT, BOT, NAPOT, or conscious Neural Rendering. The SAN comparison becomes testable only when the biological and artificial models are reduced to the same declared input, state-dependent transformation, output, recurrence, and held-out reconstruction task.

This yields the retained engineering program called **Artificial Neurology**: construct systems in which units receive structured partial conditions, maintain local state, emit time-bearing outputs, participate in recurrent multiscale control, learn from embodied consequences, and expose enough internal variables for causal testing. The goal is not to call any recurrent network conscious. It is to build a falsifiable architecture in which the SAN causal chain can be implemented, ablated, and compared.

The First Draft’s prospective sentient-AI branch remains a positive research question rather than an automatic inference. If an artificial system implemented the joined SAN architecture, maintained a perturbation-sensitive self / world / body model, used that model in recurrent embodied control, and showed content-specific loss and matched rescue under interventions, it would become a stronger candidate for consciousness research than a behaviorally similar black box. Those results would still not prove experience or settle moral status, but they would make the original artificial-SAN hypothesis experimentally sharper instead of deleting it.

#### 4.12 Conservation and correction boundary

Lossless revision does not require repeating a known error as an active conclusion. It requires the paper to keep the idea visible and state what happened to it. Accordingly:

- claims supported as established components remain labeled established;
- SAN joins and mechanisms remain labeled synthesis or hypothesis;
- metaphors such as tomography, volumetric television, quasicrystals, Sonic loops, ink / canvas, and fractal organization remain visible with operational translations;
- overly universal anatomical, frequency, deterministic-release, and consciousness claims remain visible as corrected or retired formulations;
- historical influence and collaboration language remains in the record without manufacturing coauthorship, endorsement, or independent validation.

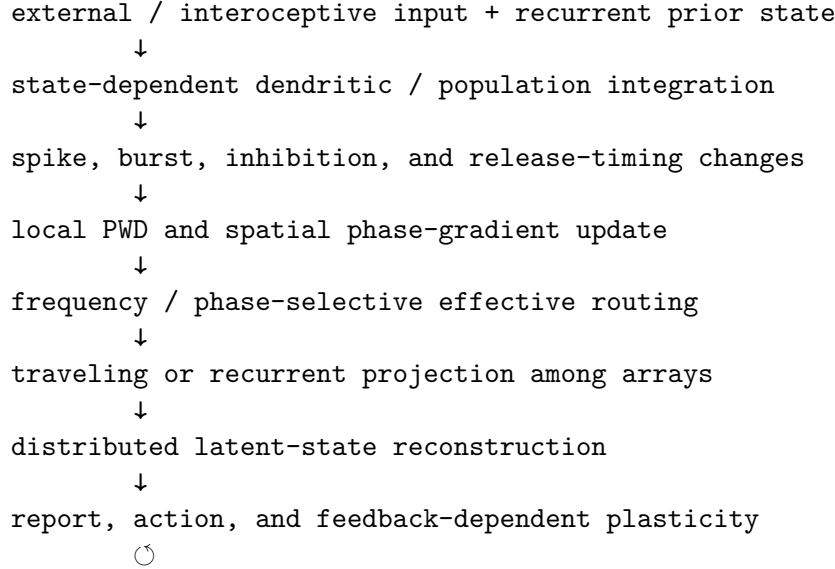
Two Draft 2 “retractions” require an additional source correction. The First Draft did not report completed neural tomography; Appendix A.7 explicitly said that literal oscillatory tomography had not yet been validated. Nor did it present “oscillation alone” as its complete binding or consciousness operator: its positive account joined phase relations with receiver and molecular state, inhibition, recurrence, differentiation, feedback, and multiscale effective connectivity. Those stronger paraphrases are therefore not retained as quotations from Micah. What remains open to test is the actual joined SAN mechanism and its proposed phenomenological consequence.

The complete Draft-1-to-Draft-12 concept ledger appears in Appendix D. It is part of this paper’s revision record and is an acceptance gate for any successor draft.

## 5. Core causal architecture

### 5.1 Input → transformation → output → feedback

The minimal NAPOT cycle is:



The arrows name causal commitments, not merely a narrative. A valid experiment should identify which arrow it tests and should not promote a result beyond that arrow.

For embodied vision, the loop has a concrete external closure: motor command and efference copy predict a change; movement changes the body and its retinal / proprioceptive consequences; those signals update distributed body and world variables; and the update changes the next policy. A decoder of body part without this perturbable action-reafference loop establishes representation, not the GCS composite.

A brief flash supplies the simpler feedforward-and-return example retained from the First Draft. Photoreceptors transform the event into parallel, convergent, and divergent retinal outputs; those outputs travel through optic and thalamic routes into early visual cortex; lateral and recurrent activity then changes which features and locations remain accessible to downstream populations. Feedback can return in a different spectral regime from feedforward activity [57]. SAN's added proposal is not that gamma alone turns the flash into one conscious field. It is that distributed receivers compare the arriving and returning activity with their own state, re-express consequential differences, and jointly update a testable latent estimate. A valid experiment must distinguish that receiver-relative recurrent update from feedforward feature decoding, common input, arousal, eye movement, and scalar synchrony.

### 5.2 There is no internal spectator

Projector and reconstruction language can invite a homunculus. The model posits no internal observer. Every downstream array is itself a transforming receiver whose state is available only through its effects on later activity and behavior.

Receiver relativity concerns access, not existence. A physical difference between states can exist without an observer or nervous system. For receiver  $R$ , let  $p_R(y \mid x)$  denote the channel from physical state  $x$  to a receiver-accessible outcome  $y$ . The difference between  $x_a$  and  $x_b$  is informationally available to  $R$  only if the induced outcome distributions are discriminable and can change a later state or action. A difference that

the channel maps to indistinguishable outcomes is inaccessible to that receiver, not absent from the world. This keeps the operational claim separate from the metaphysical claim that an observer creates physical distinctions.

The anatomical cortical homunculus does not violate this rule. It names coarse and overlapping somatotopic organization; it is neither a continuous miniature body nor a homuncular spectator. SAN treats such maps as candidate constraints within the distributed embodied state.

Distributed and bottleneck readout architectures remain empirical competitors. Evidence that distributed, perturbation-sensitive partial representations jointly predict behavior better than a specified bottleneck model would reject that bottleneck for the tested endpoint; it would not prove the metaphysical absence of a spectator.

### 5.3 Excitation and inhibition are both informative

An inhibited unit is not simply missing data. Relative to an expected tonic pattern, suppression, delay, or phase reset can alter which downstream coincidences occur. SAN therefore treats excitation, inhibition, and silence-in-context as possible signed constraints on routing. The claim does not imply that every absence is meaningful; information requires a receiver model and a counterfactual expectation.

The same boundary applies to surprise and meaning. Conditional surprise can be defined as  $I_t = -\log_2 p(y_t | y_{<t}, a_{<t})$ , and Bayesian information gain as  $D_{\text{KL}}[q(s_t | y_{\leq t}) || q(s_t | y_{<t})]$ . A rare or model-changing event can score highly on either quantity without being semantically relevant to the organism. SAN therefore keeps statistical surprise, posterior change, receiver accessibility, learned value, and causal use as distinct variables. "Meaningful" requires an explicit task, receiver, and consequence; it is not a synonym for low probability, salience, or phase difference.

### 5.4 Multimodal and multiscale integration

Different modalities enter through distinct pathways but can influence common recurrent states. Integration is not defined as simultaneous activation alone. It requires that cross-modal signals contribute nonredundant constraints to a latent variable or action. At multiple scales, a coarse phase for region  $R$  may be defined by a weighted circular mean,

$$\begin{aligned} Z_R(t) &= \frac{1}{\sum_{i \in R} w_i} \sum_{i \in R} w_i e^{i\phi_i(t)}, \\ r_R &= |Z_R|, \\ \Phi_R &= \arg Z_R. \end{aligned} \tag{22}$$

The nonnegative weights are fixed a priori or estimated on training data. The magnitude  $r_R(t)$  measures within-region phase concentration;  $\Phi_R(t)$  is meaningful only when that concentration and signal quality exceed preregistered thresholds. When local node  $i$  is compared with a regional phase, use the leave-one-out quantity  $Z_{R \setminus i}$ ; otherwise part-whole coupling creates a mathematical dependence. Same-band comparisons require a common reference and leakage-aware source estimates. Cross-frequency comparisons require an explicit  $n : m$  phase relation rather than subtracting phases with different cycle counts. This is a coarse-graining operation, not proof that the same computation exists unchanged from molecules to brains.

Nor does recurrence of a graph motif at several scales establish scale invariance. A scale-invariant claim must declare a transformation  $\mathcal{S}_\lambda$  and the quantity  $Q$  expected to obey a specified scaling law, for example  $Q(\mathcal{S}_\lambda x) = \lambda^\alpha Q(x)$  over a bounded range with a fitted exponent, uncertainty, and out-of-range failure



conditions. A claim of locality must likewise name its neighborhood rule: anatomical adjacency, spatial support of a field, conduction delay, or an explicit interaction kernel. Without those declarations, “multiscale” means only that separately measured levels are related; it does not imply exact self-similarity, a single law at every level, or unconstrained propagation.

The SAN **global spectral map** is therefore operationalized as a nested collection  $\{T_s, D_s\}_{s=1}^S$ , not one unitary electromagnetic field. Each scale has a declared spatial support, reference, estimator, resolution, and coupling interface. A local dendritic or cellular event can influence this collection through ordinary synaptic / axonal projection, recurrent excitation and inhibition, common neuromodulatory drive, and, where independently supported, ephaptic coupling. Neural transmembrane currents contribute to extracellular potentials [78], and weak extracellular electric fields can influence nearby cortical activity in bounded preparations [76,77]. Those findings admit an ephaptic term; they do not establish brain-wide wireless broadcast.

Let  $I_k^m$  be a declared transmembrane-current source and  $G_k$  a conductivity- and geometry-aware forward kernel. A minimal observation model is

$$V_e(\mathbf{x}, t) = \sum_k (G_k * I_k^m)(\mathbf{x}, t) + \epsilon(\mathbf{x}, t). \quad (23)$$

For a cellular intervention and matched control, define a field-phase footprint only where analytic phase is reliable:

$$\delta\Phi_E(\mathbf{x}, t) = \text{wrap}\left(\Phi_E^{\text{do}(i)}(\mathbf{x}, t) - \Phi_E^{\text{control}}(\mathbf{x}, t)\right). \quad (24)$$

“Global” requires reproducible source-localized effects across a predefined distributed region set, with distance / time attenuation, volume-conduction, source-leakage, common-input, and conventional synaptic propagation controls. Until then, the supported term is **local-to-distributed field perturbation**. Magnetic mediation is not inferred from an extracellular-voltage measurement.

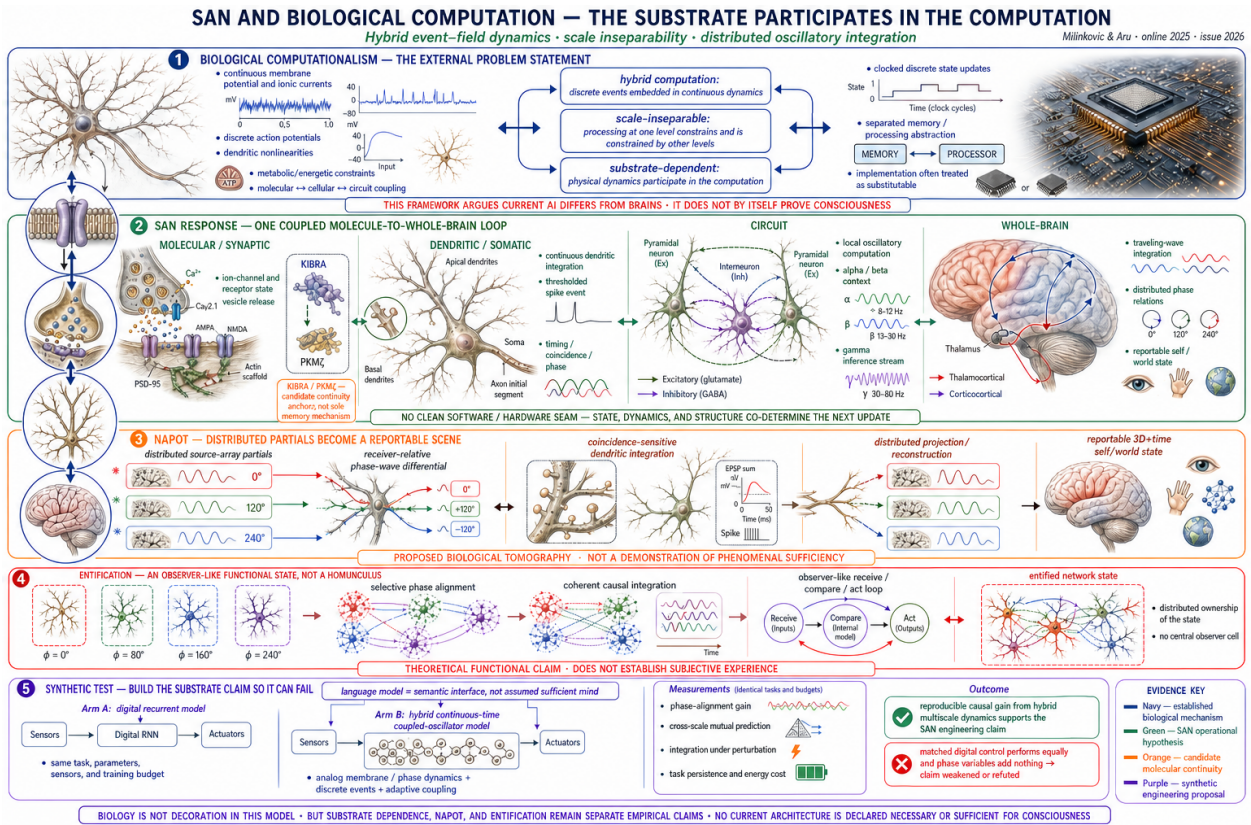
The firefly-swarm analogy [79] illustrates the topology, not the evidence: changing or removing one coupled member can reorganize a collective state if coupling, gain, receiver state, and thresholds permit it. It does not imply that every cell detects every event, that one neuron has a measurable whole-brain field effect, or that the response is instantaneous or deterministic.

The multiscale figure also places SAN beside Milinkovic and Aru’s biological-computationalism proposal that biological computation is hybrid, substrate-dependent, and coupled across scales [87]. That proposal is an external comparator and motivation for controlled synthetic tests; it is not evidence that SAN, NAPOT, entification, or biological consciousness has been established.

**Figure 6.** Multiscale biological-computation schematic. Molecules, membranes, cells, circuits, and whole-brain measurements can all constrain computation, but the figure does not assert that every scale performs the same operation, that scale invariance has been shown, or that multiscale participation is sufficient for consciousness. Cross-scale claims require explicit transformations, neighborhood rules, and intervention-sensitive endpoints.

## 5.5 Memory and rapid routing

SAN separates slower structural memory from faster functional routing. Sparse synaptic, dendritic, and channel states constrain a space of possible transformations; object-, body-, or task-relative reference-frame



**Figure 11: Multiscale biological computation in SAN**

state can condition which structural hypotheses are eligible. Receiver-relative phase, frequency, amplitude, and explicitly declared waveform or release variables then describe a fast residual around that state. Molecular turnover does not by itself erase synaptic potentiation: the CaMKII replacement and KIBRA-PKM $\zeta$  results provide two preparation-specific maintenance mechanisms [80,81]. Phase relationships may help select a transient path through the eligible space. Repeated successful paths may bias slower plasticity, but this paper does not assume a simple one-to-one conversion between phase locking and long-term memory.

This two-layer decomposition is a SAN synthesis, not a result attributed to Hawkins or the Thousand Brains Project. Sparse structure alone, reference-frame structure plus sparse state, dynamic residuals alone, and the combined model must be compared under matched information, state dimension, parameters, training budget, and compute. A fast residual contributes only if it improves held-out prediction or causal control above the strongest locked structural / reference-frame comparator.

The historical statement that a pattern “alternates” between synaptic configuration and propagated wave is therefore a testable state-to-trajectory mapping, not literal conversion of a static object into a wave. A structural match changes the distribution of measured outputs  $p(\mathbf{u}_i^{\text{out}} \mid m_i, M_i, q_i)$ ; recurrent outputs can in turn change fast receiver state and, on a slower timescale, plastic structural state. Both directions require longitudinal or causal evidence.

Neural Rendering adds a temporal cycle to the decomposition:

$$\begin{aligned}
 &\text{structural preset} \\
 &\rightarrow \text{pattern match} \\
 &\rightarrow \text{phasic output} \\
 &\rightarrow \text{multiscale projection} \\
 &\rightarrow \text{tonic-state update} \\
 &\rightarrow \text{new receiver context.}
 \end{aligned} \tag{25}$$

The phasic pattern need not simply disappear. It may be rejected, remain transient, recruit an attractor, become a maintained tonic expectation, or drive slower plasticity. These alternatives must be distinguished. In this manuscript, *dissipation* means only measured contraction, mixing, or attenuation of a declared differential. It does not mean thermodynamic entropy production, universal equilibration, or guaranteed consensus. Topology, coupling sign, balance, stochastic forcing, attractor basin, and biochemical update rules can preserve, amplify, redirect, or continually drive the mismatch. “Informational absorption” is supported only when the post-event tonic state gains held-out predictive information about the perturbation or target and the phasic residual changes in the preregistered direction. A more structured or predictive post-update state is compatible with this computational use; it is not evidence of decreased thermodynamic entropy.

This division yields a concrete prediction: after controlling for activity magnitude, the temporal persistence and recurrence of a task-specific PWD pattern should predict later structural or functional stabilization better than a single high-rate event. The prediction remains open until a longitudinal plasticity experiment tests it.

## 6. Formal model

### 6.1 Delayed stochastic phase dynamics

At a mesoscopic level, let each node  $i$  have latent phase  $\theta_i$ , amplitude  $A_i$ , intrinsic angular frequency  $\omega_i$ , angular-frequency drive  $u_i$ , and state-dependent edge coupling  $K_{ij}$ . A deliberately minimal phenomenological model in the delayed noisy Kuramoto-Sakaguchi family [44-46] is:

$$d\theta_i = \left[ \omega_i + \sum_j K_{ij}(M_i, t) \sin(\theta_j(t - \tau_{ij}) - \theta_i(t) - \psi_{ij}) + u_i(t) \right] dt + \sigma_i dW_i(t). \quad (26)$$

Here  $\theta$  is in radians;  $\omega$ ,  $u$ , and  $K$  have units of radians per unit time; and an initial history is specified on  $[-\tau_{\max}, 0]$ . If a physical input rather than an angular drive is modeled, it must enter through a declared phase-response curve. The Brownian processes have a declared cross-node covariance rather than being assumed independent silently. Observed analytic phase is linked to latent phase by

$$\phi_i(t) = \text{wrap}(\theta_i(t) + \nu_i(t)), \quad (27)$$

with an explicit measurement-noise model. Each analysis must state whether  $A_i$  and  $M_i$  are observed exogenous covariates or receive their own dynamics. The stochastic term prevents “deterministic” from becoming an unsupported biological claim. Signed or state-dependent coupling is only a reduction of richer excitatory and inhibitory dynamics. This equation is useful if it generates testable estimates; it is not presented as a biophysical derivation.

### 6.2 Routing gate

For a training-defined local ensemble  $R$ , define a candidate fast state

$$\mathbf{z}_i(t) = [\Delta\phi_i(t), \Delta\omega_i(t), \Delta \log A_i(t), \text{CV}_i(t), w_i(t), e_i(t)], \quad (28)$$

where  $\Delta\phi_i$  is a wrapped receiver-relative phase deviation,  $\Delta\omega_i$  is an instantaneous-frequency deviation from a leave-one-out ensemble reference,  $\Delta \log A_i$  is a standardized amplitude residual,  $\text{CV}_i$  is a declared inter-event or spike-count variability measure,  $w_i$  is one preregistered waveform or transmission variable, and  $e_i$  is a declared downstream effect or pathway class. Each component requires its own units, reliability gate, likelihood, and matched null. This vector is a proposed measurement interface, not a claim that one neuron broadcasts a complete learned memory.

The historical token / ink language is operationalized as the defined joint output vector

$$\mathbf{u}_i^{\text{out}}(t) = [\mathbf{z}_i(t), b_i(t)], \quad (29)$$

where  $b_i$  is the receiver-defined event from Section 3.6. The latent structural match  $m_i$  from Section 3.2 is not inserted into the emitted vector. Instead, the hypothesis is that  $p(\mathbf{u}_i^{\text{out}} \mid m_i, M_i, q_i)$  differs across matched structural states. The output is not asserted to be a lossless compression or unique inverse of the dendritic state. A study must test how much of the triggering structural state, if any, is decodable from the measured output after nuisance and alternative-code controls.

For an independently oriented or experimentally assigned edge, let  $E_{j \rightarrow i, t} = 1$  denote a preregistered downstream transfer event - for example, a receiver response or conditional information-transfer event inside a fixed post-input window. Define a candidate gate as a probability

$$g_{ij}(t) = \Pr(E_{j \rightarrow i, t} = 1 \mid \mathcal{H}_t) = \ell\left(\beta_0 + \beta_1 \cos \Delta\phi_{ij}(t) + \beta_2 A_i(t) A_j(t) + \beta_3^\top \mathbf{M}_i(t) + \beta_4^\top \mathbf{c}_{ij}(t)\right), \quad (30)$$

where  $\ell(\cdot)$  is a logistic link,  $\mathcal{H}_t$  is the allowed history, and  $\mathbf{c}_{ij}$  contains nuisance and control variables. Prediction of  $E$  does not identify a causal route. The phase term is supported observationally only if it improves held-out prediction after amplitude, power, firing rate, connectivity, evoked response, and common-input controls.

To close the formal chain, let the gated incoming message and state transition be

$$\begin{aligned} \mathbf{m}_i(t) &= \sum_j g_{ij}(t) H_{ij}(\mathbf{q}_j(t - \tau_{ij})), \\ \mathbf{q}_i(t + \Delta t) &= T_i(\mathbf{q}_i(t), \mathbf{m}_i(t), \mathbf{u}_i(t)), \end{aligned} \quad (31)$$

and let a downstream latent estimate be  $D(\mathbf{q}_{1:n}(t))$ . The edge set, array boundaries, and fixed versus learned parameters must be declared before the confirmatory test; data-driven parcellations are learned inside training data and checked against preregistered alternatives. Other links and transition families may be compared; the logistic gate is not a defining law of SAN.

At multiple scales, a deliberately phenomenological rendering differential can be written

$$dD_s = \left[ -\Lambda_s D_s + F_s^{\text{syn}}(D, M) + F_s^{\text{field}}(D, V_e) + \sum_{r \neq s} B_{sr} D_r + U_s(t) \right] dt + \Sigma_s dW_s. \quad (32)$$

The terms separate relaxation, synaptic / axonal propagation, eligible field coupling, cross-scale interaction, input, and noise. A result must estimate or ablate them rather than folding every path into “oscillation.” The rendered latent is a candidate operator

$$\hat{\mathbf{h}}(t) = \mathcal{R}(\{T_s, D_s, M_s, P_{r \rightarrow s}\}_{s,r}), \quad (33)$$

trained without test leakage and compared with equal-input recurrent models. This expression formalizes nested Neural Rendering; it is not a derivation of phenomenology from field equations.

### 6.3 Phase frustration as a diagnostic objective

For an independently fixed edge set with nonnegative normalized weights  $w_{ij} \geq 0$ , define the standard phase/XY-style mismatch objective [44,46]

$$\mathcal{F}(t) = \sum_{(i,j) \in E} w_{ij} [1 - \cos \Delta\phi_{ij}(t)]. \quad (34)$$

$\mathcal{F}$  is called *phase frustration*. It is dimensionless unless explicitly calibrated to a physical energy. Edge set, weights, offsets, temporal window, and predicted direction of contrast must be fixed on independent

or training data; otherwise a decrease can be selected tautologically. A confirmatory test must specify a quantitative contrast such as  $\mathbb{E}[\mathcal{F} \mid \text{correct}] - \mathbb{E}[\mathcal{F} \mid \text{matched error}]$  and its smallest effect of interest. Lower  $\mathcal{F}$  means better agreement with the modeled offsets, not lower thermodynamic free energy, and global zero-lag synchrony is not the target. Until a directional contrast is preregistered, phase frustration is exploratory.

## 6.4 Dynamic reconstruction

Let  $\mathbf{p}_t$  contain the array projections, PWDs, gradients, and recent history. A recurrent reconstruction can be written

$$\hat{\mathbf{s}}_t = R_{\Theta}(\mathbf{p}_t, \hat{\mathbf{s}}_{t-1}, \mathbf{a}_{t-1}), \quad (35)$$

where  $\mathbf{a}_{t-1}$  is the previous action and  $\Theta$  is estimated only on training data. A claim that the reconstructed state guides action additionally requires a predefined policy or readout

$$\mathbf{a}_t \sim \pi(\hat{\mathbf{s}}_t, \mathbf{c}_t) \quad (36)$$

and a mediation or perturbation test showing that changes in  $\hat{\mathbf{s}}_t$  account for the predicted action change beyond sensory, decisional, and motor covariates. A NAPOT model must be compared with nested alternatives:

- $R_{rate}$ : rate / event-count features only;
- $R_{power}$ : rate plus band power;
- $R_{sync}$ : rate, power, and scalar coherence;
- $R_{conn}$ : the above plus static connectivity;
- $R_{rec}$ : a flexible recurrent model without spatial phase gradients;
- $R_{raw}$ : a flexible sequence / graph model receiving the same raw multichannel time series, geometry, and history as NAPOT, with the same architecture family, training budget, and capacity-matching rule but without hand-engineered NAPOT features; and
- $R_{NAPOT}$ : the complete model including edge-conditioned PWD and gradient variables.

The primary statistical object is the preregistered out-of-sample increment

$$\Delta_{NAPOT} = \mathcal{L}(R_{best \text{ baseline}}) - \mathcal{L}(R_{NAPOT}), \quad (37)$$

for a minimized loss  $\mathcal{L}$  fixed before evaluation. Baseline selection occurs inside inner cross-validation; one locked champion is evaluated on the untouched test set, or all test-set model comparisons receive a declared multiplicity adjustment. The smallest effect of interest, equivalence margin, and subject-level confirmatory rule are fixed in advance. If  $R_{raw}$  matches NAPOT, handcrafted PWD features may still improve interpretability or sample efficiency, but predictive performance alone does not validate the claimed neural mechanism. A favorable point estimate without uncertainty, permutation control, held-out replication, and a causal test is not evidence for literal reconstruction.

## 6.5 Identifiability conditions

Tomographic language is invalid when many incompatible latent states produce the same observed projections and no independent constraint resolves them. In addition to the rank / Jacobian and stability conditions in Section 3.7, a dynamic system must satisfy an appropriate local observability-rank condition over the tested horizon. At minimum, the analysis must state:

1. the latent variable and how it is measured independently;
2. the projection features and spatial geometry;
3. which parameters are fixed, learned, or nonidentifiable;
4. how leakage from labels into feature construction is prevented;
5. what surrogate data destroy the claimed structure while preserving simpler statistics; and
6. what null result would reject the reconstructive interpretation;
7. the admissible latent subspace and remaining gauge / equivalence freedom;
8. why the inferred neural operation is not merely the experimenter's flexible decoder; and
9. the likelihood appropriate to each feature family - circular or wrapped distributions for phase, count likelihoods for events, and Gaussian residuals only where justified - with joint covariance modeled across correlated projections.

## 6.6 Relation to NDCA

Earlier SAN material used *Nonlinear Differential Continuous Approximation* (NDCA) as an umbrella term. In this paper NDCA is not treated as a completed mathematical theory. It can name the general idea that continuous nonlinear dynamics generate discrete, receiver-dependent events and approximations. It must not be cited as a derivation until its variables, equations, and theorems are specified independently of the present phenomenological model.

## 7. Empirical analysis specification

### 7.1 Study classes

The theory should be tested in increasing order of inferential strength.

1. **Reanalysis of public observational data.** Establish whether the candidate variables are measurable and add reproducible information.
2. **Prospective observational replication.** Freeze acquisition, preprocessing, features, and endpoints before collecting a confirmatory dataset.
3. **Causal perturbation.** Alter relative timing while matching simpler activity statistics.
4. **Cross-scale recording and perturbation.** Test the proposed micro-to-mesoscopic arrows.
5. **Engineering benchmark.** Determine whether an explicitly phase-structured architecture gains the predicted computational capability under fair controls.

An observational decoder can justify moving to a perturbation experiment; it cannot substitute for one.

### 7.2 Dataset requirements

The first reanalysis should use a public dataset with:

- enough spatial samples for the proposed geometry: at least  $d + 1$  non-collinear local samples for a  $d$ -dimensional gradient, with spacing relative to wavelength, coverage, and design-matrix conditioning reported;
- sufficient temporal resolution for preregistered bands;
- trial-resolved stimulus, task-state, report, or action labels;
- enough trials or continuous samples for nested train / validation / test separation;
- stable channel and artifact metadata;
- preferably anatomical connectivity or electrode localization; and
- a license that permits redistribution of derived features and code.



For a  $d_s$ -dimensional linear reconstruction, the dataset must also supply at least  $d_s$  independent constraints and pass the rank and conditioning checks in Section 3.7. Two spatial sites can estimate a one-dimensional phase difference; they cannot establish a two-dimensional gradient.

The dataset must be chosen before inspecting which one produces the largest SAN-compatible effect. If several datasets are screened, every screened dataset and null result must remain in an all-runs manifest.

No recording modality observes the complete SAN operation. EEG and MEG provide macroscopic electrical or magnetic mixtures; ECoG samples population field activity nearer the cortical surface; LFP and spike recordings provide increasingly local but still geometry- and reference-dependent signals [78]. Voltage-sensitive-dye imaging has resolved millimeter-scale waves propagating through retinotopic V1 in awake primates [108]. In simultaneous EEG-fMRI, BOLD sampled every 2.08 seconds has been fitted with a hidden Markov model to identify whole-brain network states and their transition probabilities across wakefulness and non-REM sleep, with EEG supplying the independent sleep-stage labels [109]. These modalities therefore supply different observation operators, resolutions, and confounds rather than interchangeable tomographic slices. A cross-modal SAN result must predeclare the latent variable, align clocks and anatomy, model each forward observation process, and show that the inferred receiver-to-route relation survives within-modality replication and cross-modal prediction.

### 7.3 Preprocessing and estimator discipline

Phase estimates are unusually vulnerable to analytic choices. The preregistration must therefore fix or systematically vary:

- reference scheme and volume-conduction controls;
- line-noise removal and artifact rejection;
- filter family, order, bandwidth, edge handling, and padding;
- criteria for excluding low-amplitude phase estimates;
- Hilbert, wavelet, state-space, or cycle-by-cycle phase estimator;
- spatial interpolation and gradient estimator;
- source / sink / spiral / planar wave classifier;
- conduction-delay estimation;
- temporal leakage boundaries; and
- trial, session, and subject grouping in cross-validation.

Robustness must be shown across a bounded multiverse fixed before the test set is opened. A result that exists only at one post hoc band or filter is exploratory. For prediction timing and closed-loop claims, phase must be estimated causally in real time or with a state-space method / fixed group-delay correction using strictly pre-outcome windows. Symmetric zero-phase filters and retrospective Hilbert phase are sensitivity analyses only because they can leak future activity.

### 7.4 Feature families

The analysis should construct the following features without using held-out labels:

| Family         | Candidate variables                                  |
|----------------|------------------------------------------------------|
| Rate / event   | spike count, high-gamma event rate, evoked amplitude |
| Power          | band-limited amplitude and power by site             |
| Scalar timing  | pairwise phase locking, coherence, phase-lag index   |
| Static network | anatomical weight, distance, functional connectivity |

| Family                | Candidate variables                                                                  |
|-----------------------|--------------------------------------------------------------------------------------|
| Wave geometry         | local gradient, direction, speed, curvature, source / sink / spiral identity         |
| Receiver-relative PWD | edge-conditioned, delay-constrained phase deviation and coincidence-window occupancy |
| Cross-scale           | local-to-regional phase relation and cross-frequency interaction                     |
| Recurrent history     | previous latent estimate, action, and projection state                               |

PWD and wave features must be compared against a hierarchy of nulls rather than one destructive surrogate: time shifts preserving autocorrelation; phase perturbations preserving amplitude and selected cross-channel dependence; spatial nulls preserving distance, hemisphere, degree, and sensor covariance; edge / delay shuffles; and label permutations inside valid exchangeability blocks. The report must say which marginal each null preserves and destroys.

## 7.5 Primary endpoint

The primary endpoint is a preregistered, nested, out-of-sample improvement of  $R_{NAPOT}$  over a baseline champion selected inside inner cross-validation for one independently measured latent or behavioral variable. One locked champion is evaluated on the untouched test set; alternatively, all locked test-set comparisons receive multiplicity correction. The report must include:

- the loss or information metric and its direction;
- subject- or session-level effect estimates;
- uncertainty intervals;
- correction for the number of confirmed bands, outcomes, and models;
- a permutation or surrogate null distribution;
- calibration as well as discrimination where predictions are probabilistic; and
- a held-out replication or a clear “not yet replicated” label.

The effect must not be described as tomography unless the projection / reconstruction conditions in Section 3.7 are also met.

## 7.6 Model comparison and ablation

The decisive comparison is not NAPOT versus no model. It is NAPOT versus the strongest adjacent account given the same data and capacity. Required ablations are:

1. remove spatial gradients while retaining local phase;
2. remove direction and delays while retaining scalar coherence;
3. remove phase while retaining rate and power;
4. remove recurrent history;
5. randomize the spatial embedding while preserving the graph degree distribution;
6. randomize phase while preserving the amplitude spectrum;
7. compare with  $R_{raw}$  using the same raw inputs, geometry, history, architecture family, optimization budget, and declared capacity rule; and
8. test whether eye movement, arousal, movement, or evoked waveform explains the increment.

If a simpler model performs equivalently within a preregistered smallest effect of interest, the extra SAN mechanism is not supported for that endpoint.

## 7.7 Reproducibility package

A completed empirical paper should include:

- a machine-readable dataset manifest with stable identifiers and licenses;
- a clean environment specification;
- deterministic seeds where algorithms permit them;
- raw-to-feature and feature-to-result scripts;
- hashes for inputs, derived data, code, figures, and tables;
- one documented clean replay command;
- all favorable, null, failed, and excluded runs with reasons; and
- a machine-readable result file whose scope matches the prose claim.

No simulated or expected result in this manuscript is an observation.

## 8. Falsifiable predictions and experiments

### 8.1 Prediction matrix

No single row uniquely confirms SAN. Before a confirmatory program begins, it must freeze a scope tuple - preparation, task, band, measurement class, phase-reliability threshold, primary metric, smallest effect of interest, equivalence margin, multiplicity family, and required independent replications. The SAN delta bears on the preregistered conjunction, not a generic positive phase effect.

| ID        | Prediction                                                                   | Also predicted by                                                 | SAN-specific conjunctive signature                                                                                               | Result that counts against the scoped claim                                                      |
|-----------|------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>P1</b> | Edge-conditioned PWD and wave geometry contain incremental task information. | Phase coding, traveling-wave decoding, flexible sequence models.  | Increment above rate, power, coherence, connectivity, and equal-information $R_{raw}$ , replicated under phase / space nulls.    | Equivalence to the locked champion within the preregistered margin in the required replications. |
| <b>P2</b> | Relative phase causally changes transfer.                                    | Communication through coherence and receiver-excitability models. | Preferred-phase reversal with receiver state, directed downstream transfer, and the predicted latent-state or behavioral change. | No phase-by-receiver-state effect or no downstream / latent consequence.                         |
| <b>P3</b> | The optimal coincidence window changes with receiver state.                  | Dendritic integration and inhibitory-window models.               | Independently localized window shift quantitatively predicts P2 transfer and latent update.                                      | One fixed window is equivalent, or the shift fails to predict transfer.                          |
| <b>P4</b> | Feedforward / feedback wave-intersection geometry predicts update time.      | Predictive-wave and recurrent-processing accounts.                | A preregistered non-additive interaction residual at the predicted locus / time mediates the latent update.                      | Additive evoked timing explains the endpoint equally well.                                       |

| ID         | Prediction                                                                                    | Also predicted by                                                                                               | SAN-specific conjunctive signature                                                                                                                                                                                                                                                                     | Result that counts against the scoped claim                                                                                                                          |
|------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P5</b>  | Structural geometry and independently known direction / delay predict wave flow and behavior. | Connectome-wave models.                                                                                         | Structure→wave association plus receiver-specific transfer and latent update; causal language requires intervention.                                                                                                                                                                                   | Structure does not predict wave flow, or wave flow adds no endpoint information.                                                                                     |
| <b>P6</b>  | Persistent task-specific PWD patterns predict later stabilization.                            | Timing-dependent plasticity and recurrent-state accounts.                                                       | Persistence defined before analysis predicts retention above rate / power / repetition and participates in the reconstructed route.                                                                                                                                                                    | No incremental retention relation within the scoped equivalence margin.                                                                                              |
| <b>P7</b>  | Microtiming perturbation produces a predicted mesoscopic phase / routing change.              | Cross-scale and oscillatory-coupling models.                                                                    | A preregistered micro-to-meso transfer function plus downstream latent / action consequence.                                                                                                                                                                                                           | Transfer function or downstream consequence fails in the bounded preparation.                                                                                        |
| <b>P8</b>  | A tonic / phasic Neural Rendering architecture improves rapid rerouting / reconstruction.     | Sparse sequence memory, oscillator, complex-valued, state-space, and gated recurrent networks.                  | The combined structural / reference-frame / tonic / phasic model has an advantage on one preregistered primary task after matching information, state dimension, temporal precision, bandwidth, parameters, data, compute, and seeds.                                                                  | No advantage, no structural-to-wave decodability, or a simpler matched baseline reproduces it.                                                                       |
| <b>P9</b>  | Joint phase-structured projections contribute to a self-related model.                        | Predictive self-model, agency, interoception, and metacognition accounts.                                       | Joint self and world variables generalize across conditions; projection perturbation changes self / other or agency attribution in the prederived dimension beyond perceptual / decision / motor controls.                                                                                             | Stimulus variables explain the result, or no self-related causal increment remains.                                                                                  |
| <b>P10</b> | A distributed embodied state is updated by active visual-motor-reafferent feedback.           | Body-schema, multisensory ownership, corollary-discharge, predictive-processing, and recurrent-vision accounts. | Independently localized body-part patterns cross-generalize across selected perception / imagery / action conditions; GCS phase variables add above equal-input controls; closed-loop delay shifts the predicted timing / phase relation and a phase-specific rescue restores the body-state endpoint. | General imagery, visual features, eye / EMG activity, movement state, or a raw recurrent baseline explain the endpoint; no predicted timing shift or rescue remains. |

| ID         | Prediction                                                                                                                 | Also predicted by                                                                                                  | SAN-specific conjunctive signature                                                                                                                                                                                                                                             | Result that counts against the scoped claim                                                                                                             |
|------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P11</b> | A bounded pattern-recognition event produces a measurable local-to-distributed field-phase and network-state perturbation. | Biophysical LFP generators, ephaptic coupling, synaptic propagation, neural-field models, and coupled oscillators. | A preregistered cellular intervention predicts a source-localized phase footprint, distance / time attenuation, receiver-state interaction, cross-scale transfer, and downstream latent / action change after synaptic, volume-conduction, common-input, and leakage controls. | No reliable footprint remains, controls fully explain it, it fails to transfer scales, or no downstream consequence follows in the bounded preparation. |

## 8.2 Experiment A: incremental phase-gradient information

Use a public multielectrode or ECoG dataset with a task that independently defines stimulus identity, remembered location, motion direction, or choice. Fit the nested models in Section 6.4. The primary test is the held-out difference between the full model and the strongest baseline. A secondary analysis asks whether errors are concentrated when phase gradients are weak, unstable, or directed away from the hypothesized receiver.

The experiment supports P1 only if the increment survives subject-level replication, estimator sensitivity, phase-randomized surrogates, and controls for evoked waveform and movement. It does not by itself support causation or consciousness.

## 8.3 Experiment B: phase-specific causal routing

Identify two upstream populations converging on a downstream population. Deliver closed-loop stimulation with phase condition randomized and with frequency, pulse count, total charge, and input energy fixed **before** treatment. Estimate preferred open and closed offsets in an independent receiver-phase localizer. Define a transfer endpoint  $Y$  and response window before the experiment. The total causal estimand is

$$\mathbb{E}[Y \mid do(\delta = \delta_{\text{open}})] - \mathbb{E}[Y \mid do(\delta = \delta_{\text{closed}})]. \quad (38)$$

Rate, power, and local evoked response may be post-treatment mediators, so they are measured and modeled rather than post hoc matched away. A controlled direct timing effect requires an independently calibrated factorial intervention. The design must state consistency, interference, common-input, and stimulation-specificity assumptions; include artifact and sham controls; and use a crossover or rescue condition. The decisive interaction reverses the preferred offset after the receiver’s inhibitory or integration state is independently altered, while producing the predicted downstream transfer and latent-state change. A generic phase effect supports communication-through-coherence, not the SAN composite.

## 8.4 Experiment C: wave-intersection and latent update

In a task with separable prior and sensory evidence, estimate feedforward and feedback propagation on individual trials. Before examining reports, predict the space-time locus at which the waves intersect. Define

a nonlinear interaction residual as the joint response minus the sum of responses to isolated feedforward and feedback conditions. Perturb the two sources factorially and test whether the preregistered residual at the predicted locus / time precedes the latent percept or decision change. Any PWD used here remains indexed to specified edges; a spatial locus does not have one undefined “PWD.”

Alternative explanations include a local evoked potential, a common pacemaker, volume conduction, linear superposition, and motor preparation. The interaction analysis must outperform those models and generalize across stimuli. A simple temporal correlation between a wave and report is insufficient.

## 8.5 Experiment D: literal reconstruction test

Choose a latent variable with independently known geometry, such as remembered two-dimensional location or body pose. Define several array projections and train the forward / measurement operators inside a development set. Before intervention, derive from each operator’s range and nullspace the specific error direction expected when that biological projection is perturbed.

Computational feature deletion and neural intervention are reported separately. Literal NAPOT requires the biological perturbation to produce the prederived operator-specific downstream error, not merely graceful degradation, while the full phase-structured model outperforms the equal-input  $R_{raw}$  baseline. If arbitrary decoders reconstruct the target equally well from nonspatial features, or if neural perturbation does not follow the prederived error geometry, tomography is unnecessary.

## 8.6 Experiment E: tonic / phasic Neural Rendering network

Construct a recurrent network in which each unit or population has a sparse structural state, optional object / body / task reference-frame state, a declared tonic operator, phasic / high-phasic event rule, amplitude, phase, event-interval CV, a receiver window, and delayed coupling. The primary ablation ladder is: structural sparse state only; structural plus reference frame; tonic state only; phasic residual only; and the combined Neural Rendering model. Also compare rate-only, gated recurrent, attention-based, state-space, generic oscillator, and complex-valued baselines. Preregister one primary task and metric; other tasks are exploratory:

- multimodal binding with distractors;
- rapid context switches without weight updates;
- occluded-object or partial-view reconstruction;
- embodied pose reconstruction from overlapping Gaussian-basis “dendritic splats”;
- discrimination of genuine frequency-state transitions from nonsinusoidal harmonics;
- prospective motion extrapolation; and
- recovery after node or edge perturbation.

Match trainable parameters, state dimension, temporal precision, input bandwidth, optimizer budget, data, wall-clock or compute budget, confirmatory seeds, and a declared energy proxy. Ablate sparse structure, reference-frame conditioning, phase, frequency residual, amplitude residual, waveform or release variable, tonic operator, phasic threshold, CV, gradient, delay, and recurrence independently. A new category can be allocated only by the locked novelty rule in Section 3.2. A computational advantage would establish engineering utility under the tested conditions, not biological identity or sentience.

## 8.7 Experiment F: self-related latent state

Use a task that independently varies world evidence and a self-related variable, such as agency under perturbed action feedback, self / other attribution, interoceptive prediction, or metacognitive sensitivity.

Predefine a joint latent state containing both classes and test cross-condition generalization. Then perturb a projection predicted to constrain the self-related dimension.

Evidence for P9 requires an operator-specific change in self-related report or policy after perceptual, attentional, mnemonic, decisional, confidence, and motor variables are controlled. Success would support a self-model mechanism, not phenomenal self-awareness.

## 8.8 Experiment G: embodied vision and GCS

First localize body-part representations using actual movement, actual touch, and viewed body parts. Then test held-out transfer to imagined movement, imagined touch, one’s own body, another person’s body, purely verbal body-part thought, and matched non-body imagery. Use high-resolution fMRI for spatial localization and MEG, ECoG, or laminar recording where available for timing. Record eye movement, surface EMG, respiration, and autonomic state. Train every spatial map, phase offset, and decoder inside the development data.

The observational endpoint is not activation of a “body area,” and body-part decoding alone cannot satisfy P10. The primary confirmatory tuple is frozen as: active right-index-finger movement in a closed-loop virtual-hand preparation; transfer from movement / touch localizers to held-out visual delay trials; trialwise perceived finger / hand-position error as the continuous body-state latent; and cross-validated  $\Delta R^2$  from edge-conditioned PWD / wave features beyond magnitude, rate, power, coherence, recurrent history, and the equal-information raw graph / sequence baseline. Ownership, agency, confidence, other body parts, and other imagery conditions are secondary until a separate preregistration promotes one of them. Failure to transfer beyond task, language, attention, or motor preparation counts against that scoped claim.

The confirmatory virtual-hand task factorially varies visual delay and active versus passive movement while holding appearance constant; rotation, gain, body-like versus object-like appearance, and synchronous versus asynchronous touch remain secondary manipulations. Measure perceived position as primary and correction behavior, ownership, agency, and confidence separately. Before outcome analysis, derive the predicted motor-command  $\rightarrow$  efference-copy / forward- model prediction  $\rightarrow$  sensory comparison  $\rightarrow$  embodied-state-update ordering. A delay  $\Delta t$  must shift a carrier at frequency  $f$  by

$$\Delta\phi(f) = 2\pi f\Delta t \pmod{2\pi}, \quad (39)$$

with circular ambiguity, minimum amplitude / phase-quality thresholds, and an independently measured delay. Include matched open-loop visual-feature and movement-state controls.

The strongest GCS test uses a laminar probe or another independently validated layer-resolved method in the declared granular target while perturbing motor-related feedback. Conventional surface ECoG and MEG may supply timing but not layer identity. It compares the provisional 2025 table [66] directly with the literature-backed primate-visual feedforward-theta / gamma and feedback-alpha / beta comparator [9,57]. A phase-specific causal test must match average stimulation energy, include alpha / beta, random-phase, sham, peripheral-sensation, 50/60-Hz and stimulation-artifact, and report-timing controls, and include a predeclared phase-shifted rescue. Support requires a pre-report change in perceived-position error, the predicted delay-dependent offset shift, and rescue. A generic gamma-power, broadband high-gamma / spiking, or arousal effect does not pass. Success would support a bounded embodied-state update, not a literal dendritic screen, phenomenal consciousness, or the claim that one cortical map is the self.

## 8.9 Experiment H: tonic / phasic scale-up and field perturbation

Use a preparation in which one dendritic branch, neuron, or tightly bounded cell group can be perturbed while intracellular or juxtacellular activity and a dense extracellular array are recorded simultaneously. Predefine the structural pattern, tonic operator, phasic / high-phasic event rule, current-source or forward model, reference, band, phase estimator, spatial region set, distance bins, time window, and minimum resolvable effect. Estimate the counterfactual  $\delta\Phi_E$  footprint relative to sham and matched-output controls.

Track the event across three bounded levels: branch / cell, local assembly or column, and one declared distributed network. At each level test whether the structural match is decodable from the output, whether the phasic differential attenuates, disperses, amplifies, or becomes part of a maintained tonic state, and whether the next receiver changes in the predicted direction. A verbal claim that the signal “scales up” does not pass without a predeclared transfer function and uncertainty.

Separate direct extracellular-field contribution, conventional synaptic / axonal propagation, shared drive, and volume-conducted or source-leaked observation. Eligible controls include synaptic blockade where physiologically and ethically appropriate, spatially matched current injection, source-reversal tests, conductivity-aware forward simulation, distance-dependent attenuation, and receiver-state manipulation. Support for P11 requires a reproducible footprint plus downstream transfer or latent / action consequence. A detectable local potential alone supports an observation model, not Neural Rendering; distributed correlation alone does not identify ephaptic causation.

## 8.10 Falsification policy

SAN should be narrowed rather than protected by post hoc reinterpretation. The following outcomes would be material failures:

1. the preregistered number of independent datasets within a frozen scope place the PWD / gradient increment inside the equivalence margin;
2. apparent effects disappear under phase-preserving power controls or leakage-safe validation;
3. causal relative-phase perturbations fail across receivers for which the model made quantitative predictions;
4. the receiver-window interaction is absent;
5. no independently measured latent variable is identifiable from multiple projections beyond generic recurrent baselines; or
6. the formal variables fail the preregistered reliability threshold in the bounded recording modalities and preparation named by the program.

A failure of literal tomography could leave a weaker oscillatory-routing hypothesis intact. A failure of phase-specific routing would remove the central NAPOT delta for the tested scope; a failure of the receiver-state interaction would reject that modulation claim; and a failure of rank / observability or operator-specific intervention would reject literal tomography without automatically rejecting all oscillatory routing.

## 9. Relationship to adjacent theories and exact prior-art boundaries

The mechanism, formal variables, and proposed tests are now in view. The terminology can therefore be compared without making another theory carry SAN’s explanation. The relevant question is not whether SAN uses phase, coherence, or propagating activity. It is which operation each term denotes, which variables it measures, and which causal consequence it predicts.



## 9.1 Predictive coding

Predictive coding predates SAN and provides an explicit prior / prediction versus sensory-error architecture [4,21,22]. SAN does not replace that architecture. It proposes candidate timing and spatial variables through which some recurrent predictions and errors may be routed. A predictive coding model without PWD or gradients is therefore a baseline, not an opponent that can be dismissed for different vocabulary.

## 9.2 Synchrony, binding, and communication through coherence

Synchrony-based integration, temporal correlation, and communication through coherence establish that aligned or reliably related activity can alter effective communication [5-10]. Coherence asks how consistently signals covary in phase or frequency. The PWD asks a later causal question: what does the measured arrival difference do at a particular stateful receiver? A coherent relation can maintain an expected route without introducing a new differential, while a consequential PWD can appear as increased alignment, a stable non-zero offset, structured desynchronization, inhibition, or a waveform and release change. The discriminating test is whether edge-conditioned, delay-constrained PWD variables and receiver windows explain or control a latent-state update beyond scalar synchrony.

## 9.3 Phase-of-firing and theta-gamma codes

Phase-of-firing and theta-gamma coding show that relative timing can carry information and ordered items [11,12]. They answer where a spike or event falls within a cycle and whether that position carries information. PWD adds the receiver, route delay, expected relation, local state, and downstream consequence. SAN's phrase "coincidence is a bit" describes the point at which continuous receiver-specific integration produces a discrete effective event that networks can compose across delayed projections. "Bit" here names a binary event indicator; its entropy and mutual information are measured rather than assigned in advance. SAN's priority claim concerns this receiver-relative consequential operator and its place in the joined reconstruction architecture, not the general fact that neural timing carries information.

The First Draft also cited Romain Brette's 2015 *Philosophy of the Spike: Rate-Based vs. Spike-Based Theories of the Brain* as a bridge away from rate-only description [1]. The legitimate bridge is preserved: spike timing, waveform, and event relations can carry distinctions hidden by a mean-rate summary. The stronger First Draft annotation that this source establishes timing relative to oscillatory phase is not imported automatically from the title; phase-relative coding is supported by the separate phase-of-firing literature [11,12]. PWD then asks the further receiver-, expectation-, route-, state-, and consequence-specific question defined above.

## 9.4 Traveling-wave and connectome models

Traveling waves, rotating waves, connectome harmonics, and topology-constrained propagation all have independent histories [14-17,31]. Alamia and VanRullen showed before mature NAPOT that communication delays in a hierarchical predictive-coding model can generate forward and backward alpha traveling waves [31]. Benigno and colleagues demonstrated in a complex-valued recurrent network that topography, recurrence, and distance-dependent delays can generate waves and closed-loop visual prediction [32]. Mohanta and colleagues reported human intracranial relationships among traveling waves, predictions, errors, and model updating in a non-peer-reviewed preprint [33]. These adjacent results make "waves support prediction or model update" non-distinctive. Koller and colleagues' connectome-instrength model [15] is a particularly important constraint: wave direction need not be attributed to a SAN-specific law when network topology and delay explain it. NAPOT instead predicts what structured wave flow does at a calibrated receiver and whether it makes a causal, identifiable contribution to a multi-projection reconstruction.

A traveling wave therefore describes organized change across space, while a PWD describes a consequential difference at an addressed receiver. A PWD may initiate, ride on, intersect, redirect, or terminate a traveling pattern; a traveling pattern may also pass through a recorded field without producing the predicted receiver-specific operation. The terms can apply to the same event, but they do not denote the same explanatory object.

| Term           | Primary scientific question                                                            | Additional question introduced by PWD                                                                                         |
|----------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Phase coding   | Does position within an oscillatory cycle carry information?                           | Does the delayed arrival differ from the receiver's expected relation, and does that difference change the receiver or route? |
| Coherence      | How consistently are signals related in phase or frequency?                            | Is the relation consequential at an identified stateful receiver beyond scalar coherence and common-input controls?           |
| Traveling wave | How does an organized dynamical pattern propagate through space?                       | What receiver-relative transformation occurs as the pattern arrives, intersects another pattern, or changes direction?        |
| PWD            | Which receiver-relative difference changes access, routing, reconstruction, or action? | The construct itself requires the addressed receiver, expected relation, differential, and downstream consequence.            |

## 9.5 Recurrent processing, global workspace, and integrated information

Recurrent-processing, global-workspace, and integrated-information theories address different conditions for conscious access or experience [28-30]. SAN is not established as a replacement for any of them. Its current testable target is a mechanism for distributed routing and model update. Phase-structured activity could in principle implement a workspace-like broadcast, recurrent local processing, or other dynamical organization, but none of those resemblances establishes IIT's cause-effect structure or  $\Phi$ .

Massimini and colleagues found that TMS-evoked activity propagated beyond the stimulated site during quiet wakefulness but remained local during NREM sleep [18]. Ferrarelli and colleagues reported reduced effective connectivity during midazolam-induced loss of consciousness [19]. These exact results do not establish PCI, regional percentage scores, or NAPOT's PWD and reconstruction claims.

## 9.6 Cortical columns and predictive object models

Hawkins and Blakeslee's 2004 memory-prediction framework is earlier prior art for the general prediction atom [24]. Hawkins and Ahmad's active-dendrite sequence model and Ahmad and Hawkins's sparse-pattern analysis are earlier prior art for their specific branch-local and sparse-pattern mechanisms [69,70]. Hawkins, Ahmad, and Cui's later 2017 paper proposed an exact allocentric / grid-cell-like coordinate mechanism in cortical columns [23], and Hawkins and colleagues subsequently proposed multiple distributed object models and a modified hierarchy [71]. The current Thousand Brains Project implements sensorimotor, reference-frame, modular, consensus, and compositional principles while distinguishing that implementation from HTM [72]. Those exact atoms and implementations remain their contributions; none is treated as an author-wide floor over every spatial, predictive, dendritic, or distributed theory.

SAN's independent chronology must therefore remain visible at the same resolution. Its 2011 source contains the predictive-self and whole-network-observer atoms [35]; its 2012 sources contain the recursive feedback and predictive-projector architecture [37,38,65]; and its April 2017 sources contain distributed features plus spatial / temporal context, motion prediction, coincidence events, difference transmission, and a time-varying

3D/4D internal matrix [39-42]. Under an atom-by-atom and causal-architecture comparison, those dates preserve Micah priority on that bounded SAN composite; the later exact grid-cell-like coordinate implementation remains Hawkins and colleagues' distinct delta. The mature 2022 SAN increment is the further conjunction of a persistent sparse or reference-frame-conditioned state with a fast receiver-relative phase / frequency / amplitude/ transmission residual and recurrent reconstruction [2,43]. Its biological truth remains unsupported until the locked comparator and perturbation tests pass, but an uncompleted test does not erase the dated architecture.

## 9.7 Cellular agency and bioelectric organization

Cellular learning and bioelectric pattern regulation have long histories, including Michael Levin's work on reprogrammable bioelectric circuits [26]. SAN's COT extension should therefore be tested as a specific operation, not as ownership of cellular agency. The relevant 2022 SAN sequence [43] links receptor / configuration state, oscillatory computation, structural response, and selection; its truth remains independent of chronology.

## 9.8 Constructive memory

Memory as reconstruction rather than literal retrieval has a history at least as old as Bartlett [25], followed by extensive constructive-memory and predictive-processing work. SAN can propose a particular recurrent phase implementation without claiming the general reconstructive insight. Static synaptic structure and dynamic context may jointly generate a present state, but the stronger claim that memory contains no storage requires separate evidence and is not made here.

## 9.9 Body maps, embodiment, and active vision

Somatotopy, corollary discharge, motor imagery, visual body-selective cortex, multisensory body ownership, and recurrent vision are named prior components [50-55,57]. They are not author-level floors that terminate comparison, and SAN does not claim to originate them. The bounded GCS delta is the proposed conjunction in which dendritic and population-level partial fields are re-expressed by edge-conditioned phase structure, recurrent visual-motor-reafferent exchange updates a specified body-related latent, and the exact layer / frequency ordering survives an experiment designed to distinguish it from feedforward-gamma / feedback-alpha-beta organization.

Hedger and colleagues first posted a preprint on 2024-10-22 and later reported connective-field estimates of repeated somatotopic gradients in dorsolateral visual cortex, aligned with retinotopic position and, more ventrally, visually derived body-part preferences [56,64]. Orlov and colleagues had already reported body-part topography with motor influence in occipitotemporal cortex in 2010 [53], so the bare body-map atom is convergent with named prior. That component-level result is not an author-wide floor and does not settle the causal-architecture comparison. Blumberg's public 2024-01-04 source predates the Hedger preprint on the body-map / directional-self / distributed-rendering combination [58], while the earlier 2011-2022 SAN genealogy supplies the predictive-self, recursive visual-motor feedback, spatial rendering, coincidence, and NAPOT / PWD ancestors. On that bounded SAN delta the three-lens classification is **Micah earlier**. Hedger et al.'s dual-source connective-field method, tiled-gradient result, and aligned-map evidence remain their target-specific empirical and methodological delta.

The study does not establish direct synaptic connectivity, active touch, ownership, motor feedback, GCS, dendritic splats, PWD, causal NAPOT reconstruction, self-awareness, or consciousness. It therefore neither validates the complete SAN composite nor defeats Blumberg's priority on the earlier visual-motor-reafferent and phase-reconstruction architecture.

## 10. Developmental provenance, 2011-2022

### 10.1 Why provenance is included

The dated genealogy explains how the present composite developed and supplies citable source anchors. It is not a substitute for experiments, and it does not imply that every component was novel when stated. The appropriate unit is a claim atom or bounded causal composite, not an author’s entire body of work.

A terminology date and a causal-architecture date are different claims. The later appearance of the names NAPOT, PWD, or COT does not erase an earlier functional atom whose input, transformation, output, and feedback structure is already present. Conversely, an earlier functional ancestor does not license backdating a later term or formalism. Earlier work by another author on one component is credited at that component and is not converted into an author-wide floor over a different atom or later bounded composite.

Evidence grades used here are:

- **A:** exact public URL, DOI, immutable commit, or archive with exact text and date;
- **A-minus:** exact public identity and matching source text / date, with an additional platform capture still desirable;
- **B:** exact content and an owner / internal date, but no preserved contemporaneous public timestamp sufficient for a strict public-priority claim; and
- **C:** retrospective or title-level routing evidence only.

### 10.2 Bounded genealogy

| Date                                | Dated SAN-side content                                                                                                | Relation to the present model                                 | Grade / boundary                                                                                                                                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2011-08-24/28</b>                | Awareness as expectation; the self as a prediction pattern; “the whole neocortex” rather than a hidden observer [35]. | Predictive self-model and distributed observer ancestor.      | <b>B.</b> The exact content and owner-recorded / upload dates are preserved and the atom is present; a platform export or archived public capture would harden the public-date proof. This evidence limitation is not a concession that the atom was absent. |
| <b>2012-07-31</b>                   | EEG peaks aligned to moments in video [36].                                                                           | Temporal-coincidence measurement ancestor.                    | <b>A-minus.</b> Not yet PWD or NAPOT.                                                                                                                                                                                                                        |
| <b>2012-08-01 local / 08-02 UTC</b> | EEG-driven returned light and sound make vision, hearing, and feeling co-activate in a “two-way conversation” [65].   | Recursive multimodal-feedback ancestor.                       | <b>A-minus.</b> Exact live YouTube metadata supplies the local / UTC boundary; separate source from the July 31 video; not yet GCS, PWD, or NAPOT.                                                                                                           |
| <b>2012-08-23</b>                   | Mirror→vision→neural / motor change→changed image→vision; “fire out of sync will not link” [37].                      | Explicit closed self-feedback chain and timing-gated linkage. | <b>A-minus.</b> Functional architecture only.                                                                                                                                                                                                                |
| <b>2012-08-23 local / 08-24 UTC</b> | The mind as a projector producing a prediction of the world, with substantial internally supplied detail [38].        | Predictive internal rendering constrained by external input.  | <b>A.</b> Exact public recording and platform timestamp; “projector” remains a functional metaphor.                                                                                                                                                          |
| <b>2012-08-23 to 08-26</b>          | Local brain groups, hemispheres, brains, and social networks discussed as connected scales [37,49].                   | Multiscale distributed-network ancestor.                      | <b>A-minus.</b> Some exact fractal / medium wording belongs to dialogue partner Ben Thomas and must be credited to him.                                                                                                                                      |

| Date                           | Dated SAN-side content                                                                                                                                              | Relation to the present model                          | Grade / boundary                                                                                                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2017-04-11</b>              | Spatial computation unifies lines / features into an object concept and predicts motion; a time-varying 3D/4D ionic / electromagnetic matrix is proposed [39].      | Spatial-predictive object / world representation.      | <b>A.</b> Public dated episode; not yet literal tomography.                                                                                                                                              |
| <b>2017-04-11 to 04-22</b>     | Cells as coincidence detectors; “coincidence is a bit” [39,41].                                                                                                     | Receiver-defined coincidence-event ancestor.           | <b>A.</b> Peter Tse [47] and earlier cortical coincidence-detection work [48] supply named components; the bounded delta is the later information-theoretic reading, not ownership of the detector atom. |
| <b>2017-04-13</b>              | Dendrites perform substantial local computation inside feedback circuits [40].                                                                                      | EDS and nonlinear receiver ancestor.                   | <b>A.</b> Dendritic computation itself is earlier prior art.                                                                                                                                             |
| <b>2017-04 to 12</b>           | Distributed representations converge through recurrent cortical / thalamic passes; feedback-connected field interactions are discussed [39-41].                     | Distributed recurrent rendering ancestor.              | <b>A.</b> Exact NAPOT and ephaptic formulations are later.                                                                                                                                               |
| <b>2017-06-30</b>              | Excited and inhibited regions jointly define thought and self-boundaries; inhibition carries contextual information [42].                                           | Signed constraint and inhibition-as-information claim. | <b>A.</b> Does not imply every silence is a message.                                                                                                                                                     |
| <b>2017-04-11 → 2022-08-23</b> | Calculate the difference between the present neural pattern and a desired pattern and transmit that difference [39].                                                | Dated causal atom for a content-carrying differential. | <b>A.</b> The difference-and-transmission operation is a 2017 priority anchor in its own right; only the exact phase-specific formalism and phase-wave-differential name are anchored in 2022 [43].      |
| <b>2022-06 to 08</b>           | Public repository records formalize oscillatory rendering / tomography, NAPOT / COT precursors, phase-wave language, and the volumetric-rendering framework [2,43]. | Mature terminology and composite.                      | <b>A.</b> Claim-specific commits control dates; repository creation alone cannot date every later sentence.                                                                                              |
| <b>2022-10-24/25</b>           | Extracellular ionic gradients, local field potentials, ephaptic effects, and synchronization are placed in one causal implementation [43].                          | Explicit field-coupling extension.                     | <b>A.</b> Claim-specific commits 34e8c8d... (2022-10-24) and ad9d23a5... (2022-10-25); SAN does not claim origination of ephaptic coupling.                                                              |

The chronology supports a 2011→2012→2017→2022 developmental spine: predictive self / world model; recursive feedback and projection; spatial, coincident, distributed computation; then mature phase-tomographic vocabulary. It does not license backdating “NAPOT,” “PWD,” or “COT” to 2011.

### 10.3 The hybrid TMS node

The August 2012 TMS passage is a mixed explanation, not an all-or-nothing authorship unit [37,49]. Rothman’s August 18 interview with Tononi is the best-supported near-exact source match for the practical-number and cross-being language [34], but the promised attachment itself has not been recovered. The source reconstruction is:

Tononi-Massimini TMS / EEG program  
→ Micah Blumberg’s bounded measurement / deployment extrapolations. (40)

| Atom in the 2012 discussion                                                       | Attribution in this paper                                                                         |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| TMS plus EEG / electrodes and state-dependent local versus widespread propagation | Earlier Tononi-Massimini research [18].                                                           |
| Practical numerical measurement and broad comparison across beings                | Present in Rothman’s 2012 Tononi interview [34]; not claimed as SAN’s invention.                  |
| Linear 0-100 percentage normalization                                             | Micah Blumberg’s dated historical speculative extrapolation; not a validated measurement advance. |
| Separate values for different brain regions                                       | Micah Blumberg’s dated historical speculative extrapolation; not a validated regional score.      |
| Elephant, friend / family, and shopping-mall deployment examples                  | Micah Blumberg’s dated historical speculative application extrapolations.                         |
| “Turn off / reboot the brain”                                                     | Inaccurate paraphrase of perturbation, removed as an innovation claim.                            |

The later perturbational complexity index [20] is not backdated to SAN. This attribution is a model for the whole paper: name an exact earlier atom, preserve a real later delta, and never turn an earlier component into an author-level shield or a later composite into ownership of the component.

### 10.4 Post-2022 embodiment refinement

The early genealogy closes in 2022 because that is when the mature NAPOT / PWD / COT vocabulary became public. Later records refine rather than backdate it. A repository commit public on 2024-01-04 preserves discussions of cortical body-map magnification, the self as a body-attached directional construct for action, and distributed synaptic configurations as “pixels” in an internal rendering [58]. A Micah-labeled repository dialogue public on 2024-10-21 already describes gamma “in the middle” balancing incoming sensory signals against thoughts / decisions, calls this a “consideration sandwich,” and connects it to proprioceptive feedback [67]. A repository chapter public on 2025-01-21 then calls Layer-2/3 gamma the middle layer merging top-down beta, incoming sensory alpha / beta disruptions, and proprioceptive feedback [68]. It is useful genealogy but not needed for the earlier priority anchor. Its compressed statement that Layer-5 PV interneurons carry proprioceptive feedback is recovered as two joined operations rather than treated as a discarded idea: established ascending pathways carry peripheral afference into sensorimotor circuits, and locally recruited Layer-5 PV+ interneurons can gate and time the excitatory subnetwork through which that evidence affects motor preparation and correction [97-100].

The exact capitalized **Gamma Consideration Sandwich** name and expanded sensory / top-down/ proprioceptive section are public in repository draft1.md by 2025-02-21 local time / 2025-02-22 UTC [59].

A February-May working-diagram lineage adds a more specific layer / frequency reconciliation [66]. The two primary Excalidraw files have been recovered, hashed, and privately verified. Their working-file

metadata does not establish a public publication date, so the diagram-specific date remains provisional; the files are not needed to date the public GCS name or broad composite.

The 2026 embodiment specification is the first source in this record to fuse those components into one explicit, falsifiable composite: a dendritic receive-transform-re-express field; a distributed somatotopic / body-related latent; and a GCS visual-motor-reafferent update with a direct laminar alternative and frequency-harmonic falsifier. That exact fusion is not backdated to the earlier ancestors.

## **11. Limitations**

### **11.1 No new empirical result**

This manuscript defines a test program but reports no completed NAPOT reanalysis, prospective experiment, causal perturbation, or engineering benchmark. The equations generate hypotheses; they are not fitted results. Language such as “validated,” “confirmed,” or “proved” would be premature.

### **11.2 The formalism is phenomenological**

The delayed phase model is a tractable reduction, not a derivation from channel kinetics, dendritic morphology, spiking microcircuits, or electromagnetic field equations. Different phase estimators and neural mass models may imply different variables. A successful fit would establish utility at the modeled scale, not identity with every underlying biological process.

### **11.3 Phase is estimator-dependent**

Instantaneous phase is unstable at low amplitude and can be distorted by filtering, waveform asymmetry, edge effects, reference choice, and volume conduction. Spatial gradients can arise from evoked latency fields or common input. The controls in Section 7 are therefore constitutive, not optional.

### **11.4 Scale bridging remains open**

Cellular coincidence, local-field phase, ECoG waves, and whole-brain state are not interchangeable. The cross-scale order parameter is a proposed summary and may discard critical information. COT and BOT remain hypotheses until experiments show a causal scale bridge rather than a recurring visual analogy.

### **11.5 Tomography may remain a metaphor**

Neural activity can be distributed and reconstructive without implementing tomography. If no measurement operators or nonredundant projections can be identified, the theory should be renamed oscillatory recurrent integration. The name is subordinate to the test.

### **11.6 Content, access, report, and phenomenology differ**

Decoding stimulus content does not prove conscious access. Predicting report can be confounded by attention, memory, decision, and motor preparation. Reportable integration does not establish the qualitative character of experience. This version therefore makes no claim that NAPOT is necessary or sufficient for phenomenal consciousness. That evidential boundary does not make the paper agnostic about mechanism: Sections 3.5 and 4.10 advance a positive candidate account in which a maintained tonic context, content-specific receiver-relative PWD, distributed detection and re-expression, recurrent sustainment, and embodied consequence jointly constitute the proposed neural-rendering operator. The theory can therefore be wrong

in a specific way if content generalization, selective perturbation, and rescue fail while matched access, decision, and report models succeed.

### **11.7 Agency is graded and operational**

The minimal agent definition can be useful for comparing scales, but it may still classify systems too broadly. Protein conformational change alone does not establish learning, goals, or selfhood. Every use of *agent* should state the measured state, transformation, action, feedback, and adaptation.

### **11.8 Artificial systems and sentience**

A phase-coded network might improve routing or partial-view reconstruction without having any experience. Behavioral resemblance and architecture are not sufficient sentience tests. This paper does not provide a moral-status criterion for artificial systems.

### **11.9 Provenance and scientific merit are orthogonal**

A dated record can show when a sentence or bounded composite existed. It cannot show that the claim is correct, that every prior source was searched, or that a later author accessed the record. The scientific theory must succeed under tests that would be equally decisive if the genealogy were unknown.

### **11.10 The dendritic screen and GCS may fail**

The dendritic grid, Gaussian-splat parameterization, and GCS layer / frequency ordering are models, not observations. A dendritic field may prove useful without being a visual rendering; body-related decoding may arise from generic imagery or action preparation; and apparent 40/60/80-Hz codes may be harmonics of a nonsinusoidal 20-Hz rhythm. If matched raw models, waveform controls, and direct laminar tests explain the data without the proposed variables, these embodiment claims must be narrowed or rejected even if the broader recurrent self-model remains viable.

### **11.11 Portfolio and philosophical-translation boundary**

Separate SAN companion manuscripts develop stronger proposals about phase-wave-differential calculus, choice, altered states, machine-consciousness measurement, and the tonic canvas as a translation target for philosophical or contemplative descriptions. Their existence does not add evidence to this flagship paper. A category-theoretic diagram, process-calculus trace, or input-transform-output correspondence can prove a bounded structure-preserving translation only after the objects, arrows, observables, and failure conditions have been declared. It cannot by itself prove that a religious, psychological, phenomenological, and neural account are identical, that they share ontology, or that the neural mechanism is true. This manuscript therefore adopts only the operational tonic / phasic mechanism and leaves whole-tradition equivalence outside its claim set.

The new structural-plus-dynamic extension can also fail independently. Sparse / reference-frame models may explain rapid recognition without dynamic residuals; phase / frequency / amplitude features may add no held-out information; presynaptic waveform may affect release without carrying the represented category; and a category-allocation rule may overfit noise. Any of those outcomes narrows the conjunction rather than being redescribed as confirmation.



## 12. Conclusion

The strongest form of Self-Aware Networks is not the claim that the brain oscillates, that synchrony matters, or that distributed activity can represent content. Those are broad findings and research programs with substantial prior histories. The testable SAN proposal is a composite: receiver state defines effective timing windows; edge-conditioned phase-wave differentials and spatial gradients help route information when direction is supplied independently; structural connectivity and delay constrain propagation; learned synaptic / dendritic presets define pattern selectivity; tonic states provide role-relative predictive context; phasic and high-phasic differentials provide candidate updates; and recurrent partial projections jointly render a distributed latent self / world model. The same event may be rejected, attenuated, amplified, maintained, or incorporated into a changed tonic state. Ephaptic coupling is an eligible local term, not an assumed global broadcast. Embodiment supplies one discriminating case: the theory must explain how predicted movement and visual, proprioceptive, tactile, and internal refference update a distributed body state without a central screen or observer.

This version makes that composite measurable and vulnerable to failure. It defines nested baselines, identifiability conditions, causal perturbations, and outcomes that would narrow or reject the theory. It also places the 2011-2022 development record in its proper role: a chronology for accurate attribution, not a substitute for evidence. The immediate next step is not another analogy. It is a leakage-safe public-data reanalysis followed, if warranted, by a preregistered phase-specific causal experiment.

The framework’s vocabulary is deliberately typed. Classical neural coherence is not quantum coherence; multiscale organization is not scale invariance without a declared scaling test; locality requires a specified adjacency, field, delay, or kernel; and receiver relativity concerns which physical differences a system can discriminate and use, not whether those differences exist. These boundaries make the proposal narrower, but they also make a positive or negative result interpretable.

Publication of this theory-and-methods framework would not constitute empirical validation. The 2011-2022 record preserves the dates of distinct functional atoms and their later formal integration; it does not allow an earlier component attributed to another author to become an author-wide floor over the SAN composite.

## Data and code availability

No new data were generated and no empirical analysis is reported in this theory-and-methods paper. The development package contains deterministic synthetic implementation tests for analytic-signal extraction, edge-conditioned PWD summaries, graph-edge phase gradients, grouped splitting, and equal-input comparison behavior. These are software checks, not neural evidence. Under the stronger raw equal-input comparator, the recorded synthetic contrast is unfavorable ( $M5 - M6 = -0.0114485499$ ), which demonstrates that the test can fail and supplies no scientific support for SAN. No neural dataset has been selected. A frozen dataset manifest, complete nested models, train-only transforms, nulls, uncertainty, an all-runs ledger, and independent replay are required before a future empirical claim, not before publication of the present prospective framework. Any future analysis must use data under licenses compatible with its stated release conditions. The six included figures are explanatory schematics with governed hashes; they are not empirical measurements and do not upgrade any hypothesis to an observed result.

## Ethics statement

This theory-and-methods report describes no new experiment involving human participants or animals. Any prospective human or animal experiment proposed here requires independent ethical, institutional, safety,

and consent review before execution.

## Competing interests

The author is the originator of the Self-Aware Networks framework and maintains the associated repository and comparison corpus. This intellectual interest is disclosed so that theory testing, model selection, and provenance adjudication can be reviewed independently.

## Author contributions

Micah Blumberg: conceptualization, theory development, original historical source material, methodology, visualization concepts, and manuscript authorship. This is a single-author manuscript.

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## Appendix A. Claim ladder

| Level | Claim                                                                                                    | Current status                                                |
|-------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| A1    | Neural receivers perform state-dependent nonlinear temporal integration.                                 | Established component; not SAN-specific.                      |
| A2    | Relative phase can carry information and regulate effective communication.                               | Established component; not SAN-specific.                      |
| A3    | Spatially organized traveling waves occur and are shaped by anatomy / delay.                             | Established component; scope depends on preparation and task. |
| A4    | Edge-conditioned PWD and wave geometry add content-specific routing information beyond matched controls. | Central SAN prediction; untested in this package.             |
| A5    | Multiple recurrent phase-structured projections identify a latent self / world state.                    | NAPOT reconstruction hypothesis; untested.                    |
| A6    | The operation is causally required for a defined report or action.                                       | Open causal hypothesis.                                       |
| A7    | The operation is necessary or sufficient for phenomenal consciousness.                                   | Not claimed.                                                  |

## Appendix B. Minimum reporting checklist for a future empirical SAN test

The unchecked items below are prospective requirements for an empirical claim. They are not missing results represented as part of this theory-and-methods publication.

- ☐ Exact dataset identifier, version, license, and inclusion / exclusion criteria
- ☐ Preregistered latent variable and primary endpoint
- ☐ Recording geometry and reference scheme
- ☐ Filter / phase-estimator specification and low-amplitude exclusion rule



- ☐ Delay, PWD, gradient, and wave-classifier definitions
- ☐ Nested split that prevents trial / session / subject leakage
- ☐ Rate, power, scalar coherence, connectivity, recurrent, and nuisance baselines
- ☐ Phase-randomized, spatially shuffled, time-shifted, and label-shuffled surrogates
- ☐ Capacity / complexity matching and ablations
- ☐ Subject / session effects, uncertainty, multiplicity control, and replication state
- ☐ All-runs manifest, environment, seeds, hashes, clean replay command, and result JSON
- ☐ Claim language bounded to the tested arrow in the causal chain

## Appendix C. Comparison-corpus provenance boundary

The associated Comparison Wiki is a living provenance corpus whose totals change as duplicate containers are separated from substantive targets and as audits are re-run. Its exact dated validator snapshot belongs in release metadata rather than being used as a scientific argument. Main-text theory comparison must use dated operational mechanisms, causal architecture, and discriminating predictions. A prior component is never promoted into an author-wide floor, and a later term date does not erase an earlier dated functional atom.

The corpus uses three co-equal research lenses - mechanism equivalence, causal-architecture equivalence, and adversarial source forensics - while preserving named precursors, partials, nonmatches, unresolved cases, and genuinely opposite conclusions. Surface vocabulary or additional machinery does not by itself establish an opposite result, but resemblance is never enough to manufacture equivalence. These methods and counts are not replications, likelihood ratios, or empirical confirmations of SAN.

## Appendix D. OCA First Draft semantic-conservation ledger

This ledger makes loss detectable. A successor to this paper fails the lossless-revision gate if a row disappears without an in-paper disposition. *Preserved* means the proposition remains active at an appropriate claim level. *Corrected* means the original proposal remains visible but a narrower form replaces it. *Historical* means the item remains part of the genealogy or metaphor set rather than an active empirical conclusion.

| ID     | First Draft construct or content branch                    | Draft 12 in-paper disposition                                                                         | Primary location |
|--------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------|
| OCA-01 | BOT: biological oscillatory integration across scales      | Preserved as the widest graded-agency hypothesis; does not assign consciousness to cells or tissues   | 4.6, 4.8         |
| OCA-02 | COT: cells as oscillatory information-processing agents    | Preserved with receptor, intracellular, transcriptomic, energetic, and morphology variables separated | 4.6, 4.7         |
| OCA-03 | NAPOT: arrays receive, transform, project, and reconstruct | Preserved and operationalized with latent targets, comparators, and falsifiers                        | 3.7, 4.8, 5, 6   |
| OCA-04 | Coincidences as bits, including BTSP and Hebbian learning  | Preserved as receiver-defined effective events; not a universal literal binary code                   | 3.6, 4.7, 4.8    |
| OCA-05 | Traveling waves as phase-wave differentials                | Preserved with PWD distinguished from a whole wave and made receiver-relative                         | 3.4, 4.3, 4.7    |

| ID     | First Draft construct or content branch                    | Draft 12 in-paper disposition                                                                                         | Primary location |
|--------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------|
| OCA-06 | NDCA                                                       | Preserved as a nonlinear continuous state-update family, not a completed universal derivation                         | 3.4, 6.6         |
| OCA-07 | Oscillatory solution to binding                            | Preserved as integration plus structured differentiation, not synchrony alone                                         | 4.2, 4.8, 4.11   |
| OCA-08 | Functional connectivity stitches regions into a global map | Preserved as a dynamic, testable routing and latent-state claim                                                       | 4.8, 5.4         |
| OCA-09 | Inhibitory interneurons create choice bifurcations         | Preserved as distributed route selection; universal anatomical claims bounded                                         | 4.10, 5.3        |
| OCA-10 | Global spectral frequency map and coordinate system        | Preserved as a relational spectral / phase-response-address hypothesis                                                | 4.8              |
| OCA-11 | Mental ink and tonic canvas                                | Preserved; tonic / phasic are functional roles, not fixed bands                                                       | 3.5, 4.10        |
| OCA-12 | Ephaptic coupling                                          | Preserved as a candidate contributor with synaptic / common-input controls                                            | 4.1, 4.8         |
| OCA-13 | Entification                                               | Preserved as compositional behavioral unity; not proof of phenomenal unity                                            | 4.6, 4.11        |
| OCA-14 | Consciousness through oscillatory dissipation              | Preserved as a testable relaxation / reconfiguration hypothesis; no thermodynamic proof claimed                       | 4.11             |
| OCA-15 | Neural rendering as a 3D volumetric television             | Preserved as a decoder-tested distributed reconstruction metaphor                                                     | 3.7, 4.8, 5.4    |
| OCA-16 | Phasic-tonic absorption of rendered patterns               | Preserved as background / context plus receiver-relative update                                                       | 3.5, 4.10        |
| OCA-17 | Latent-diffusion analogy for oscillation and LTP           | Preserved as an iterative-refinement analogy; identity with diffusion models is denied                                | 4.11             |
| OCA-18 | Multiscale fractal computation                             | Preserved as repeated receive-transform-project organization; mathematical fractal dimension not assumed              | 4.6              |
| OCA-19 | Deterministic wave-based consciousness                     | Corrected: microstate constrains distributions, but empirical stochasticity and consciousness sufficiency remain open | 4.7, 4.11        |
| OCA-20 | Sonic the Hedgehog loops                                   | Preserved with the operational translation “re-entrant thalamocortical refinement loop”                               | 4.9              |
| OCA-21 | Thalamocortical loop as central hub                        | Preserved and expanded into the stateful thalamic-heartbeat mechanism                                                 | 4.9              |

| ID     | First Draft construct or content branch                                               | Draft 12 in-paper disposition                                                                                                                                                  | Primary location |
|--------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| OCA-22 | Dendrites as vector embeddings and spikes as tokens                                   | Preserved with a formal many-to-one state-dependent map; lossless inversion denied                                                                                             | 4.7              |
| OCA-23 | Quasicrystal analogy                                                                  | Historical metaphor preserved; literal neural quasicrystallinity requires its own measurement                                                                                  | 4.8              |
| OCA-24 | Immuno-neural interplay                                                               | Preserved under BOT / graded biological agency; not used as evidence of NAPOT or consciousness                                                                                 | 4.6, 4.8         |
| OCA-25 | Agentic co-regulation across scales                                                   | Preserved as nested reciprocal control                                                                                                                                         | 4.6, 4.9         |
| OCA-26 | Addressable coefficient of variation                                                  | Preserved as a multivariate receiver-profile hypothesis; scalar uniqueness denied                                                                                              | 4.7              |
| OCA-27 | Gamma Consideration Sandwich                                                          | Preserved; the compressed Layer-5/PV carrier wording is recovered as ascending body-state input plus local PV-mediated gating, while the joined GCS operation remains testable | 3.9, 4.10, 8.8   |
| OCA-28 | Four trillion dimensions per millisecond                                              | Historical capacity metaphor preserved; effective dimension must be measured                                                                                                   | 4.8              |
| OCA-29 | Molecular bridge: channels, calcium, vesicles, KIBRA / PKM $\zeta$ , protein turnover | Preserved with the observed conditional-probability model separated from SAN's hidden-microstate determinism hypothesis                                                        | 4.7, 5.5         |
| OCA-30 | COT receptor learning and transcriptomic activation                                   | Preserved as hypotheses requiring cell-specific perturbation and state measurement                                                                                             | 4.7              |
| OCA-31 | ATP-based cellular reinforcement                                                      | Preserved as a later biochemical energy-and-selection hypothesis; ATP is not treated as a literal machine-learning reward token                                                | 4.7              |
| OCA-32 | BTSP-based physical linkage among neurons / columns                                   | Preserved while distinguishing structural from transient effective connectivity                                                                                                | 4.8              |
| OCA-33 | Core, matrix, and TRN thalamic functions                                              | Preserved with nucleus / species / task qualifications                                                                                                                         | 4.9              |
| OCA-34 | PFC-MD/AM-thalamic maintenance, attention, and reinforcement loop                     | Preserved as a specific hypothesis requiring causal, pathway-resolved tests                                                                                                    | 4.9              |
| OCA-35 | Hippocampal-entorhinal / Papez co-regulation                                          | Preserved as memory / context participation in recurrent selection                                                                                                             | 4.9              |
| OCA-36 | Sensory-organ-to-layered-cortex recurrent flow                                        | Preserved without forcing a universal layer order on every modality                                                                                                            | 4.9              |

| ID     | First Draft construct or content branch                | Draft 12 in-paper disposition                                                                                                                                                                                                          | Primary location |
|--------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| OCA-37 | Peter Tse and criterial causation                      | Preserved as a precursor bridge, not as the origin of SAN constructs                                                                                                                                                                   | 4.10, 9          |
| OCA-38 | 1/f relation as tonic / phasic support                 | Recovered: spectral slope, amplitude, duration, persistence, and information are separated while the scale-spanning magnitude question remains a declared SAN hypothesis                                                               | 4.10             |
| OCA-39 | Alpha / beta-gamma top-down / bottom-up control        | Recovered: fixed band / layer assignments remain historical hypotheses, while maintained context -> phasic update -> effective drive -> action -> returned evidence is preserved as the positive circuit- and task-dependent mechanism | 4.10             |
| OCA-40 | Miller stenciling versus SAN ink / canvas              | Preserved as a functional comparison with chronology handled separately from scientific support                                                                                                                                        | 4.10, Appendix C |
| OCA-41 | Global loss-of-consciousness correlates                | Preserved as effective-connectivity evidence; necessity / sufficiency for phenomenology not claimed                                                                                                                                    | 4.11             |
| OCA-42 | Perceptron / deep-learning comparison                  | Restored as the contrast motivating richer temporal, dendritic, and recurrent units                                                                                                                                                    | 4.11             |
| OCA-43 | Perceptron-to-Metatron / Artificial Neurology          | Restored with the two Metatron meanings separated and implementation status explicit                                                                                                                                                   | 4.11             |
| OCA-44 | AI applications, BCI, and potentially sentient systems | Preserved as an engineering and ethics program, not an achieved sentience claim                                                                                                                                                        | 4.11, 11.8       |
| OCA-45 | Micah's New Law / signal-dissipation connection        | Preserved as a cross-paper hypothesis; no derivation from neural data claimed                                                                                                                                                          | 4.11             |
| OCA-46 | AGI / Michael Miller collaboration framing             | Historical wording preserved as part of Draft 1's history; current paper remains Micah-authored unless formal coauthorship is approved                                                                                                 | Appendix E       |
| OCA-47 | Influential voices and works                           | Preserved in the in-paper influence and source register below                                                                                                                                                                          | Appendix E       |
| OCA-48 | Public archive, book, podcast, and related-work routes | Preserved as provenance / reading routes, not independent scientific validation                                                                                                                                                        | 10, Appendix E   |

| ID     | First Draft construct or content branch                       | Draft 12 in-paper disposition                                                                                                                                                                                                                    | Primary location |
|--------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| OCA-49 | Constructive / destructive interference and “pruning” wording | Typo / compression repaired by restoring the intended fast interference -> GSTP / GSTD coupling redistribution -> possible slower LTP / LTD or structural-change chain                                                                           | 4.8              |
| OCA-50 | Deterministic synaptic-release wording                        | Retroactive correction repaired: state-dependent release probability is the measurement model; unmeasured causal microstate remains a distinct SAN hypothesis with a residual-variance test                                                      | 4.7              |
| OCA-51 | 1/f, magnitude, amplitude, and duration wording               | Retroactive correction repaired: distinct measurements are preserved, and the original strength-plus-persistence and cross-scale-law questions remain positively stated and testable                                                             | 4.10             |
| OCA-52 | Bat, electrical valence, and building-PWD action bridge       | Retroactive correction repaired: 2017 send / return ancestry, 2020/2022 bat observables, 2022 valence / PWD synthesis, and the non-backdated 2026 receiver-drive formalization are separately preserved                                          | 4.10             |
| OCA-53 | Layer-5/PV proprioceptive wording                             | Retroactive correction repaired: the first public source’s compressed carrier sentence is resolved by its own later pathway account and by circuit evidence into ascending body-state input, Layer-5 motor output, and local PV+ timing / gating | 3.9, 4.10, 10.4  |
| OCA-54 | Alternating Transmission Protocol and the two ATP meanings    | Retroactive omission repaired: the 2017 feedback-confirmed / loss-tolerant protocol and target-minus-current BCI residual are restored separately from the later ATP-energy reinforcement hypothesis                                             | 4.7, 10          |

| ID     | First Draft construct or content branch                       | Draft 12 in-paper disposition                                                                                                                                                                                                                                                                                                                                                                   | Primary location             |
|--------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| OCA-55 | Tonic canvas, phasic distinction, and the Theory of Redness   | Retroactive correction repaired: LFP / band measurements remain evidence-limited, while SAN's positive low-information context -> content-specific differential -> distributed detection/re-expression -> recurrent embodied consequence account is restored and given a discriminating test                                                                                                    | 3.5, 4.10, 11.6              |
| OCA-56 | PWD vocabulary and adjacent-term comparison order             | Structural correction locked: the receiver problem and SAN operation are explained before PWD is named; phase coding, coherence, and traveling-wave comparisons occur later and identify the additional receiver, expectation, route, state, and consequence variables                                                                                                                          | Abstract, 1, 3.4, 9          |
| OCA-57 | Positive figure mechanism versus bounded evidence / test role | Visual correction remains lossless only when the positive SAN operation is explained before its empirical boundary: the captions now preserve BOT sampling-to-update-to-propagation and GCS context-to-drive-to-action-to-return; Figures 2A-2C compose local reception, population reconstruction, and recurrent nesting; Figure 5 is explicitly one neural assay within the wider COT program | Figure status, 3.7, 3.9, 4.1 |

## Appendix E. First Draft influence, source, and collaboration register

The First Draft named a broad intellectual neighborhood. To preserve that history without implying endorsement or equivalence, this revision records those routes here. The scientific claims of this paper continue to depend on claim-specific primary citations, not on appearing in this register.

**Historical computational and systems influences:** Alan Turing; Claude Shannon; Norbert Wiener; John von Neumann; Donald Hebb; Marvin Minsky; Valentino Braitenberg; Douglas Hofstadter; Stephen Wolfram; Yann LeCun; Demis Hassabis; David Silver; Albert-László Barabási; and the perceptron, multilayer-network, reinforcement-learning, sparse-representation, predictive-coding, and dynamical- systems traditions.

**Neuroscience, consciousness, and biological-control influences:** Santiago Ramón y Cajal; Eric Kandel; György Buzsáki; Steven Strogatz; Peter Ulric Tse; Francisco Varela; Wolf Singer; Pascal Fries; Karl Friston; Jeff Hawkins; Olaf Sporns; Earl K. Miller; Michael Levin; Jon Lieff; Antonio Damasio; Stanislas Dehaene; Patricia Churchland; Christof Koch; Francis Crick; Giulio Tononi; Marcello Massimini; Michael Graziano;

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**Physics, mathematics, and philosophy influences retained from the First Draft:** Albert Einstein; Niels Bohr; Erwin Schrödinger; Max Planck; John Bell; David Bohm; Louis de Broglie; Eugene Wigner; Roger Penrose; Basil J. Hiley; Paul Dirac; Isaac Newton; Jacob Bekenstein; Leonard Susskind; Kip Thorne; Freeman Dyson; Netta Engelhardt; Ivette Fuentes; Sabine Hossenfelder; David Chalmers; Daniel Dennett; Kuramoto-model, nonlinear-dynamics, network-science, thermodynamic, relational, and decoherence traditions. Their inclusion records influence or comparison, not agreement with SAN.

**Authorship and collaboration history:** The First Draft described a prospective Agentic General Intelligence collaboration involving Michael S. P. Miller and *Building Minds with Patterns*. Retaining that sentence is part of lossless history. It does not by itself establish contribution to this revision or present coauthorship. Draft 12 remains authored by Micah Blumberg unless a formal contribution review and author approval changes the byline.

**Source-artifact boundary:** The First Draft extraction also contains the sentence “Can you convert this to LaTeX format inside a markdown container?” immediately between its choice discussion and Section 8. That sentence is preserved here as part of the source history but is classified as an authoring / conversion instruction, not a SAN proposition. Removing it from the scientific narrative therefore does not retire any mechanism, prediction, example, or priority claim.

**Micah source routes retained from the First Draft:** the 2022 Self Aware Networks GitHub corpus; the 2011, 2012, and 2017 public video / podcast genealogy; *Bridging Molecular Mechanisms and Neural Oscillatory Dynamics*; the original OCA DOI; the Neural Lace Podcast; SelfAwareNetworks.com; and the author’s related SAN, COT, BOT, NAPOT, PWD, Super Dark Time, Super Information Theory, and informational-dissipation publications. These are first-party provenance sources. They establish what Micah publicly articulated at a date; they do not independently validate the biology.

**Historical interactive route:** The First Draft’s Key References section also linked a custom *Self-Aware Networks GPT* at <https://chatgpt.com/g/g-FA3lrTWTq-self-aware-networks-gpt>. The route is retained as source history and may change or become unavailable. It is not a claim-bearing citation, an independent validator, or a dependency of the present theory and methods.

**First Draft historical reading and source record (complete record-level conservation):** The earlier compact author-year census was not lossless: it omitted titles, publication details, explanatory annotations, source routes, and the original license text. The 113 records below restore OCA-P266 through OCA-P378 inside this paper. LaTeX display commands are normalized for reading, but each stable ID remains bound to the immutable First Draft paragraph hash in oca-draft1-paragraph-inventory.json. This is a historical source register, not a claim that every item is verified, cited as present evidence, equivalent to SAN, or accurate in every bibliographic detail. The current numbered References remain the claim-bearing evidentiary bibliography; CR-005 requires identity, locator, and claim-use audit before any historical item is promoted into that list.

## E.1 Phase Wave Differentials

- **OCA-P266.** Lisman, J.E. & Jensen, O. (2013). “The  $\theta$ - $\gamma$  neural code,” *Neuron*, 77(6), 1002-1016.
- **OCA-P267.** Proposes that information is encoded by gamma spikes’ phase relative to theta, illustrating phase differences as information units.

- **OCA-P268.** Montemurro, M.A. et al. (2008). “Phase-of-firing coding of natural visual stimuli in primary visual cortex,” *Current Biology*, 18(5), 375-380.
- **OCA-P269.** Shows that precise firing phase relative to ongoing rhythms carries visual information beyond firing rate.
- **OCA-P270.** Palmigiano, A. et al. (2017). “Flexible information routing by transient synchrony,” *Neuron*, 94(5), 1036-1051.e6.
- **OCA-P271.** Illustrates how dynamic phase adjustments can route information selectively across networks.

## **E.2 Deterministic, Wave-Based Consciousness**

- **OCA-P272.** Friston, K. (2010). “The free-energy principle: a unified brain theory?” *Nature Reviews Neuroscience*, 11(2), 127-138.
- **OCA-P273.** Frames brain function as deterministic inference minimizing free energy, echoing SAN’s phase realignment mechanisms.
- **OCA-P274.** Acebrón, J.A. et al. (2005). “The Kuramoto model: a simple paradigm for synchronization phenomena,” *Reviews of Modern Physics*, 77(1), 137-185.
- **OCA-P275.** Provides mathematical proof that coupled oscillators self-synchronize, underpinning SAN’s self-organizing conscious states.
- **OCA-P276.** Breakspear, M. (2017). “Dynamic models of large-scale brain activity,” *Nature Neuroscience*, 20(3), 340-352.
- **OCA-P277.** Reviews deterministic dynamical systems models that reproduce cognitive phenomena via oscillatory networks.

## **E.3 Agentic Brain Dynamics**

- **OCA-P278.** Edelman, G.M. (2003). “Naturalizing consciousness: a theoretical framework,” *PNAS*, 100(9), 5520-5524.
- **OCA-P279.** Introduces Neural Darwinism - dynamic neuronal populations that cooperate and compete - paralleling SAN’s multi-scale agency.
- **OCA-P280.** Hawkins, J., Ahmad, S. & Dubinsky, D. (2017). “A Theory of How Columns in the Neocortex Enable Learning the Structure of the World,” *Frontiers in Neural Circuits*, 11, 81.
- **OCA-P281.** Treats each cortical column as an autonomous agent learning complete models, resonant with SAN’s column-based arrays.
- **OCA-P282.** Minsky, M. (1986). *The Society of Mind*. Simon & Schuster.
- **OCA-P283.** Proposes mind as many simple agents; conceptually underpins SAN’s view of neurons and circuits as interacting agents.

## **E.4 Oscillatory Binding and Conscious Integration**

- **OCA-P284.** Singer, W. (1999). “Neuronal synchrony: a versatile code for the definition of relations?” *Neuron*, 24(1), 49-65.
- **OCA-P285.** Reviews how gamma synchrony binds features into coherent percepts, foundational to SAN’s binding-by-oscillation.
- **OCA-P286.** Fries, P. (2015). “Rhythms for cognition: communication through coherence,” *Neuron*, 88(1), 220-235.
- **OCA-P287.** Elaborates how phase alignment gates inter-areal communication, enabling dynamic binding and selection.
- **OCA-P288.** Uhlhaas, P.J. & Singer, W. (2010). “Abnormal neural oscillations and synchrony in schizophrenia,” *Nature Reviews Neuroscience*, 11(2), 100-113.



- **OCA-P289.** Demonstrates that disrupted synchrony correlates with cognitive fragmentation, highlighting binding's role in consciousness.

## E.5 Micah's New Law of Thermodynamics

- **OCA-P290.** Strogatz, S.H. (2003). *Sync: How Order Emerges from Chaos in the Universe, Nature, and Daily Life*. Hyperion.
- **OCA-P291.** Illustrates universal synchronization phenomena, supporting the signal-dissipation principle of Micah's Law.
- **OCA-P292.** Wolfram, S. (2002). *A New Kind of Science*. Wolfram Media.
- **OCA-P293.** Argues that simple computational rules iterated yield complexity, aligning with Micah's view of equilibration as algorithmic.
- **OCA-P294.** Friston, K.J., Levin, M., Sengupta, B. & Pezzulo, G. (2015). "Knowing one's place: a free-energy approach to pattern regulation," *Journal of The Royal Society Interface*, 12(105), 20141383.
- **OCA-P295.** Extends the Free Energy Principle to biological pattern regulation, echoing Micah's Law of iterative difference minimization.

## E.6 Further Reading

- **OCA-P296.** Blumberg, Micah (2025). *Super Dark Time*. Figshare Journal Contribution. DOI: [doi:10.6084/m9.figshare.28284545](https://doi.org/10.6084/m9.figshare.28284545).
- **OCA-P297.** Blumberg, Micah (2025). *Micah's New Law of Thermodynamics: A Signal-Dissipation Framework for Equilibrium and Consciousness*. Figshare Journal Contribution. DOI: [10.6084/m9.figshare.28264340](https://doi.org/10.6084/m9.figshare.28264340).
- **OCA-P298.** Blumberg, Micah (2025). *Super Information Theory*. Figshare Journal Contribution. DOI: [doi:10.6084/m9.figshare.28379318](https://doi.org/10.6084/m9.figshare.28379318).
- **OCA-P299.** Blumberg, Micah (2025). *Bridging Molecular Mechanisms and Neural Oscillatory Dynamics: Explore how synaptic modulation and pattern generation create the brain's seamless volumetric three-dimensional conscious experience*. Available online at Amazon: <https://www.amazon.com/dp/B0DL4701875>, ASIN: B0DL4701875.
- **OCA-P300.** These platforms expand upon the foundational research, offering broader perspectives on consciousness frameworks and computational neuroscience.
- **OCA-P301.** SVGN.io News features many articles with similar content from the same author as this paper: <https://www.svgn.io/p/a-new-book-out-today-bridging-molecular>.
- **OCA-P302. Self Aware Networks Online Archive:** Comprehensive time-stamped notes and original research materials spanning over a decade are available in the Self Aware Networks GitHub repository.
- **OCA-P303.** This archive provides detailed documentation of the evolution and refinement of foundational theories, including Super Dark Time (also previously referred to as Quantum Gradient Time Crystal Dilation and Dark Time Theory),
- **OCA-P304.** Micah's New Law of Thermodynamics, Neural Array Projection Oscillation Tomography (NAPOT), and Self Aware Networks theory of mind. Accessible at: <https://github.com/v5ma/selfawarenetworks>.
- **OCA-P305. The Neural Lace Podcast:** Explore discussions and analyses regarding consciousness, neuroscience advancements, neural synchronization, EEG-to-WebVR integration, and theoretical physics. The podcast content provides further insight into the conceptual background and implications of the theories presented in the cited works. Find episodes of the Neural Lace Podcast via this old link: <http://vrma.io>
- **OCA-P306. Supplementary Websites and Resources:** Further materials, related projects,

and additional context for the research presented can be accessed via the following websites: [selfawareneuralnetworks.com](http://selfawareneuralnetworks.com), [selfawarenetworks.com](http://selfawarenetworks.com)

## E.7 Influential Voices

- **OCA-P307.** Warm thanks to authors, writers, scientists, mathematicians, or theorists like: Steven Strogatz,
- **OCA-P308.** Peter Ulric Tse, Stephen Wolfram, György Buzsáki, Dario Nardi, Luis Pessoa, Grace Lindsay,
- **OCA-P309.** Oliver Sacks, Michael Graziano, Michael Levin, Michael Gazzaniga, Jeff Hawkins, Nicholas Humphrey,
- **OCA-P310.** Valentino Braitenberg, Douglas Hofstadter, David Eagleman, Olaf Sporns, Jon Lieff, Donald Hebb,
- **OCA-P311.** Netta Engelhardt, Ivette Fuentes, Jacob Bekenstein, Sabine Hossenfelder, Albert Einstein, John Bell,
- **OCA-P312.** David Bohm, Roger Penrose, Earl K. Miller, Eugene Wigner, Basil J. Hiley, John von Neumann,
- **OCA-P313.** Louis de Broglie, Claude Shannon, Alan Turing, Norbert Wiener, Santiago Ramon y Cajal, Eric Kandel,
- **OCA-P314.** Antonio Damasio, Stanislas Dehaene, Patricia Churchland, Christof Koch, Francis Crick,
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- **OCA-P317.** foundational insights into computation, oscillatory synchronization, and higher-order cognition.
- **OCA-P318.** Their work has significantly shaped the wave-computational perspective laid out here or at least
- **OCA-P319.** influenced my thinking on the topic.

## E.8 Influential Works

- **OCA-P320.** Strogatz, S. H. (2003). *Sync: The Emerging Science of Spontaneous Order*. Hyperion.
- **OCA-P321.** Strogatz, S. H. (1994). *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Westview Press.
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- **OCA-P336.** Schrödinger, E. (1944). *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press.
- **OCA-P337.** Hossenfelder, S. (2020). *Superdeterminism: A Guide for the Perplexed*. arXiv preprint arXiv:2010.01324.
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- **OCA-P342.** Neuroscience and Free Energy Principle
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- **OCA-P344.** Predictions not commands: Active inference in the motor system. *Brain Structure and Function*, 218(3), 611-643.
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**License boundary:** OCA-P377 and OCA-P378 preserve the First Draft’s CC BY-SA 4.0 statement as historical source content. This Version 2.2 preprint candidate proposes CC BY 4.0, consistent with the published Version 2 release. Micah Blumberg retains the final license choice at deposit time; historical preservation is not a silent provider-side license change. # Version-history note

Version 2 narrows the broad first SAN draft into an operational theory-and-methods framework. It adds the phase→gate→message→state→policy chain, edge-conditioned PWD, explicit coincidence events, equal-information comparators, randomized causal estimands, multi-projection identifiability gates, the embodied-vision / GCS experiment, and the 2011-2022 provenance spine. These revisions define a more discriminating proposal; they do not report an empirical confirmation. This revision also separates classical neural and quantum coherence, formalizes receiver-relative access without observer-created physics, requires an explicit transformation for scale-invariance claims and an explicit neighborhood rule for locality, and distinguishes surprise from semantic and causal meaning.

Version 2.1 is the illustrated revision. It preserves the Version 2 scientific claims, references, models, predictions, and limitations while adding eight author-directed explanatory schematics with claim-bounded captions. This version expands the NAPOT sequence into a population-level overview, a bounded cellular receive-integrate-project motif, and a proposed cross-scale nesting diagram. The new figures do not add empirical results. The published Version 2.0 record at [doi:10.5281/zenodo.21329627](https://doi.org/10.5281/zenodo.21329627) remains immutable.

Version 2.2 Draft 13 is the release-framing successor to the lossless Draft 12 integration. It preserves Draft 12 as a rollback baseline and restores the First Draft's micro / meso / macro OCA organization; full COT / BOT / NAPOT relationship; molecular-to-PWD chain; coefficient-of-variation address; dendritic token model; functional-connectivity, BTSP, binding, ephaptic, and spectral-address claims; thalamic, PFC, hippocampal-entorhinal, core / matrix / TRN, layered-sensory, and Sonic-loop accounts; tonic / phasic,  $1/f$ , Peter Tse, choice, Miller, and GCS bridges; entification, quasicrystal, high-dimensional, deep-learning, Perceptron-to-Metatron, Artificial Neurology, influence, and source-history branches. Claims that were too universal or insufficiently supported remain visible but are corrected and bounded rather than deleted. Appendix D is the semantic-conservation acceptance gate for subsequent revisions. Draft 13 changes only release-status and license framing; the scientific, genealogical, evidentiary, and semantic-conservation content remains lossless from Draft 12.